

SECTION 26 05 00 - BASIC ELECTRICAL REQUIREMENTS

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Work included in this Section: All materials, labor, equipment, services, and incidentals necessary to provide and install the Electrical Work as shown on the drawings and as specified hereinafter, including, but not limited to the following:
 - 1. Electrical and telecommunications service provisions as outlined on the drawings, including temporary power for construction.
 - 2. Main switchboard, distribution panels, branch panels, transformers, circuit breakers, and feeders.
 - 3. Branch circuit wiring, wiring devices and connections to all equipment requiring electrical service.
 - 4. Lighting fixtures completely lamped, including switches, raceways and wiring.
 - 5. Emergency egress/exit illumination system.
 - 6. Central Emergency Lighting Inverter.
 - 7. Telecommunications system.
 - 8. Fire Alarm system.
 - 9. Mechanical equipment power connections, and motor starters where noted.
 - 10. Low voltage lighting control system and programming.
 - 11. Emergency Generator and Automatic Transfer Switches.
 - 12. Photovoltaic (PV) System.
 - 13. Battery Energy Storage System (BESS).
 - 14. All required incidental work, such as roof flashing, electrical testing, title 24 acceptance testing, and temporary power.
 - 15. Any other electrical work as might reasonably be implied as required, even though not specifically mentioned herein or shown on the the drawings.
 - 16. It is the intent of the drawings and specifications that systems be complete and, except as otherwise noted, be ready for operation.

1.2 RELATED WORK

- A. Division 1 - General Requirements
- B. Division 9 - Finishes
- C. Division 23 - Mechanical
- D. Section 07270 - Firestopping

1.3 INCORPORATED DOCUMENTS

- A. Requirements of the General Conditions, Supplementary Conditions, and Division 1 Sections apply to all work in this Section, unless modified herein.
- B. Published specifications, standard tests or recommended methods of trade, industry or government organizations apply to work of this Section where cited by abbreviations noted below, unless modified herein.
 - 1. 2022 California Code of Regulations.
 - 2. 2022 California Building Standards Administrative Code, Part 1, Title 24, C.C.R.

3. 2022 California Building Code (CBC), Part 2, Title 24, C.C.R. (Based on 2021 International Building Code with 2022 California Amendments).
4. 2022 California Electrical Code (CEC), Part 3, Title 24, C.C.R. (Based on 2017 National Electrical Code with 2022 California Amendments).
5. California Energy Code, Part 6, Title 24, C.C.R.
6. 2022 California Fire Code (CFC), Part 9, Title 24, C.C.R. (Based on 2021 International Fire Code with 2022 California Amendments).
7. 2022 California Green Building Standards (CALGreen) Code, Part 11.
8. American Society of Civil Engineers 7-16 (ASCE/SEI), Minimum Design Loads for Buildings and Other Structures.
9. Underwriters' Laboratories, Inc. (UL).
10. Local Utility Company regulations.

C. All State and Municipal Codes and Ordinances.

1.4 CONDITIONS AT SITE:

- A. Visit to site is required of all bidders prior to submission of bid. All will be held to have familiarized themselves with all discernible conditions and no extra payment will be allowed for work required because of these conditions, whether specifically mentioned or not.
- B. Lines of other services that are damaged as a result of this work shall promptly be repaired at no expense to the Owner to the complete satisfaction of the Owner.

1.5 QUALITY ASSURANCE

- A. Conformance:
 1. All work shall conform to the applicable requirements of Article 1.03 above.
 2. The Contractor shall notify the Architect, prior to submission of bid, about any part of the design, which fails to comply with abovementioned requirements.
 3. If after contract is awarded, minor changes and additions are required by aforementioned authorities, even though such work is not shown on the drawings or covered in the specifications, they shall be included at Contractor's expense.
- B. Coordination:
 1. The Contractor shall become familiar with the conditions at the job site, and with the drawings and specifications and plan the installation of the electrical work to conform with the existing conditions and that shown and specified so as to provide the best possible assembly of the combined work of all trades.
 2. The Contractor shall work out in advance all "tight" conditions, involving all trades and if found necessary, supplementary drawings shall be prepared by this Contractor, for the Architect's approval, before work proceeds in these areas. No additional costs will be considered for work, which must be relocated due to conflicts with the work of other trades.
 3. The Contractor shall coordinate and verify all backbox, device, lighting fixture, or equipment mounting requirements with the devices or equipment to be installed, prior to rough in.

1.6 SUBMITTALS

- A. Product Data:
 1. Comply with the provisions of Section 01 33 00 - Submittals.
 2. Within 15 days after award of the Contract, submit:

- a. Complete electrical, lighting, and signal systems material list of all items proposed to be furnished and installed under this Division. Provide manufacturers data sheets for all devices, raceways, fixtures, equipment, and related products to be used for the Division 26, 27, and 28 work.
- b. Manufacturers' specifications and other data required demonstrating compliance with the specified requirements.
- c. Manufacturers' recommended installation procedures which, when approved by the Architect, shall become the basis for inspecting and accepting or rejecting actual installation procedures used on the work.
3. Shop Drawings: Furnish shop drawings and/or equipment cuts for the following:
 - a. Light fixtures including lamps and ballasts
 - b. Main Switchboard, distribution panels, branch panel boards, and transformers. Panel board submittals shall include diagrams of the circuit breaker arrangements in the panels. Arrange circuit breakers in panels exactly as shown on the panel schedules in the construction documents.
 - c. Scaled drawings of equipment layout at all exterior equipment pad and interior electrical and telecom rooms.
 - d. Fire alarm system.
 - e. Telecommunications system.
 - f. Disconnect switches.
 - g. Low voltage lighting control system
 - h. Central Emergency Lighting Inverter.
 - i. Arc flash, Short Circuit, and Protective Device Coordination Study.
 - j. Photovoltaic system as indicated on the drawings (Bid Alternate).
 - k. Battery energy storage system as indicated on the drawings (Bid Alternate).
 - l. Emergency Generator and Automatic Transfer Switch as indicated on the drawings.
4. Test Reports:
 - a. Factory Tests: As specified for specific equipment.
 - b. Field Tests: Performance tests as specified for specific equipment.
 - c. Megger Tests: As specified under TESTING.
 - d. When series rated circuit breakers are used, provide a letter from the manufacturer of the equipment confirming that U.L. series rating exists for all protective devices. State the available fault current from the Utility Company and indicate that the overcurrent devices exceed the available fault current at the respective point of protection.
 - e. Special Seismic Certification documentation as per CBC Section 1616A and ASCE/SEI 7-16 requirements for all equipment defined as 'critical' with an importance factor of 1.5 in Paragraph 1.10.M.3 of this Section.
 - f. Manufacturer's Seismic Certification or Project-Specific Design of Supports and Attachments for all other equipment and fixtures as per CBC Section 1616A and ASCE/SEI 7-16 requirements.
5. Maintenance and Operating Manuals:
 - a. Systems Description: Description of operating procedures.
 - b. Controls: Diagrams and description of operation of each system.
 - c. Equipment: Manufacturer's brochures, ratings, certified shop drawings, maintenance data, and parts lists with part numbers. Mark each sheet with equipment identification number and actual installed condition.
 - d. Materials and Accessories: Manufacturer's brochures, parts lists with part numbers, and maintenance data where applicable. Mark each sheet with identification number of system and location of installation.
 - e. The Maintenance and Operation Manual shall be presented in a bookmarked PDF file with tabbed sections as stated below. Provide all information in each section as stated below.
 - 1) 26 2400:

- (a) Insert the approved submittals for the main switchboard and panelboards, and for medium voltage switchgear and transformers if specified herein and/or indicated on the drawings.
 - (b) Provide the names, addresses and telephone numbers of the manufacturer and the two closest manufacturer's representatives of the equipment.
- 2) 26 5101:
 - (a) Insert the approved submittals for the light fixtures.
 - (b) Highlight the lamp type that was installed for each light fixture.
 - (c) Provide the names, address and telephone numbers of the manufacturer and the closest manufacturer's representative for each light fixture.
- 3) 26 5101:
 - (a) Insert the approved submittals for the motion sensing light control equipment.
 - (b) Insert all operating instructions.
 - (c) Provide the names, address and telephone number of the manufacturer and the closest manufacturer's representative of the equipment.
 - (d) Include the manufacturer's recommended maintenance of the equipment.
- 4) 26 5700:
 - (a) Insert the approved submittals for the low voltage lighting control equipment.
 - (b) Insert all operating instructions.
 - (c) Provide the names, address and telephone number of the manufacturer and the closest manufacturer's representative of the equipment.
 - (d) Include the manufacturer's recommended maintenance of the equipment.
- 5) 27 0000:
 - (a) Insert the approved submittals for the telecommunications system.
 - (b) Provide the names, address and telephone number of the manufacturer and the closest manufacturer's representative of the equipment.
 - (c) Include the manufacturer's recommended maintenance of the equipment.
- 6) 28 3101:
 - (a) Complete the "Record Of Completion" entirely.
 - (b) In the "Download File" indicate the exact equipment that the Monitor Modules are monitoring. i.e. fire sprinkler flow switches, tamper switches, etc..
 - (c) Simplify the Download File so that it coincides with the submitted and approved fire alarm single line diagram.
 - (d) Provide the names, address and telephone number of the manufacturer and the closest manufacturer's representative of the equipment.
 - (e) Include the manufacturer's recommended maintenance of the equipment.
 - (f) Insert an abbreviated data sheet that states how to test, reset and silence the fire alarm system.
 - (g) Insert the name and telephone number of the Central Station that receives the alarms, and the proper sequence to follow during an alarm.
- 7) 26 32 00 and 26 36 00
 - (a) Insert the approved submittals for the generator, fuel tank, base tank alarm panel, automatic transfer switch, and remote annunciator.
 - (b) Provide the names, address, and telephone numbers of the manufacturer and the two closest manufacturer's representatives of the equipment.
 - (c) Insert all operating instructions.
 - (d) Include the manufacturer's recommended maintenance of the equipment.
- 8) For all other systems specified herein and/or indicated on the drawings:
 - (a) Insert the approved submittals for each system.
 - (b) Insert all operating instructions for each system.
 - (c) Provide the names, address and telephone number of the manufacturer and the closest manufacturer's representative for each system.
 - (d) Include the manufacturer's recommended maintenance of each system.

- 9) 26 0800:
 - (a) Insert all systems testing results.
- 6. Record Documents: "As-builts": As specified under Paragraph 3.2 of this Section.

1.7 DELIVERY, STORAGE AND HANDLING

- A. Protection: Use all means necessary to protect the materials of this Section before, during, and after installation and to protect the work and materials of all trades.
- B. Delivery and Storage: Deliver all materials to the job site in their original containers with all labels intact and legible at time of use. Store in strict accordance with approved manufacturers' recommendations.
- C. Replacements: In the event of damage, immediately make all repairs and replacements necessary to the approval of the Architect and at no additional cost to the Owner.
- D. This Contractor shall personally, or through an authorized representative, check all materials upon receipt at jobsite for conformance with approved shop drawings and/or plans and specifications.

1.8 SCHEDULING/SEQUENCING

- A. Place orders for all equipment in time to prevent any delay in construction schedule or completion of project. If any materials or equipment are not ordered in time, additional charges made by equipment manufacturers to complete their equipment in time to meet the construction schedule, together with any special handling charges, shall be borne by this Contractor.
- B. The Contractor shall coordinate production and delivery schedule for all Owner-supplied equipment with the equipment suppliers to ensure that all Owner-supplied equipment is delivered to site in coordination with the construction schedule and in such a manner as to cause no delays in completion of the Contract as scheduled.

1.9 REQUIREMENTS

- A. The contract drawings indicate the extent and general arrangements of the conduit wiring systems, etc. If any departures from the contract drawings are deemed necessary by the Contractor, details of such departures and the reasons therefore shall be submitted as soon as practicable, and within thirty-five (35) days after award of the electrical contract.
- B. Unless material list and data is received as a complete and all-inclusive submittal within the stipulated time all items shall be provided as specified, with no deviations permitted.
- C. Any and all additional costs incurred by the substitution of electrical material or equipment, or installation thereof, whether architectural, structural, plumbing, mechanical or electrical, shall be borne by the Contractor under this Section.
- D. Burden of proof of equality of any substitution for a specified product is the responsibility of this Contractor.
- E. Where required by Architect to ascertain equality of substitute product, Contractor may be requested to provide the specified item and the submitted substitution for comparison, at no additional cost to the Owner.

1.10 SEISMIC CERTIFICATION AND INSTALLATION OF EQUIPMENT

- A. See Architectural and Structural Drawings and Specifications for description of Occupancy Group and Seismic Design Category applicable to this project.
- B. Provide Special Seismic Certification per CBC Section 1616A and ASCE/SEI 7-16 for all equipment and components defined as critical with an importance factor 1.5 in Paragraph 1.10.M.3 of this Section.
- C. Special Seismic Certification shall require either certification through approved analytical method or approved shake table testing in accordance with Section 13.2.5 of ASCE/SEI 7-16 or experience data in accordance with Section 13.2.6 of ASCE/SEI 7-16.
- D. Manufacturer's Seismic Certification or Project-Specific Design of Supports and Attachments for all other equipment and fixtures as per CBC Section 1616A and ASCE/SEI 7-16 requirements.
- E. Provide seismic restraints per applicable code and as specified or indicated. Design restraints to prevent permanent displacement in any direction caused by lateral motion, overturning, or uplift.
- F. Rigidly Supported Equipment, Conduits, and Raceways.
- G. Lighting:
 - 1. Fasten lighting fixtures in suspended ceilings to ceiling grid system or otherwise support from the structures as specified herein and as per details indicated on the Drawings. Comply with National Electric Code (NEC) Article 410.
 - 2. Suspension systems for light fixtures shall allow fixtures to swing a minimum of 45 degrees from the vertical in all directions without contacting obstructions.
 - 3. Free-swinging suspension systems shall have a safety wire or cable attached to the fixture and structure at each support. The wire shall be capable of supporting four times the weight of the lights.
 - 4. Point-source fixtures: provide slack wires to structure at two diagonal corners.
 - 5. Troffer fixtures: provide hold-down clip at each fixture corner, and slack wires to structure at two diagonal corners.
 - 6. Supports for pendants: Provide diagonal seismic wire restraints per Code.
- H. Components supported by chains or simply suspended from above are not required to meet lateral seismic force requirements and seismic relative displacement requirements provided that they cannot be damaged or cannot damage any other component when subject to seismic motion. They must have ductile or articulating connections to the structure at the point of attachment.
- I. Electrical Cabinets:
 - 1. Electrical cabinet design shall conform to National Electrical Manufacturers Association (NEMA) 250 and NEMA ICS6 standards. Cutouts in the lower shear panel that do not appear to have been made by the manufacturer and significantly reduce the strength of the cabinet are not permitted unless analysis demonstrates that the remaining strength is sufficient.
 - 2. Single freestanding cabinets shall have a minimum of four anchor bolts designed and specified with one anchor located at each corner.

3. Multiple sections of cabinets or enclosures located adjacent to each other shall be bolted together. Minimum acceptable bolting is three bolts in the front and back along the adjacent vertical faces - 6 bolts total.
 4. Multiple cabinets bolted together to form a section or line-up shall have at a minimum two anchors specified for each cabinet, one at the front and one at the rear.
 5. Base anchorage shall be installed through anchor points designated by the Manufacturer. The largest bolt diameter for the anchor hole provided in the equipment shall be provided.
 6. A latch or fastener to prevent opening during an earthquake event and damaging the cabinet and internal components shall secure all doors.
 7. Slide-out components in electrical control panels, etc., shall have a latching mechanism to hold contents in place.
 8. Attached cabling shall have adequate slack or flexibility between the cabinets and surrounding structure supporting the conduit to preclude severing of the cabling due to differential seismic displacements.
- J. The design load shall include the effects of loading on the equipment imposed by attached utility or service lines that are also attached to separate structures.
- K. The attachment of additional external items is not permitted unless such items have either been provided by the Manufacturer, or analysis shows that their effects are supported by design.
- L. Conduit and their connections shall be constructed of ductile materials unless otherwise approved by the Architect. Conduits and their connections constructed of non-ductile materials (e.g., cast iron, no-hub pipe and plastic) shall have brace lengths reduced to one-half that allowed for ductile material.
1. All trapeze assemblies supporting conduit shall be braced to resist CBC design forces considering the total weight of the elements on the trapeze.
 2. Seismic restraint spacing shall be in accordance with hanger spacing.
- M. Critical Equipment:
1. Design with importance factor of 1.5.
 2. Provide Special Seismic Certification for all equipment and components and their installation per CBC and ASCE/SEI requirements.
 3. Critical Equipment shall include the following:
 - a. Fire Alarm system equipment.
 - b. Central Emergency Lighting Inverter.
 - c. Emergency Generator and Automatic Transfer Switches if specified herein and/or indicated on the drawings.
 - d. Emergency power system panels and transformers as indicated on the drawings.
- N. Seismic Design Submittals: For all Critical Equipment included in paragraph 26 05 00.1.10.M.3.
1. The Manufacturer of each item of critical equipment shall arrange for the testing or analysis by an approved agency of each component and assembly and its mounting system or anchorage.
 2. The Manufacturer shall submit a Certificate of Compliance for each item for approval by the Architect and by the Authority Having Jurisdiction.
 3. Based on Manufacturer's approved submittal, Contractor shall retain the services of a State of California registered Structural Engineer to prepare final installation details and drawings for equipment supports and attachments.
 4. Submit drawings of the equipment showing dimensions, support equipment, connections, and the proper anchorage locations.
 5. Equipment weight and weight distribution (e.g., center of gravity in elevation and plan).
 6. Thickness of sheet metal bases.

7. Seismic Vibration Isolation Devices: Manufacturer's product information indicating class and type. Indicate load ratings as published manufacturer's data or shop drawings. Indicate proper orientation of devices on plan.
 8. Inertia bases and support frames.
 9. Specific details of restraints including anchor bolts and welds and maximum load at each location.
- O. Independent Supports: An independent means of secure support shall be provided for all wiring methods in non-fire-rated assemblies. Where independent support wires are used, they shall be distinguishable by color, tagging, or other effective means.

1.11 GUARANTEE

- A. This Contractor shall guarantee that all work executed under this Section will be free from defects of materials and workmanship for a period of one (1) year or as per the General Conditions of this project, whichever is longer. Dates shall be from the date of final acceptance of the building. The contractor shall further guarantee that he will, at his own expense, repair and replace all such defective work, and all other work damaged thereby, which becomes defective during the term of the guarantee. Such repair or replacement shall be guaranteed for one (1) year from the date of repair or replacement.

1.12 PERMITS AND INSPECTIONS

- A. This Contractor shall arrange for and obtain all required permits and inspections.
- B. Do not allow or cause any of the work to be covered or enclosed until it has been tested and/or inspected.

1.13 IDENTIFICATION

- A. Switchgear, switchboards, distribution panels, and feeder circuit breakers therein, panels, disconnect switches, motor starters, transformers, motor disconnect switches, cabinets, and other apparatus used for the operation of, or control of circuits, appliances or equipment, shall be properly identified by means of engraved laminated plastic descriptive nameplates mounted on apparatus using stainless steel screws. Nameplates shall have white letters with black background and be submitted to the Architect for approval. Cardholders in any form are not acceptable.
- B. Provide p-touch style labeling of circuit designations for all receptacles on the project.
- C. Each branch circuit of panel boards to have a permanently fixed number with load directory, mounted under celluloid on inside of cabinet door, showing circuit numbers and typewritten description of equipment supplied by breakers. Where changes are made to existing panelboards, newly typewritten circuit directories shall be prepared to replace existing directories.
- D. Provide label on all motors: "Caution. Automatic equipment. May start at any time."
- E. Provide silk-screened or engraved identification labels on all switch box covers identifying specific loads that are not readily apparent to the user, including electroscreens, projection screens, exhaust fans, audio-visual controls, etc.. Submit proposed labels to Architect for approval prior to manufacture of labels.

- F. Provide identification of all pull boxes, junction boxes, and conduit stub-ups on the project as outlined below:
1. For Power Feeders:
 - a. Stencil cover with identifying circuit number.
 - b. Lettering 1" high.
 - c. Color of lettering black.
 - d. Place lettering on cover in neat manner; run parallel to long sides of box.
 2. For branch circuits, grounding, communication, signal, and control systems boxes and blank conduit stub-outs:
 - a. Paint inside back of each j-box, front of each cover, and ends of each blank conduit stub-out with identifying system color as listed below:
 - 1) 120/208-volt Blue
 - 2) Telephone/Data Grey
 - 3) Ground system Green
 - 4) Fire Alarm Red
 - 5) Lighting control Orange/White
 - 6) Audio/Visual Yellow
 - 7) Security White
 - 8) Emergency Power 120V Blue/Red

PART 2 - PRODUCTS

2.1 GENERAL

- A. Refer to applicable Division 26, 27, and 28 Sections for complete products specifications.

2.2 MATERIALS

- A. Materials of the same type or classification, used for the same purpose, shall be the product of the same manufacturer.

2.3 ACCEPTABLE MANUFACTURERS

- A. Materials shall be of make mentioned elsewhere in this specification. All materials shall be the best of their several kinds, perfectly new and approved by the Underwriters' Laboratories.
- B. Where material, equipment, apparatus or other products are specified by manufacturer, brand name, type or catalog number, such designation is to establish standards of desired quality, style and utility and shall be the basis of the bid. Materials so specified shall be furnished under the contract unless changed by written approval of the Architect. Where two or more designations are listed, choice shall be optional with this Contractor, but this Contractor must submit his choice for final approval.

2.4 POSTED OPERATING INSTRUCTIONS

- A. Furnish approved operating instructions for systems and equipment where indicated in the technical sections for use by operation and maintenance personnel. The operating instructions shall include wiring diagrams, control diagrams, and control sequence for each principal system and equipment. Print or engrave operating instructions and frame under glass or in approved laminated plastic. Post instructions as directed. Attach or post operating instructions adjacent to each principal system and equipment including startup, proper adjustment, operating, lubrication, shutdown, safety precautions, procedure in the event of equipment failure, and other

items of instruction as recommended by the manufacturer of each system or equipment. Provide weather-resistant materials or weatherproof enclosures for operating instruction exposed to the weather. Operating instruction shall not fade when exposed to sunlight and shall be secured to prevent easy removal or peeling.

2.5 CATALOGED PRODUCTS/SERVICE AVAILABILITY

- A. Materials and equipment shall be current products by manufacturers regularly engaged in the production of such products. Products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period shall include applications of equipment and materials under similar circumstances and of similar size. The 2-year period shall be satisfactorily completed by a product for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures. Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6,000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished. The equipment items shall be supported by service organizations which are reasonable convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Examine the areas and conditions under which the work of this Section will be installed. Correct conditions detrimental to the proper and timely completion of the Work. Do not proceed until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Drawings:
 - 1. The general arrangement and location of wiring and equipment is shown on the electrical drawings and shall be installed in accordance therewith, except for minor changes required by conflict with the work of other trades.
 - 2. The Contractor shall coordinate and verify all backbox, device, lighting fixture, or equipment mounting requirements with the devices or equipment to be installed, prior to rough in.
 - 3. Drawings indicate the circuit and panel which supplies each device or fixture. Provide and install conduit and conductors to make all connections from panel to nearest device and from first device to additional devices on same circuit. Conduit size and fill shall satisfy NEC requirements. Two or three different phases supplied by a 3-phase panel may share a single neutral only if circuit positions are adjacent in the panel. Do not exceed 4 #12 or 3 #10 conductors in a 1/2" conduit, 7 #12 or 5 #10 in a 3/4" conduit, and 11 #12 or 9 #10 in a 1" conduit, unless otherwise noted. Provide common handle-tie on breakers for multi-wire branch circuits (with common neutral), per NEC. If more than three current carrying conductors are installed in one conduit, conductor size shall be increased as required per NEC. Do not share neutrals for branch circuit runs to electronic equipment or where noted on the drawings.
 - 4. Drawings indicate the location of all light switches. Where fixtures in a room are controlled by more than one switch, the same lower case letter is drawn adjacent to a switch and each fixture controlled by that switch. Where no lower case letter is adjacent to a switch, all fixtures in the room are controlled by that switch. Provide and install conduit and wire from fixture to switch and between fixtures as required to accomplish switching shown. Do not route branch circuit wiring for light fixtures through switch boxes. Where dimming controls are specified, provide required dimming control wiring in addition to power wiring

- from control device to all controlled light fixtures. Provide separate conduit for dimming control wiring unless otherwise indicated on the drawings.
5. Drawings indicate location of all signal outlet boxes. Provide and install conduit system as required and complete system wiring, unless otherwise noted.
 6. Control wiring is generally not shown on the plans. Contractor shall refer to control diagrams and provide and install all wiring and raceways required to make all interconnections.
 7. All branch circuit wiring No. 12 or No. 10 as noted, all control wiring No. 14, except as noted next to "slash marks" on the drawings, or as noted under "Wire," as specified herein.
 8. All dimensions, together with locations of doors, partitions, etc. are to be taken from the Architectural Drawings, verified at site by this Contractor.
 9. Maintain "as-built" records at all times, showing the exact location of concealed conduits and feeders installed under this contract, and actual numbering of each circuit. Upon completion of work and before acceptance can be considered, this Contractor must forward to the Architect, updated CAD plans, corrected to show the electrical work as actually installed.
 10. All standard 20A branch circuit conductors shall be #12 minimum for up to 75 linear circuit feet, #10 minimum for up to 150 linear circuit feet, and #8 for runs longer than 150 feet.

- B. Measurements: Before ordering any material or closing in any work, verify all measurements on the job. Any differences found between dimensions on the drawings and actual measurements shall be brought to the Architect's attention for consideration before proceeding.

3.3 FIELD QUALITY CONTROL

- A. All workmanship shall be first class and carried out in a manner satisfactory to and approved by the Architect.
- B. This Contractor shall personally, or through an authorized and competent representative, constantly supervise the work and so far as possible keep the same foreman and workmen on the job throughout.

3.4 COORDINATION

- A. In electrical rooms, Contractor is responsible for installation of electrical equipment within the space provided. Contractor shall provide 1/4" scale plans of electrical room layouts, and elevations of steel channel supports (where used or required) of electrical equipment for review and approval prior to any installation or rough-in

3.5 INSTALLATION/APPLICATION/ERECTION

- A. All electrical raceways and devices shall be installed concealed (for raceways) and/or flush mounted (for devices), unless otherwise noted. Provide cut-in boxes and "fish" flexible MC or flex conduit and wire through existing walls to remain, unless shown otherwise on plans. Cut and patch to facilitate such installation to match adjacent and original finish.
- B. All cutting, repairing and structural reinforcing for the installation of this work shall be done by the General Contractor in conformance with the Architect's requirements.

3.6 EMERGENCY POWER SOURCES

- A. All code required emergency source circuits (i.e. from EM Lighting Inverter) shall be installed in separate raceways (from normal power), per 2017 NEC 700.10(B), or the applicable code at the time of permitting.

3.7 TEMPORARY LIGHTING AND POWER

- A. Provide and install temporary lighting and power systems for the duration of construction, of adequate size to accommodate the required lighting and power loads. Coordinate with other trades to insure adequate sizing.
- B. Provide distribution equipment as required to support all construction activities.

3.8 FIRE STOPPING AND FIRE RATED PENETRATIONS

- A. All electrical equipment mounted in, on, or through fire rated construction shall be installed to maintain the fire rating of the construction.
- B. Provide fire rated pads (or other suitable assembly) around all electrical junction boxes in fire rated walls/ceilings/floors to maintain the fire rating.
- C. Provide fire rated construction around all recessed light fixtures and/or panel board / cabinets mounted flush in fire rated walls to maintain the fire rating. Coordinate depth of construction with other trades to avoid conflicts.
- D. Conduit sleeves shall be provided as a means of routing cables through fire-rated walls or floors. Openings in sleeves and conduits used for system cables and those which remain (empty) spare shall be sealed with an approved fireproof, removable sagging material. Sleeves which pass vertically from floor to floor shall be sealed in a similar manner using an approved re-enterable system. Additional penetrations through rated assemblies necessary for passage of tel/data wiring shall be made using an approved method and permanently sealed after installation of cables.

3.9 ADJUSTING AND CLEANING

- A. All electrical equipment, including existing equipment not "finish painted" under other sections, shall be touched up where finished surface is marred or damaged.
- B. All equipment, lighting fixtures, etc., shall be left in clean condition, with all shipping and otherwise unnecessary labels removed there from.

3.10 SCHEDULES

- A. Coordination: Coordinate installation of electrical items with the schedule for other work to prevent unnecessary delays in the total Work.

3.11 WARNING SIGN MOUNTING

- A. Provide the number of signs required to be readable from each accessible side, but space the signs a maximum of 30 feet apart.

3.12 PAINTING OF EQUIPMENT

- A. Factory Applied: Electrical equipment shall have factory-applied painting systems which shall, as a minimum, meet the requirements of NEMA ICS 6 corrosion-resistance test, except equipment specified to meet requirements of ANSI C37.20 shall have a finish as specified in ANSI C37.20.
- B. Field Applied: Paint electrical equipment as required to match finish or meet safety criteria. Painting shall be as specified in the respective equipment section.

3.13 TESTS

- A. Testing and inspection: See Section 26 08 00 - Testing.

END OF SECTION

SECTION 26 08 00 - TESTING

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Work Included in This Section: All materials, labor, equipment, services, and incidentals necessary to perform the testing and inspection of the electrical work, including but not limited to the general systems noted below:
1. Grounding system.
 2. Lighting system.
 3. Distribution system.
 4. Fire Alarm system.
 5. Lighting control system.
 6. Lighting emergency inverter system.
 7. Telecommunications system.
 8. Title 24 Acceptance Testing.
 9. Test additional work where specified in other Sections of these specifications or where indicated on the drawings (provide all materials, labor, equipment, services, and incidentals necessary to perform the testing and inspection of this Electrical Work):
 - a. Emergency Generator and Automatic Transfer Switches.
 - b. Photovoltaic system.
 - c. Battery Energy Storage System.
 10. Any other electrical work as might reasonably be implied as required, even though not specifically mentioned herein or shown on the drawings.
 11. All work shall comply with Sections 26 05 00 and 26 27 00.
 12. In addition to the general system tests and inspections indicated above, the Contractor shall perform the following inspections and tests. The Contractor shall provide all material, equipment, labor, and technical supervision to perform such tests and inspections:
 - a. System Grounding.
 - b. Switchgear, Switchboards, Distribution Panels, Panelboards.
 - c. Feeders.
 13. The purpose of these tests is to assure that all tested electrical equipment is operational and within industry and manufacturer's tolerances and is installed in accordance with design specifications.

1.2 APPLICABLE CODES, STANDARDS, AND REFERENCES

- A. All inspections and tests shall be in accordance with the International Electrical Testing Association - Acceptance Testing Specifications ATS-2021 (referred to herein as NETA ATS-2021).

1.3 QUALIFICATIONS

- A. Qualifications of the Testing Firm shall be as listed in NETA ATS-2021.

PART 2 - PRODUCTS

2.1 THIS ARTICLE DOES NOT APPLY TO TESTING.

PART 3 - EXECUTION

3.1 GENERAL

- A. Final test and inspection to be conducted in presence of the Authority having Jurisdiction (AHJ) or Inspector of Record (IOR). Test shall be conducted at the expense of, and managed by, the Contractor, at a mutually agreed time. Submit written test report of all tests, with test result values and overall outcome.
- B. All portions of the electrical installation shall be inspected and tested to ensure safety to building occupants, operating personnel, conformity to code authorities and Contract Documents, and for proper system operation.

3.2 INSPECTIONS AND TESTS

- A. Tests: Field tests shall be performed and reports submitted, as per Section 26 05 00, Part 1.
 - 1. Final Inspection Certificates: Prior to final payment approval, deliver to the Owner, with a copy to the Architect, signed certificates of final inspection by the appropriate local authority having jurisdiction.
 - 2. Grounding System:
 - a. All ground connections shall be checked and the entire system shall be checked for continuity. The resistance of grounding electrodes in the system shall be measured using a 3 point fall-of-potential method. The maximum ground resistance shall be three ohms. If the measured ground resistance exceeds three ohms, install (1) additional ground rod, bonded and interconnected with the grounding electrode system.
 - b. Ground tests shall meet or exceed the requirements of the National Electric Code.
 - 3. Lighting Systems:
 - a. The interior and exterior lighting systems shall be checked for proper local controls and operation of entire installation, including the operation of the low voltage lighting control system.
 - 4. Power Distribution System:
 - a. Test main switchboard, distribution boards, panel boards, and transformers for grounds and shorts with mains disconnected from feeders, branch circuits connected and circuit breakers closed, all fixtures in place and permanently connected and grounding jumper to neutral lifted and with all wall switches closed.
 - b. Test each individual circuit at each panelboard with equipment connected for proper operation. Inspect the interior of each panel.
 - c. Check verification of color coding, tagging, numbering, and splice make-up.
 - d. Verify that all conductors associated with each circuit are in same conduit.
 - e. Demonstrate that all lights, jacks, switches, outlets, and equipment operate satisfactorily and as called for.
 - f. Perform megger tests of all new distribution system feeders prior to energizing. All Cables failing megger tests or with evidence of damage shall be removed and replaced in their entirety (no splices), at no cost to the Owner. Damaged cables may not be field repaired without specific approval of the Architect.
 - 5. Fire Alarm System: Verify that all equipment, components, and devices function as specified. Refer to Section 28 3101 for additional testing requirements.

6. Lighting Control System: Verify that all equipment, components, and devices function as specified. Refer to Section 26 5101 for additional testing requirements.
 7. Telecommunications System: Verify that all equipment, components, and devices function as specified. Refer to Section 27 0000 for additional testing requirements.
 8. Where the following systems are specified herein and/or indicated on the drawings, verify that all equipment, components, and devices function as specified and meet all additional testing as described in related individual Sections of this specification:
 - a. Emergency Generator and Automatic Transfer Switches.
 - b. Photovoltaic System.
 - c. Battery Energy Storage System.
- B. Title 24 Acceptance Testing: Contractor shall complete the requirements for Title 24 Acceptance Testing, as per CA Title 24, Part 6.
1. Perform testing requirements as per Title 24 Lighting Acceptance requirements. Testing shall include construction inspection of installed controls, occupancy / motion sensor testing, manual daylighting controls testing, automatic time switch controls testing, and demand response system interface, as applicable.
 2. Complete and submit all required forms for complete Acceptance Testing.
 3. Acceptance tests must be performed or overseen by certified Acceptance Test Technicians.
 4. Obtain required review and approval of Acceptance Forms to allow final certificate of occupancy to be granted.

END OF SECTION

SECTION 26 24 00 - SERVICE AND DISTRIBUTION SYSTEM

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Work Included in This Section: All materials, labor, equipment, services and incidentals necessary to install the electrical work as shown on the drawings and as specified hereinafter, including but not limited to the work listed below.
- B. Underground service distribution conduits and cables where noted for power and telecommunications services, including utility company coordination.
- C. Temporary power for construction.
- D. Concrete pad and ground rods for Utility installed padmounted transformer.
- E. Main switchboard, Distribution Panels, Transformers, Distribution System, Panel Boards, Grounding, and Overcurrent Protective Devices.
- F. All required incidental work, such as excavating, backfilling, testing, and temporary power.
- G. Any other electrical work as might reasonably be implied as required, even though not specifically mentioned herein or shown on the drawings.
- H. All work shall comply with Sections 26 05 00 and 26 27 00.

1.2 RELATED WORK

- A. Division 09 - Finishes
- B. Division 23 - Heating, Ventilating, and Air Conditioning

1.3 SUBMITTALS

- A. Comply with the provisions of Section 26 05 00 - Submittals.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Refer to Section 26 05 00, Part 2 - Products
- B. All new equipment shall match existing.
- C. List of Equipment Manufacturers:
- D. Switchboards and Motor Control Centers
 - 1. Eaton-Cutler Hammer, General Electric, Industrial Electric Manufacturing, Siemens, Schneider-Square D.

- E. Panelboards and Distribution Panel
 - 1. Same manufacturer as Main Switchboard.
- F. Dry-type Transformers
 - 1. Same manufacturer as Main Switchboard.

2.2 MATERIALS

- A. Provide and install conduits for primary cables by utility company, concrete pad and grounding for utility company transformer, and conduit for secondary service to main switchboard. Comply with all Utility Co. requirements.
- B. Furnish and install telecommunications service conduits and pullboxes; install conduits to main point-of-entry backboard as indicated on the drawings. All work shall conform to utility company requirements and to Section 26 27 00.
- C. Grounding:
 - 1. Provide and install grounding system as noted on the drawings.
 - 2. Grounding electrode conductor: bare stranded copper type, #4/0 minimum.
 - 3. Install ground wires in rigid conduit.
 - 4. All grounding electrode conductor connections "thermite" or "cad-weld" welded.
 - 5. Use approved pressure type solderless connector or use fusion welding for all connections to and bonding of grounding electrode system. All connections shall be visible, readily accessible for testing purposes. Grounding electrode conductor between the grounding electrode and service equipment: Minimum #4/0.
 - 6. Furnish and install solid copper or copper-clad 5/8" x 10'-0" ground rod(s). Where multiple ground rods are shown, install a minimum of 20'-0" apart. Install ground rods in accessible boxes with covers. Furnish and install 2-#4/0 bare copper cables between multiple ground rods and main switchboard ground bus.
 - 7. Terminate grounding conduits at equipment with ground bushing, with ground wire connected through bushing.
 - 8. Provide No. 12 stranded (green) THHN conductor from outlet box to ground screw of every receptacle.
 - 9. Ground all isolated sections of metallic raceways.
 - 10. Provide #12 minimum stranded (green) THHN conductor sized per NEC, or as noted, connected continuously throughout branch circuit for all circuits, bonded to panel ground bus, and to all electrical devices and equipment enclosures.
 - 11. Grounding electrode installed as follows:
 - a. Place #4/0 bare copper cable in foundation trench; tensioned, supported in such a manner that it cannot be less than two (2) inches from bottom or side of concrete when foundation concrete is poured; not less than one hundred feet of conductor. Embed in foundation with a loop at approximate center, brought out at top of foundation at location of building service equipment for connection to service equipment and for bonding to other parts of the grounding electrode system.
 - b. Use approved pressure type solderless connector or use fusion welding for all connections to grounding electrode. Connection visible, readily accessible for testing purposes. Grounding electrode conductor between the grounding electrode and service equipment: Minimum #4/0.
 - c. Connect grounding electrode system to metallic water service entry metallic cold water pipe (if available) with nonferrous clamp and bare copper cable (sized as required) in conduit. Connection shall be accessible for inspection.

- d. Connect grounding electrode system to effectively grounded building steel as indicated on the drawings. Use exothermic weld, connection shall be accessible for inspection.
 - e. After installation, test system using the three-point fall of potential method only. Record results and submit to Architect for approval. If resistance to ground exceeds three ohms, install an additional ground rod, bonded and interconnected to the grounding electrode system.
 - f. Connect ground bar of separately derived systems (e.g all dry-type transformers) to effectively grounded building steel at the closest possible accessible location, or if building is concrete, or the steel is not effectively grounded, to the main switchboard ground bus: Use #4/0 copper conductor for all connections.
- D. Main Switchboard, Distribution Switchboards, and Distribution Panels:
- 1. General: Switchboard shall be group-mounted type, metal enclosure with ground bus and insulated full capacity neutral bus.
 - 2. Equipment:
 - a. The switchboard shall be braced for a short circuit current of 65,000 amps minimum, and for 100,000 amps when indicated on the drawings. Bracing shall be per NEMA and UL standards.
 - b. The switchboard shall comply with all the requirements of the Utility Company.
 - 3. The switchboard shall be pad-mounted, self-supporting, dead-front and rear, front-operated, front-connected, distribution type. Nema 1 (indoor) or Nema 3R (outdoor). The enclosure shall be 90 inches high, made of cold rolled steel on a structural shape, or formed, steel frame and shall be mounted on two 3-inch, 5-pound continuous channel iron sills, which shall be closed at the ends between the two channels.
 - 4. This contractor is responsible for the complete installation of the switchboard within the space provided (both vertical and horizontal) and shall verify and/or coordinate all dimensions prior to ordering equipment. Proper allowances should be included to allow complete installation and erection.
 - 5. The switchboard shall be a minimum of 24 inches deep and shall be constructed of National Electrical Code (NEC) gauge steel.
 - 6. The switchboard shall be provided with a cable pull section at the top of the switchboard. Provide a minimum 12 inches of vertical clearance between the cable terminal lugs bolted to the switchboard busses and the top and bottom of the switchboard enclosure. Horizontal pull sections and gutters shall be kept free and clear of busses. Where busses cross vertical pull sections, the busses shall be insulated.
 - 7. All connections between bus bars shall be of a bolted type using Belleville washers. Clamps will not be accepted. All bus bars shall be accurately formed, and all holes shall be made in a manner which will permit bus bars and connections to be fitted into place without being forced.
 - 8. The design of all current-carrying devices or parts of the switchboard shall conform to the standard specified in the related sections of Underwriters' Laboratories, Inc. (UL) No. UL-891 and National Electric Manufacturer's Association (NEMA) Standard PB-2, except as these characteristics may be modified herein.
 - 9. Bus bars, connection bars and wiring on the back of the switchboard shall be arranged so that maximum accessibility is provided for cable connections from the front.
 - 10. Ampere ratings for rectangular bus bars shall be in accordance with the temperature rise standard of National Electric Manufacturer's Association (NEMA) and the Underwriters' Laboratories, Inc. (UL).
 - 11. The enclosure shall be chemically cleaned by parkerizing, bonderizing or phosphorizing as a unit after all welding has been completed. The enclosure shall then be painted with a rust-resisting primer coat of paint and shall be finished with a coat of light gray, baked enamel.
 - 12. Each section shall be bussed for the full connected load of that section. Extend bussing to spare circuit breaker "Spaces." Drill busses for future circuit breakers, and provide

breaker connector hardware where indicated on the drawings or where required for ready installation of future circuit breakers.

13. Provide copper bus bars and connections with silver-plated contact surfaces.
14. The contact surfaces and studs of all devices to which bus connections are made shall also have silver-plated surfaces.
15. Provide a 200% rated neutral bus for switchboards supplied with 200% rated feeders (incoming or outgoing). Refer to single line riser diagram for feeder ratings.
16. Locate ground bus, with a cross-section equal to at least 25 percent of the capacity of the main bus rating, in the back of the switchboard and extend bus throughout the length of the switchboard assembly. Ground each housing of the assembly directly to this bus.
17. Rigidly support all bus and connection bars and current transformers.
18. Fit all nuts and connections with locking devices to prevent loosening.
19. Provide load connections with solderless lugs. Factory-install all devices shown on Drawings as specified herein.
20. Provide a bonding strap from the equipment ground bus to the neutral bus.
21. Distribution Panels shall comply with all relevant requirements of the above paragraphs - minimum 12" deep, for floor or wall-mounting.
22. The main switchboard or distribution panel shall include space for future electric vehicle charging station connection circuit breaker(s), with a label to identify the space(s) as "EV Capable", per California Green Building Standards Code.

E. Panelboards:

1. Surface (or flush where indicated on the drawings) mounted, with branch circuits as indicated on the drawings.
2. Enclosures: code gauge galvanized sheet steel with welded full flange end pieces, stretcher- leveled steel trim, backpan and door.
3. Bussing of copper with silver-plated contact surfaces.
4. Provide a 200% rated neutral bus for panels supplied with 200% rated feeders (incoming or outgoing). Refer to single line riser diagram for feeder ratings.
5. Trims on surface-mounted cabinets secured with nickel-plated screws with cup washers, bottom of all trims to have lugs for resting on cabinet flange.
6. Panels shall be 20 inches minimum in width, provided with approved gutter space, barriers and adjustable supports. Doors mounted with concealed hinges provided with combination spring latch and lock. Doors and trims and surface mounted cabinets primed and finished with one coat baked on gray enamel. All visible panel enclosures and covers in finished (occupied) areas shall be painted to match adjacent wall finish.
7. Breakers on same phase to be aligned horizontally. Each panel provided with quantity (5) spare handle locks. Install handle locks on all breakers serving fire alarm equipment.
8. Each branch circuit of panelboards to have a permanently fixed number with one word directory, mounted under celluloid on inside of cabinet door, showing circuit numbers and typewritten description of outlets controlled by breakers. Color code mains and each breaker terminal, same as conductor insulation.
9. Each panel shall be equipped with a copper ground bus.
10. All panels shall be fully bussed to accept future circuit breakers, with breaker hardware provided where indicated on the drawings.
11. Panel board submittals shall include diagrams of the circuit breaker arrangements in the panels. Arrange circuit breakers in panels exactly as shown on the panel schedules in the construction documents - no deviations permitted.

F. Circuit Breakers:

1. General: Circuit breakers shall be molded case rated for 480 or 240 volts, multiple or single pole and amperage rating as shown on the drawings, bolt on, manually operated with "de-ion" arc chutes.
2. For all circuit breakers rated 1,200 Amps or higher, provide an arc energy reduction measure in compliance with NEC 240.87(B), to reduce arc clearing time.

3. Main circuit breaker shall be rated to interrupt the available short circuit current - 65,000 amps RMS unless otherwise indicated on the drawings.
4. Distribution circuit breakers shall be rated for the amps interrupting capacity noted on the drawings or U.L. series rated with the main circuit breaker.
5. Branch circuit breakers shall be rated for the amps interrupting capacity or U.L. series rated with the distribution and main circuit breakers, General Electric type THQB or equal, minimum 10,000 A.I.C for 120/208 volt.
6. Where mechanical equipment is U.L. listed for overcurrent protection with fuses or HACR type circuit breakers, provide fuses where a fused switch is shown. Where the overcurrent protection is a circuit breaker provide HACR, (Heating, Air-Conditioning and Refrigeration) type.
7. Provide switch rated type "SWD" circuit breakers where the circuit breaker is used as a switching device in a panelboard.

G. Dry-Type Transformers:

1. Ventilated type.
2. Dry-type general distribution transformers shall meet the California Title 24 requirements for energy efficiency standards and DOE 10 - CFR, Part 431 (2016) for energy efficient transformers.
3. Transformer shall be 3 phase, 60 Hertz. Primary winding shall be Delta connected and secondary winding shall be Wye connected. The temperature rise at rated voltage and full load shall not exceed 150 degrees C with a 220 degrees C U.L. Component Recognized Insulation System. The windings shall be aluminum or copper.
4. The higher voltage winding shall have quantity (6) 2.5% taps - (2) FCAN and (4) FCBN. Set secondary voltage for 120/208V.
5. Transformer terminals shall be front connected for ease of installation and maintenance.
6. Where the transformers are installed outdoors provide weatherproof drip cover, rodent screen and a weathertight rating of the enclosure.

H. Magnetic starters: shall be rated in accordance with latest published NEMA standards for size and horsepower rating, Eaton-Cutler Hammer A-200 series or equal. Provide with overload sensor in each phase, hand-off-auto switch, red "run" pilotlight, in NEMA 1, NEMA 4X, or NEMA 3R enclosure or in motor control center where indicated. Coil shall be rated 120 VAC. Starters shall be across-the-line nonreversing unless otherwise noted.

1. Contacts: Across-the-line magnetic starters shall be equipped with double break silver alloy contacts. All contacts shall be replaceable without removing power wiring or removing starter from panel. The starter must have straight-through wiring.
2. Coils: Coils shall be of molded construction. All coils shall be replaceable from the front without removing the starter from the panel.
3. Overload Relays and Thermal Units: Overload relays shall be the melting alloy type with a replaceable control circuit module. Thermal units shall be of one-piece construction and interchangeable. The starter shall be inoperative if the thermal unit is removed.

2.3 ARC FLASH EVALUATION STUDY:

- A. An arc flash evaluation study shall be performed to determine, in coordination with the Owner's safety policy, the required personal protective equipment (PPE) when working on energized equipment.
- B. The arc flash evaluation study shall comply with all NFPA 70E and OSHA requirements for calculating and identifying incident energy levels and the corresponding PPE that would be required in each instance.

- C. The calculated incident energy levels and recommended PPE for each location shall be summarized in a tabulated form listing location, circuit identification, and PPE. Discrepancies, problem areas, or inadequacies shall be promptly brought to the Architect's attention.
- D. Arc flash calculations shall be based on values of fault current magnitudes identified in the short-circuit analysis and the associated clearing times of the over current protective devices determined by the coordination study. The settings recommended by the coordination study shall be the basis of arc flash calculations.
- E. Calculation methods shall comply with IEEE Standard 1584 "IEEE Guide for Performing Arc-Flash Hazard Calculations".
- F. Per IEEE Standard 1584, a maximum arc time of two seconds shall be utilized to limit incident energy values.
- G. Recommended settings of all protective equipment based on the short circuit and equipment coordination study shall be implemented prior to attaching arc flash hazard labels to the equipment.
- H. All electrical equipment shall be field marked to indicate where a flash hazard exists in compliance with NEC 110-116. Labels shall be submitted for approval prior to application. Labels shall be clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.
- I. Arc Flash Study Report:
 - 1. The results of the arc flash study shall be summarized in a final report. Five (5) bound copies of the final report shall be submitted to the Architect.
 - 2. The report shall include the following sections:
 - a. Description, purpose, basis, written scope, and a single-line diagram of the portion of the power system which is included within the scope of study.
 - b. Tabulations of circuit breaker, fuse, and other equipment ratings versus calculated arc flash energy levels, and commentary regarding same.
 - c. Tabulation of appropriate tap settings for relay seal-in units.
 - d. Tabulation of arc flash study incident energy levels and PPE requirements.

2.4 SHORT-CIRCUIT ANALYSIS, PROTECTIVE DEVICE EVALUATION, AND COORDINATION STUDY

- A. Scope of Services: Provide a current and complete short-circuit study, equipment interrupting or withstand evaluation, and protective device coordination study for the electrical distribution system..
 - 1. Study shall include all portions of electrical distribution system. Normal and emergency system connections and those which result in maximum fault conditions shall be adequately covered in the study.
 - 2. The study shall be performed by Emerson Network Power, or equal. Study shall be prepared and signed by a California registered Electrical Engineer.
 - 3. In the case of additions or modifications to existing distribution systems, the scope of the Study shall include all new portions of the distribution system, and all existing devices upstream of the distribution system modifications all the way to the facility main service switchboard.
- B. Submittals:

1. Submit Study to Architect for review prior to receiving final acceptance of distribution equipment shop drawings or prior to release of equipment for manufacture. If formal completion of Study may cause delay in equipment manufacture, acceptance from Architect may be obtained for preliminary submittal of sufficient study data to ensure that selection of device ratings and characteristics will be satisfactory.
- C. Short-Circuit Study:
1. The study shall be in accordance with applicable ANSI and IEEE Standards.
 2. The study input data shall include the utility company's primary short-circuit single-and three-phase contribution, with the X/R ratio, the resistance and reactance components of each branch impedance, motor and generator contributions, base quantities selected, and all other applicable circuit parameters.
 3. Short-circuit momentary duties and interrupting duties shall be calculated on the basis of maximum available fault current at each switchgear bus, switchboard, distribution switchboard, panelboard, and other significant locations through the system.
- D. Equipment Evaluation Study:
1. An equipment evaluation study shall be performed to determine the adequacy of circuit breakers, controllers, surge arresters, switches, and fuses by tabulating and comparing the short-circuit ratings of these devices with the available fault currents. Any problem areas or inadequacies in the equipment shall be promptly brought to the Architect's attention.
- E. Protective Device Coordination Study:
1. A protective device coordination study shall be performed to select or to check the selections of power fuse ratings, protective relay characteristics and settings, ratios and characteristics of associated voltage and current transformers, and low-voltage breaker and fuse trip characteristics and settings.
 2. The coordination study shall include all voltage classes of equipment and protective devices. The phase and ground overcurrent protection shall be included, as well as settings for all other adjustable protective devices.
 3. The time-current characteristics of the specified protective devices shall be plotted on appropriate log-log paper. The plots shall include complete titles, representative one-line diagram and legends, significant motor starting characteristics, complete parameters of transformers, complete operating bands of low-voltage circuit breaker trip curves, relay curves and fuse curves. The coordination plots shall indicate the types of protective devices selected, proposed relay taps, time dial and instantaneous trip settings, ANSI transformer magnetizing inrush and withstand curves per ANSI C37.91, cable damage curves, symmetrical and asymmetrical fault currents. All requirements of the current National Electrical Code shall be adhered to. Reasonable coordination intervals and separation of characteristics curves shall be maintained. The coordination plots for phase and ground protective devices shall be provided on a complete system basis. Sufficient curves shall be used to clearly indicate the coordination achieved to each main breaker or fused device, each feeder breaker, and each primary protective device.
 4. The selection and settings of the protective devices shall be provided separately in a tabulated form listing circuit identification, IEEE device number, current transformer ratios, manufacturer, type, range of adjustment, and recommended settings. A tabulation of the recommended power fuse selection shall be provided for all fuses in the system. Discrepancies, problem areas, or inadequacies shall be promptly brought to the Architect's attention.
- F. Study Report:
1. The results of the power system study shall be summarized in a final report. Five (5) bound copies of the final report shall be submitted to the Architect.
 2. The report shall include the following sections:

- a. Description, purpose, basis, written scope, and a single-line diagram of the portion of the power system which is included within the scope of study.
- b. Tabulations of circuit breaker, fuse, and other equipment ratings versus calculated short-circuit duties, and commentary regarding same.
- c. Protective device time versus current coordination curves, tabulations of relay and circuit breaker trip settings, fuse selection, and commentary regarding same.
- d. Fault current tabulations including a definition of terms and a guide for interpretation.
- e. Tabulation of appropriate tap settings for relay seal-in units.
- f. Tabulation of arc flash study incident energy levels and PPE requirements.

PART 3 - EXECUTION

3.1 REFER TO SECTION 26 05 00 FOR DETAILS OF WORK UNDER THIS SECTION.

3.2 INSTALLATION/APPLICATION/ERECTION

- A. Electric Service: Contact the local electric utility company service planning representative and coordinate with and arrange with the utility company for electric service to the project, including finalization of service application as required. Furnish and install all materials and labor necessary for complete installation as noted on drawings. Submit shop drawings and obtain approval from the utility company prior to fabrication. Also provide and install temporary power as required for construction operations.
- B. Telecommunications Service: Contact the local telecommunications and broadband utility company service planning representative and coordinate with and arrange with the utility company for telecommunications service to the project, including finalization of service application as required. Furnish and install all materials and labor necessary for complete installation as noted on the drawings and as required by the utility company.
- C. Excavate and trench as necessary for the electrical installation, and when the work has been installed, inspected and approved, backfill all excavations with clean earth from excavation, or imported sandy soil in maximum 8" (eight-inch) layers, moisten and machine tamp to 95% compaction, and restore the ground and/or paving or floor surfaces to their original condition.
- D. Switchboards and Distribution Panels Installation: Mount as detailed on the drawings.
- E. Motor Connections:
 1. Install motor circuits complete for all motors by other trades
 2. Furnish and install all disconnect switches, outlet boxes, etc., as required by code.
 3. All motor and temperature control low voltage wiring shall be installed and connected by Division 23 Section of specifications, unless otherwise indicated on electrical drawings.
- F. Motor Starters Installation:
 1. Deliver starters to site without thermal overload elements. Determine nameplate rating of each motor, after motor and starter installation, select thermal element rating from measured motor running current and install proper elements in starters.
 - a. Submit chart denoting motor designation, motor H.P., motor running current (N.P.), actual running current fuse/breaker size and thermal element catalog number. Take readings of motor running currents in conjunction with Division 23 - Heating, Ventilating, and Air Conditioning.

3.3 TESTS

- A. Testing and Inspection: See Section 26 08 00 - Testing.

END OF SECTION

SECTION 26 27 00 - BASIC ELECTRICAL MATERIALS AND METHODS

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Work included in this Section: All materials, labor, equipment, services, and incidentals necessary to install the electrical work as shown on the drawings and as specified hereinafter, including but not limited to the work listed below:
 - 1. Raceways, feeders, branch circuit wiring, wiring devices, safety switches and connections to all equipment requiring electric service.
- B. Any other electrical work as might reasonably be implied as required, even though not specifically mentioned herein or shown on the drawings.
- C. All work shall comply with Section 26 05 00.

1.2 RELATED WORK

- A. Division 09 - Finishes
- B. Division 23 - Motors and Mechanical Equipment Installation

1.3 SUBMITTALS

- A. Comply with the provisions of Section 26 05 00.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Refer to Section 26 05 00, Basic Electrical Requirements, Part 2 - Products.
- B. List of Equipment Manufacturers:
- C. Conduit and Conduit Fittings
 - 1. Allied Tube and Conduit, Wester Tube and Conduit, LTV Steel Tubular, National Electric Products, AFC, Republic Steel Corporation, Rome Cable Corporation, United States Steel Corporation, Killark Electric Manufacturing Company, Raco, VAW Aluminum Company, Bridgeport, Steel City, Thomas & Betts, Carlon, O.Z. Gedney, Appleton, Regal.
- D. Wire and Cable (600V)
 - 1. American Wire Company, General Wire and Cable Corporation, Okonite Company, Rome Cable Corporation, Cerrowire, American Insulated Wire, AFC Cable Systems, Essex, Simplex Wire and Cable Company, Southwire.
- E. Solderless Lugs and Grounding Connections
 - 1. Burndy Engineering Company Inc, O.Z. Gedney Company Inc, Penn Union Electric Corporation, Thomas and Betts Company Inc.
- F. Pull Boxes, Gutters, Special Cabinets

1. Schneider-Square D Company, Columbia Electric Manufacturing Company, General Electric Company, Eaton Inc.
- G. Outlet Boxes
1. Appleton Electric Company, Killark Electric Manufacturing Company, Lew Electric Fittings Company, National Electric Products Corporation, Racor, Steel City Electric Company, Carlon, Bowers.
- H. Floor Boxes
1. Steel City Electric Company, Hubbell Inc, RCI, Walker.
- I. Wiring Devices
1. Leviton, Arrow-Hart, Cooper, Hubbell, Lutron, Bryant.
- J. Conduit Racks, Hangers
1. General Electric Company, Killark Electric Manufacturing Company, Caddy, National Electric Products Corporation, Republic Steel Corporation, Rome Cable Corporation, United States Steel Corporation, VAW Aluminum Company, Superstrut, B-Line.
- K. Safety Switches (Disconnect and Fusible)
1. Schneider-Square D Company, Eaton-Cutler Hammer Inc, General Electric Company.
- L. Fuses
1. Bussman Manufacturing Company, Chase-Shawmut Company.
- M. Firestopping
1. 3M, Nelson.

2.2 MATERIALS

- A. Raceways: Only the raceways specified below shall be utilized on this project. Substitutions shall be pre-approved in writing. All bare conduit ends (stub-ups or stub-outs) shall be provided with bushed ends or manufactured insulated throat connectors:
1. Rigid Type - hot dip galvanized or sherardized steel, use on all exterior locations, below grade or in concrete slab, and to 18" on either side of structural expansion joints in floor slabs, with completely watertight, threaded fittings throughout. Compression fittings are not acceptable.
 - a. All rigid steel conduit couplings and elbows in soil or concrete or under membrane to be ½ lap wrapped with Scotch #50 tape and threaded ends coated with T&B #S.C.40 rust inhibitor prior to installation of couplings.
 - b. ½ lap wrap all rigid steel conduit stub-ups from slab or grade to 6" above finished grade level with Scotch #50 tape.
 2. In lieu of rigid steel conduit for power and control raceways and branch circuit conduits in soil or concrete slabs, "Schedule 40" PVC with Schedule 80 PVC conduit elbows and stub-ups may be used with code size (minimum No. 12) ground wire. A "stub-up" is considered to terminate 6" above the finished surface.
 - a. Schedule 80 PVC conduit shall be used in all concrete footings or foundations and to 18" of either side of footings or foundation walls.
 - b. Schedule 80 PVC conduit shall be used in all concrete masonry unit (CMU) walls or columns.
 - c. All conduit runs in concrete floor slabs (where allowed) shall be installed to comply with all applicable CBC and structural codes to maintain the structural integrity of the

- floor slab. Where conflicts occur, alternate routing shall be provided at no additional cost to the Owner.
- d. Where schedule 80 PVC is coupled to schedule 40 or other raceways with differing interior dimensions, each end shall be reamed with a reaming tool to reduce the edge profile for protection of the passing conductors during the pull.
 3. Intermediate metal conduit may be used in all exposed interior locations, except that electrical metallic tubing may be used in some locations as noted below. Utilize steel compression type fittings for all exposed conduit runs, unless otherwise noted. Die-cast zinc fittings are unacceptable.
 4. Electrical metallic tubing shall be used exposed in interior electrical and mechanical rooms, in interior unfinished spaces, and in interior concealed and furred spaces, made up with steel watertight or steel set screw type fittings and couplings. EMT shall not be used in under-building crawl spaces or other areas subject to moisture. Set screws shall have hardened points. Die-cast zinc fittings are unacceptable.
 5. Surface mounted rectangular steel raceways and boxes: use for all surface mounted installations, unless otherwise noted (all catalog numbers listed are Wiremold - equals allowed) - color Ivory, unless otherwise noted;
 - a. #V500 for branch power runs on ceilings and walls (used with V500 series straps, elbows, connectors and V5000 series low profile boxes and covers).
 - b. #2000 or 2400 low profile for larger power run requirements on ceiling or walls (used with V2000 series straps, elbows, connectors and low profile boxes and covers).
 - c. #2400D for dual service power and tel/data run requirements (used with divided V2400 boxes and covers).
 6. Surface mounted rectangular non-metallic dual service raceways; Wiremold #5400 (Ivory) or equal with all required compatible activation covers, bezels, inserts, and blank plates for a complete installation. Refer to drawings for outlet quantities in raceway and feed points. All raceway fed flush from rear with horizontal j-boxes, unless otherwise noted.
 7. Use flexible conduit for all motor, transformer and recessed fixture connections, minimum ½"; "Seal tight" type used outdoors and in all wet locations, provide with code size (minimum No. 12) bare ground wire in all flexible conduit.
 8. All conduit cuts (factory or field cut) shall be perfectly square to the length of the conduit and cut ends shall be reamed with a reaming tool to provide a smooth edge to the passing conductors and to remove all burs and scrapes. Use of a hand file is not acceptable.
 9. All electrical raceways shall be installed concealed, unless otherwise noted. Cut and patch to facilitate such installation to match adjacent and original finish. All exposed conduits, where required, shall be installed parallel to building members.
 10. All code required emergency source circuits (i.e. EM Inverter) shall be installed in separate raceways (from normal power), per 2017 NEC 700.10(B), or the applicable code at the time of permitting.
 11. Where existing conditions preclude the installation of EMT in existing walls to remain, provide and install cut-in type boxes and "fish" flexible MC or flex conduit and wire through existing walls to remain, unless shown otherwise on plans.
 12. Fasten conduits securely to boxes with locknuts and bushings to provide good electrical continuity.
 13. Provide chrome escutcheon plates at all exposed wall, ceiling and floor conduit penetrations.
 14. Support individual suspended conduits with heavy malleable strap or rod hangers; supports for ½ inch or ¾ inch conduit placed on maximum 7-foot centers; maximum 10-foot centers on conduits 1 inch or larger.
 15. Support multiple conduit runs from Kindorf B907 channels with C-105 and C-106 straps.
 16. Conduit bends - long radius.
 17. Flash conduits through roof, using approved roof jack; coordinate with General Contractor.
 18. To facilitate pulling of feeder conductors, install junction boxes as shown or required.
 19. All empty conduits on the project shall be provided with a nylon pull rope to allow pulling of future conductors intended for the specific raceway. Provide plastic wire-tie style

- nameplate tags on each end of pull rope with printed identification of conduit use and the location of the opposite end of the rope. Pull ropes for telecommunications service conduits shall meet the utility company requirements.
20. Where conduits pass through structural expansion joints in floor slab, rigid galvanized conduit shall be used 18" on either side of joint, complete with Appleton expansion couplings and bonding jumpers, or equal. All above grade expansion joint crossings shall also utilize expansion joint couplings or flex conduit transitions as required for each particular installation. Installed condition shall allow for a minimum deflection of raceway and wire (in any direction) equal to the structural expansion joint dimension (building to building). No solid conduits shall be allowed to cross expansion joints without proper provisions for building and seismic movement.
 21. Minimum cover of conduits in ground outside of building - 36 inches, unless otherwise noted.
 22. Provide and install exterior wall conduit seals and cable seals in the locations listed below. Coordinate installation and scheduling with other trades:
 - a. Conduit seals through exterior wall or slab (below grade): O.Z. Gedney series "FSK" in new cast in concrete locations, series "CSM" in cored locations.
 - b. Conduit seals through exterior wall or slab (above grade): O.Z. Gedney series "CSMI."
 - c. Cable seals at first interior conduit termination after entry through exterior wall or slab: O.Z. Gedney series "CSBI." Coordinate quantity of conductors at each location.
- B. Outlet Boxes and Junction Boxes. Verify all backbox requirements with devices to be installed prior to rough-in.
1. One piece steel knockout type drawn boxes, unless otherwise noted, sized as required for conditions at each outlet or as noted.
 2. Flush-mounted boxes equipped with galvanized steel raised covers for device mounting flush with finished surface. Provide extension rings as required on all acoustical or additional wall treatment areas to bring top of cover flush with finished surface (coordinate with architectural drawings). Devices shall be capable of being tightly mounted to boxes without distorting or bending device or mounting hardware.
 3. Boxes for fixture outlets: 4-inch octagon or larger as required, or as noted.
 4. Switch and receptacle outlets - not smaller than 4-inch-square in furred walls, with raised cover for single device; ganged where required.
 5. Outlet and switch boxes for wet locations, cast aluminum FS or FD type with cast aluminum gasketed spring lid cover. Weatherproof "Bell" type boxes are not acceptable.
 6. All connectors from conduit to junction or outlet boxes shall have insulated throats. Connectors shall be manufactured with insulated throats as integral part. Insertable insulated throats are unacceptable.
 7. Outlet boxes for telecommunications, 4" square or larger as required or noted, multi-ganged for voice, data, and other services where indicated on the drawings.
 8. Conduit Bodies: Malleable iron type, with lubricated spring steel clips over edge of conduit body, O-Z/Gedney type EW, or equal.
 9. Pull boxes: All site pull boxes shall be flush in-ground concrete, with engraved covers identifying service use (i.e. electrical, communications, etc.). Boxes shall be Nema 250, Type 6, outside flanged, with recessed cover for flush mounting, by Christy or equal, with required depth to provide box and conduit depths shown or required.
 - a. Provide concrete covers for all boxes in planted or paved areas (up to available concrete cover size).
 - b. Provide galvanized steel covers for all larger boxes (when concrete is not available), or in traffic areas. No cast iron covers.
 - c. Provide bolted covers and slab bottoms (with grouted perimeter) or vault type boxes for all electrical distribution and signal system pull boxes used for site distribution, to prevent rodent entry. No collar type boxes with dirt or gravel bottoms

- d. Provide drain hole at bottom of all vault type boxes, with loose aggregate base below, for proper drainage.
 - e. All covers to be completely flush with finished adjacent surfaces.
 - f. Provide galvanized steel H20 rated covers and installation of box rated for H20 in all traffic areas.
 - g. Provide pullboxes per utility company specifications for all electrical primary and secondary services and for telecommunications service runs. Verify exact size and type prior to order with each utility company.
- C. Wire and Cable (line voltage and signal systems):
1. 600-volt class where used for or run with line voltage power wiring, insulation color coded, minimum No. 12 AWG for power branch circuits, No. 14 for power control circuits, and wiring size and type as directed by signal system manufacturer for each signal system.
 2. All conductors shall be copper.
 3. Size and insulation type:
 - a. Standard locations: #12 to #1 AWG: THWN for wet locations and THHN for dry locations. #1/0 through #4/0 AWG: XHHW (55 Mils). 250MCM and larger: XHHW (65 Mils). All wire sizes used shall be based on a 75 degree insulation rating, unless specifically used with 90 degree rated breakers and devices.
 - b. All wiring (power and signal) installed underground between buildings, or in wet or damp locations, shall be outside listed and rated for wet locations.
 - c. High temperature and non-standard locations: Provide wire type and insulation category suitable for area of use as defined in NEC table 310-13.
 4. Conductors No. 8 and larger and as otherwise noted on drawings shall be stranded. Power conductors No. 12 and No. 10 shall be solid or stranded. Power conductors No. 14 or smaller shall be solid.
 5. Provide signal system wiring for each system to meet the system manufacturers requirements and recommendations for each device or equipment type. Signal wiring systems shall be provided with shielding and/or insulation type and cable quantities as directed by the manufacturer, and meet all NEC requirements for locations used.
 6. Install all wiring branch circuits and feeders (low voltage and line voltage) in conduit unless noted otherwise on the drawings. Contractor shall mandrel all feeders and pass a "sock" (or utilize other suitable means) through each raceway prior to pull to remove all water and construction debris. All raceways shall be completely clear of any obstructions or debris and all cut ends shall be reamed, prior to pull. Utilize pulling compound on all runs to insure minimum friction and pulling tension.
 7. Megger test all feeders prior to energizing. See section 26 08 00 for additional information.
 8. Approximately balance branch circuits about the neutral conductors in panels.
 9. Connections to devices from "thru-feed" branch circuit conductors to be made with pigtails, with no interruption of the branch circuit conductors.
 10. Neutral conductor identified by white outer braid, with different tracers of "EZ" numbering tags used where more than one neutral conductor is contained in a single raceway.
 11. Neatly arrange and "marlin" wires in panels and distribution panelboards with "T and B Ty-rap" or approved equal plastic type strapping.
 12. All wire and cable shall bear the Underwriters' Label, brought to the job in unbroken packages; wire color-coded as follows:

a. Voltage	Phasing	A	B	C	N
b. 120/208	3PH4W	Black	Red	Blue	White
c. 2083PH	3W	Black	Red	Blue	--
 13. The equipment grounding conductor shall be insulated copper; where it is insulated, the insulation shall be colored green.
 14. Label each wire of each electrical system in each pull box, junction box, outlet box, terminal cabinet, and panelboard in which it appears with "EZ" numbering tags indicating the connected circuit numbers.

15. Provide permanently affixed adhesive labels with machine printed lettering (min. 1/8" high) at junction boxes serving fixtures that are supplied by (2) electrical sources (i.e. normal and emergency lighting). Label to read "CAUTION - This light fixture is powered by (2) separate sources. The normal power source breaker and the emergency power source breaker must be turned off before servicing this light fixture."
 16. Install feeder cables in one continuous section unless splices are approved by Architect. Exercise care in pulling to avoid damage or disarrangement of conductors, using approved grips. No cable shall be bent to smaller radius than the spool on which it was delivered from the manufacturer. Color code feeder cables at terminals. Provide identifying linen tags in each pullbox.
- D. Switches: Model numbers are Hubbell, color to be selected by architect, unless otherwise noted. All switches to utilize screw terminals for wire connections - no plug-in terminations:
1. Single Pole - No. HBL1221
 2. Two Pole - No. HBL1222
 3. Three Way - No. HBL1223
 4. Momentary contact - No. HBL1557
 5. Momentary contact Keyed - No. HBL1556L
 6. Keyed, - No. HBL1221L
 7. Pilot Light (on with load on) - Hubbell No. 1221-PLC
 8. Motor Rated Double Pole (30A) - Hubbell No. 7832
 9. Motor Rated Three Pole (30A) - Hubbell No. 7810.
 10. Low voltage Data line switches - Refer to lighting control system (for compatibility)
- E. Receptacles: Mounting straps and contacts shall be one piece design, constructed of minimum .050" solid brass. Base shall be high strength, heat resistant, glass reinforced nylon. Device shall accept up to #10 wire, side or back wired with screw terminals - no plug-in terminations. Hubbell, Leviton, Pass & Seymore, or equal. Color to be selected by architect, unless otherwise noted. Numbers listed below are Hubbell:
1. 15A 3PG 125 volt duplex - No. HBL5262
 2. 20A 3PG 125 volt duplex - No. HBL5362
 3. 20A 3PG 125 volt ground fault interrupter receptacle; GFI receptacles shall conform to the 2006 UL requirements to a) interrupt power to the unit in the event of internal failure, or b) provide an audible or visual indication of internal failure of the GFI; No. GF20 or equal. Through wiring to down stream GFI designated receptacles is not acceptable.
 4. 15A 3PG 125 volt half controlled duplex receptacle - No. BR15C1(color), with permanent "controlled" marking, factory applied.
 5. 20A 3PG 125 volt half controlled duplex receptacle - No. BR20C1(color), with permanent "controlled" marking, factory applied.
 6. 15A 3PG 125 volt full controlled duplex receptacle - No. BR15C2(color), with permanent "controlled" marking, factory applied.
 7. 20A 3PG 125 volt full controlled duplex receptacle - No. BR20C2(color), with permanent "controlled" marking, factory applied.
 8. GFI Module (blank face), no indicator light, 20A - No. GFBF20 or equal.
 9. All receptacles located in exterior or wet locations shall be corrosion resistant with UV stabilized body.
 10. All receptacles in locations identified in NEC 406.12 (i.e. dwelling units, hotel/motel guest rooms, child care, preschool, K-12 schools, business office common areas, clinics, medical, and outpatient facilities, assembly area common areas, dormitory units, and assisted living units) shall be tamper resistant.
- F. Plates: Leviton, or equal, except as noted:
1. The color of all faceplates shall match the color of the devices installed under/in the faceplate, except as specifically noted otherwise.

2. For flush outlet boxes, for switches, and receptacles: nylon, color to be selected by architect, unless otherwise noted.
 3. Plates for surface-mounted outlets: galvanized steel unless otherwise noted.
 4. Weatherproof duplex receptacle plates for exterior locations with ground fault interrupter receptacles in type FS or FD boxes - Hubbell #WPFS26 or compatible equal. Verify cover compatibility with box type and device installed.
 5. Weatherproof "in-use" cover, vertical or horizontal mount, for exterior with GFCI receptacles. Die-cast metal alloy, TayMac MX series or equal with openings to match installed devices.
 6. Locking plates for duplex receptacles where noted; Pass & Seymour #WP26-L (non-weather proof).
 7. Locking plates for duplex exterior GFCI receptacles (or in wet or damp locations); Heavy duty cast aluminum flush cover with locking latch and key, Pass & Seymour #4600 with appropriate mounting plate for type of device installed. Coordinate backbox requirements and finished wall trim-out with wall installer prior to rough-in to insure an adequate and neat trim appearance upon completion.
 8. Plates for flush tele/data boxes: white nylon or as otherwise directed - provide and install at each tele/data outlet plate to match duplex power outlet plate, for jack installation under Section 27 00 00. Where the power and tele/data outlet boxes are shared the plate shall be continuous in multi-gang locations. See drawings.
- G. Equipment Disconnects: All disconnects shall be located to allow proper code required clearance in each area. Locations shown on drawings are diagrammatic only. The contractor shall coordinate exact locations in the field (with other trades) prior to rough-in to insure proper clearances.
1. Motor Disconnect Switches and Safety Switches: General Electric Company Heavy Duty Type "THD", cover interlocked with operating handle so that cover cannot be opened with switch in closed position and switch cannot be closed with cover in open position. 240V or 480V rating, single or multi-pole as required or as noted on drawings, in Nema 1 enclosure indoors or Nema 3R enclosure outdoors unless otherwise noted. Provide dual element motor circuit fuses sized as recommended by equipment manufacturer (for final equipment actually installed).
 2. Code required disconnects: Provide a local disconnect in addition to the branch circuit protection device for all equipment as required by code (whether shown or not). Disconnects shall consist of a motor rated switch (or disconnect) for all motor loads less than 3/4HP or other suitable disconnect sized to match branch circuit conductors and load current of equipment, with number of poles as required.
- H. Lugs and Connectors: Thomas and Betts "lock-tite", for No. 4 and larger wire; 3M "Scotchlock" fixed spring screw-on type wire connectors with insulator for No. 6 and smaller wire.
1. All splices shall be made up with screw-on type connectors - no plug-in or push-in style connectors acceptable. Wires shall be solidly twisted together with electricians pliers before screw-on connector is installed to ensure a proper connection in the event of wire nut failure. No exceptions.
 2. Connectors listed or labeled for "no wire twisting required" are not an acceptable substitute for actual wire twisting.
 3. Utilize porcelain type connectors in all high temperature environments (above 105 degrees Celsius).
- I. Splice Insulation: "Scotch" electrical tape with vinyl plastic backing or rubber tape with protective friction tape for interior work.
1. Splices in electrical cables of 600 volt insulation class in underground system duct shall be made only in accessible locations such as pullboxes, light pole handholes, etc., using a compression connector on the conductor and by insulating and waterproofing (for exterior and underground locations) by one of the following methods:

- a. Cast type splice insulation shall be provided by means of a molded casting process employing a thermosetting epoxy resin insulating material which shall be applied by a gravity poured method or by a pressure injected method. The component materials of the resin insulation shall be in a packaged form ready for convenient mixing after removing from the package. Do not allow the cables to be removed until after the splicing material has completely set.
 - b. Gravity poured method shall employ materials and equipment contained in an approved commercial splicing kit which includes a mold suitable for the cables to be applied. When the mold is in place around the joined conductors, the resin mix shall be prepared and poured into the mold. Do not allow cables to be moved until after the splicing materials have completely set.
- J. Identification: Refer to Section 26 05 00.
- K. Firestopping: as manufactured by 3M Fire Protection Products or equal.
 - 1. Fire-rated and smoke barrier construction: Maintain barrier and structural floor fire and smoke resistance ratings including resistance to cold smoke at all penetrations, connections with other surfaces or types of construction, at separations required to permit building movement and sound vibration absorption, and at other construction gaps.
 - 2. Systems or devices listed in the UL Fire Resistance Directory under categories XHCR and XHEZ may be used, providing that it conforms to the construction type, penetration type, annular space requirements and fire rating involved in each separate instance, and that the system be symmetrical for wall penetrations. Systems or devices must be asbestos free.

PART 3 - EXECUTION

3.1 REFER TO BASIC ELECTRICAL REQUIREMENTS - SECTION 26 05 00 FOR WORK UNDER THIS SECTION.

3.2 TESTS

- A. Testing and Inspection: See Section 26 08 00 - Testing.

END OF SECTION

SECTION 26 31 01 - PHOTOVOLTAIC SYSTEM

PART 1 - GENERAL

1.1 DESCRIPTION

- A. The solar photovoltaic system design shown in the contract documents and these specifications is a "basis of design", to be used to establish the minimum system performance, sizing, function, and quality of the required system.
- B. The solar photovoltaic system is to be provided and installed as Add Alternate No. 2 and 3. Refer to the General Conditions and Bid Alternate Descriptions for more information.
- C. Since solar products and availability actively change over short periods of time, there is no guarantee that the specified products will be available at the time of bid and construction. This contractor shall include a complete system bid to equal the basis of design shown in the contract documents, including finalized shop drawings and any substituted products and installation, design parameters, and/or modified design calculations, and equipment or conduit / wiring / protection sizing.
- D. Change orders for differences between the specified products and the products available at the time of bid and construction will not be approved. The intent is that the contractor include a complete PV system to equal the basis of design, at the time of installation.

1.2 WORK INCLUDED

- A. Work included in this Section: All final design (based on actual products to be supplied), materials, labor, equipment, services, and incidentals necessary to install a complete Photovoltaic (PV) System as specified hereinafter, including but not limited to the work listed below.
- B. Refer to the General Conditions and Bid Alternate Descriptions for more information.
- C. The system shall be utility grid connected. The contractor shall be responsible for all required utility company coordination, approval, and applications for the complete interconnection of the PV system with the utility company grid, including bi-directional utility meter.

1.3 SCOPE

- A. The system shall consist of arrays of framed photovoltaic modules (roof and carport canopy mounted), all mounting hardware, terminal boxes and combiner panels, quick-connect electrical connectors, DC wiring, DC disconnects, utility interactive inverters, AC disconnect, AC feeder, main PV system disconnect, system contactor (for microgrid operation with generator system), utility company PV system disconnect, and a complete data acquisition and monitoring system to allow the Owner to monitor and utilize the collected data over the Owner network.
- B. The PV system output shall be automatically disconnected from the distribution system by a controllable contactor device, whenever the diesel generator is activated (via the automatic transfer switch), to ensure that the PV and generator equipment are not on the same electrical bus at the same time.

- C. PV roof mounted system shall be capable of Rapid Shutdown of PV system circuits on or in buildings per NEC 690.12 (this shall not apply to the carport mounted system).
- D. The work shall include furnishing all labor, materials, and equipment necessary to form a complete installation, ready for operation to produce solar power, both in parallel with the utility company, and while islanded during microgrid operation.
- E. The installing contractor shall be responsible for adequate clearance and equipment space within the allotted roof areas and/or canopy structure areas and interior building area. All equipment and sizes / clearances shall be coordinated with the architect and Owner prior to rough-in.
- F. The system installer shall submit for and pay for the required permits and inspections with the local Authority Having Jurisdiction (AHJ) and utility company.
- G. The installer shall complete all of the required paperwork for the utility interconnection agreement contract in conjunction with the Owner's input and approval, including rate schedule (i.e. TOU or other) designations. In order for the Installer to act on behalf of the Owner, the installer (in conjunction with the Owner) shall submit to the utility company the proper authorization forms.
- H. System installation shall include the programming, set-up, and commissioning of a web based data acquisition system and interactive data application to allow public viewing of the real-time system performance and past historical performance.

1.4 WARRANTIES

- A. All solar energy equipment for electricity generation (PV modules, inverters, solar collectors, and tracking mechanisms) shall have a minimum 10-year manufacturer warranty, and shall protect against degradation of electrical generation output of more than 15% from their originally rated electrical output.
- B. Contractor shall provide a minimum 10-year guarantee to provide for no-cost repair and replacement of the system for any expenses not otherwise covered by the manufacturer.
- C. Contractor shall provide a minimum 10-year guarantee to protect the purchaser against more than a 15% degradation of electrical generation output that may occur as a result of faulty installation.
- D. Meters must have a one-year warranty to ensure against defective workmanship, system or component breakdown, or degradation in electrical output of more than fifteen percent from their originally rated electrical output during the warranty period. For meters that are integrated into the inverter, the meter warranty period must be 10 years.
- E. System installation shall be such that it does not affect the roofs and/or shade structures warranty.

1.5 APPLICABLE GUIDELINES / REGULATIONS / STANDARDS

- A. CPUC approved Electric Rule 21 - Generating Facility Interconnections

- B. UL1741 (Inverters, Converters, and Controllers for Independent Power Systems)
- C. UL1703 (Standard for Flat-Plate Photovoltaic Modules and Panels).
- D. IEEE 929 (2000) - Recommended Practice for Utility Interface of Photovoltaic (PV) Systems.
- E. IEEE 1262 (1995) - Recommended Practice for Qualifications of Photovoltaic (PV) Modules.
- F. NEC Articles 690 and 705.

1.6 QUALITY ASSURANCE

- A. All components shall be Underwriters' Laboratories labeled.
- B. Contractor Qualification - The contractor shall be approved by the PV equipment manufacturer(s) to install the PV materials, where such approvals are required.

1.7 SUBMITTALS

- A. Submit the following for approval:
 1. Completed roof and canopy mounted system shop drawings with the PV System layout (based on submitted modules and inverters).
 2. Single line Diagrams indicating all required connections and utility tie-in.
 3. Array calculations including string design, string amperage, array amperage (including short circuit currents), and DC voltages (maximum and minimum based on coldest record low and average high ambient temperatures).
 4. DC combiner boxes with fusing.
 5. DC/AC Inverter(s).
 6. Isolation Transformer (if applicable).
 7. kWh Meter and Logger.
 8. Data Acquisition System.
 9. PV System weights.
 10. PV Modules.
 11. Installation Manuals.
 12. Operation and Maintenance manuals.
 13. Web based data acquisition system and application software.
 14. Mounting hardware.
 15. Wiring (AC and DC).
 16. Disconnects.
 17. PV Weather Station including solar radiation, environmental temperature and humidity, wind direction, wind speed and module temperatures.
 18. Placards (with all Code and utility required designations).
 19. Testing and certification/commissioning results (upon completion).

1.8 DELIVERY, STORAGE, AND HANDLING

- A. All equipment and panels shall be handled with care so as not to damage the delivered products. All equipment shall be installed in new and neat condition.

- B. Appropriate protective clothing shall be worn when handling the equipment. Such clothing shall include hard hats and steel-toe boots when lifting materials to roof, and insulated gloves when working on an active system.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Acceptable system manufacturers/vendors shall have been in the business of producing and/or installing similar commercial grade solar photovoltaic systems for the last 5 years minimum (35 kW systems or higher). Manufacturers shall provide their latest line of equipment, meeting all current industry standards.
- B. The minimum performance of the submitted and installed system shall substantially match the physical and electrical equipment characteristics of the basis of design, as shown on the electrical drawings.
- C. Substitute modules and equipment are acceptable when meeting the minimum system performance matching the basis of design, requirements listed herein, and complying with all other technical aspects of the system.

2.2 MATERIALS

- A. The PV modules shall have physical and electrical characteristics which match the modules specified on the electrical drawings. Alternate modules, if allowed, shall have similar or better characteristics.
 - 1. All PV modules shall be UL 1703 listed.
 - 2. All PV modules shall be a max system voltage of 600Volt UL, or 1000Volt UL. Inverter and module max system voltages shall match.
 - 3. All PV module cell types shall be 'monocrystalline' or 'polycrystalline', and shall be similar in cell type to that of the modules specified on the electrical drawings.
- B. The following Electrical Module characteristic shall be used as a minimum standard for any design submitted.
 - 1. Power Output (Pmax) - 445 Watts; STC
 - 2. Open Circuit Voltage (Voc) - 48.9 Volts; STC
 - 3. Maximum Power Voltage (Vmp) - 40.9 Volts; STC
 - 4. Short Circuit Current (Isc) - 11.54 Amps; STC
 - 5. Maximum Power Current (Imp) - 10.89 Amps; STC
- C. Each inverter shall be by Solar Edge or equal, sized as required to support the PV module production load that shall not exceed the manufacturer's rating of the equipment. All inverters shall be CEC approved and shall be utility line interactive type with the following:
 - 1. Nominal AC Voltage shall match the available utility line voltage
 - a. Nominal AC Frequency (+ 0.5 Hz)
 - 1) 60 Hz
 - b. Line Power Factor (Above 20% rated power)
 - 1) >0.99
 - 2. Maximum Open Circuit Voltage DC
 - a. 600 VDC or 1000VDC
 - b. Inverter and PV modules max voltages shall match.

3. Peak Inverter Efficiency
 - a. >97.5% for transformerless type inverters
 4. Enclosure Environmental Rating
 - a. NEMA 3R (or NEMA 4X)
 5. Enclosure Environmental Rating
 - a. Galvaneal folded steel enclosure
 6. Sound level
 - a. <65dB(A)
 7. Array Configuration
 - a. Monopole, negative grounded or floating - Bi-polar, neutral grounded
 8. Protective Functions
 - a. Standard wakeup voltage, wakeup time delay, shutdown power, shutdown time delay, AC over / under voltage and time delays, AC over / under frequency and time delays, ground over current, over-temperature, AC and DC over current, DC over voltage
 - b. Arc-Fault Circuit Protection (Direct Current) per NEC Article 690.11
 9. User Display
 - a. Standard-LCD with on/off capability
 10. AC Disconnect
 - a. Load break rated
 11. DC Disconnect
 - a. 600 VDC or 1000 VDC load break rated
 12. Communications Software
 13. Serial communications and control software
 14. UL 1741 listed.
 15. CEC compliant seismic rating.
 16. Internal MPPT combiners to allow connections of sub-arrays at the Inverter without the use of additional equipment.
 17. Provide a placard on the Inverter per section 3.1 below.
- D. PV System to electric service equipment interface shall be as follows:
1. The PV system shall be connected on the load side of the main service disconnect per NEC Article 705.12(B).
 - a. The connection shall be made with a bolt-on circuit breaker connection located at the opposite end of the distribution section bussing from where the bussing receives its normal utility service supply. The circuit breaker shall be listed for back-feeding (i.e. it shall not contain designations for line and load side wiring).
 - b. The sum of the circuit breakers supplying the bussing (utility main, PV, and BESS) shall not exceed 120% of the main switchboard bussing rating.
 - c. The phase connections, including neutral connection (where used), shall be made on the load side of any service ground fault protection (GFP) system and shall not allow any unbalanced currents to pass outside the service ground fault protection system (where used). DO NOT connect to the load side of the GFP system if it is not listed and identified for back-fed installations (allowing currents to pass backwards through the CT's to the utility grid).
 - d. Provide a placard on the Main Switchboard to identify the two sources of power feeding the equipment. Also provide a placard identifying the inverter(s) breaker position and its intended purposes and location. Refer to section 3.1 below.
- E. All AC interconnecting feeders shall be sized to NEC Table 310.15(B)(16) (75 degree column) based on associated disconnect amperage. Conduit fill to 40% max. Include temperature derating as required for the ambient temperatures and roof conditions per NEC 310.15. Provide equipment grounding conductor in each conduit.

- F. All AC circuits to be 3-wire + ground. All grounding per NEC 690, Part V.
- G. All DC circuits and feeders sized to NEC table 310.15(B)(16) (90 degree column) based on associated disconnect amperage. Minimum ampacity shall be 156% of the rated short circuit current available to be carried on the specific conductor. Conduit fill to 40% max. Include temperature derating as required for the ambient temperatures and roof conditions per NEC. Provide equipment grounding conductor in each conduit.
- H. All DC circuits to be 2-wire + ground.
- I. All AC and DC wiring in conduit to be RHW-2, THWN-2, or XHHW-2 with ratings as follows:
 - 1. Utilize 90 degree C ampacity, wet rated conductor, for use with 90 degree listed terminals on PV DC equipment.
 - 2. Utilize 75 degree C ampacity, wet rated conductor, for use with 75 degree listed terminals on AC equipment.
- J. All exposed DC wiring to be listed 'PV WIRE', USE-2 or SE (90 degree) wet rated and sunlight resistant, 600V or 1000V rated as required.
- K. All above ground exposed conduit shall be rigid galvanized steel or aluminum with threaded fittings or painted EMT with water-tight compression fittings. All interior conduit to be EMT with steel set-screw fittings (no cast fittings).

2.3 ARRAY MOUNTING

- A. Modules shall be roof or parking canopy structure mounted, with appropriate racking hardware and structural attachments.

2.4 MISC. SYSTEM REQUIREMENTS

- A. All exterior equipment to be sunlight and UV resistant as well as rated for elevated temperatures at which they are expected to operate (on roofs in hot sunlight).
- B. Heavy duty urethane sealants shall be used for all non-flashed roof penetrations.
- C. No dissimilar metals allowed to contact (use plastic or rubber washers)
- D. No aluminum in contact with concrete or masonry materials.
- E. Use high quality stainless steel fasteners only.
- F. Structural members for PV supports should be corrosion resistant aluminum (6061 or 6063), hot dipped galvanized steel (per ASTM A 123), coated or painted steel (in non-corrosive environments only), or stainless steel (in corrosive environments).
- G. All PV modules to be installed such that they are 100% free from shade between 8am and 5pm daily.

2.5 SYSTEM ELECTRICAL OPERATION

- A. The modules shall be interconnected using cable assemblies. The pigtails shall be quick-connect electrical wiring connections rated for the application (90 degree rated).
- B. The array shall have at least one terminal box, providing a watertight entry to the raceway system leading to the combiner box and Inverter(s).
- C. Full specifications of the inverters shall be supplied as part of the system submittal.
- D. All major components of the systems and the installation procedures shall meet National Electrical Code requirements, including Article 690.
- E. The inverters shall automatically drop-off-line when normal utility power is lost to avoid un-intentional utility grid islanding effects. Drop-off to be activated by over-voltage (110%) and under-voltage (88%), and shall be adjustable. Frequency drifts outside 59.3 to 60.5 Hz for more than 10 cycles shall also activate automatic drop-off. Automatic reconnection shall not occur until the normal utility power has been stable for at least 60 seconds.
- F. The system shall be programmed to allow operation when the microgrid system is in operation (isolated from the utility grid) and shall sync to and track the grid forming BESS Inverter equipment during microgrid operation.
- G. The system shall be programmed to never allow the diesel generator and the PV system to be on the same electrical bus at the same time. When the ATS is in emergency position, provide interlock to PV contactor (as shown on the single line diagram), to open the PV contactor.
- H. When the diesel generator is not running (based on control information from the microgrid controller), the ATS shall be in the neutral position (no normal or emergency line connection) and the PV system contactor shall be closed to allow PV system operation on the same bus as the BESS (for load support and battery charging).
- I. All electrical system equipment shall be properly rated to withstand and interrupt (in the case of over current protection devices) the available fault current at the point of use.
- J. The system shall be capable of operating between a power factor of 0.9 lagging to 0.9 leading.
- K. All required overcurrent protection and electrical bussing sizes per NEC 690.
- L. Provide a grounding electrode connection from the inverter assembly to the nearest building steel per NEC Article 690 and the manufacturer's instructions. Inverters shall have GFCI protection, allowing grounding per NEC Table 250.122.
- M. The Main PV System Disconnect (in the main electric service panel room) shall be clearly labeled and located within 10 feet of the main service meter location per Utility Company requirements.

2.6 MONITORING

- A. A Data Acquisition and Monitoring System shall be provided as part of the System. The system shall allow measurement, calculation, and display of the following items (at minimum) via on-site PV Weather Station and Inverter communication (RS 485):
 - 1. Ambient temperature
 - 2. Wind speed
 - 3. Solar irradiation
 - 4. System electrical functions (instantaneous and accumulated power output (kW and kWh), AC and DC system voltage and amperage, and peak value tracking with associated time stamps).
 - 5. Pounds of Co2 emissions avoided from the generation of PV energy at the site (compared to conventional coal and gas production methods).
- B. Provide a Web based software application to allow interactive display and user requests of system performance, including historical data.
- C. Load software on owner provided web page (URL) and train owner in operation and maintenance of software and related monitoring functions.

PART 3 - EXECUTION

3.1 REQUIRED PLACARDS

- A. All placards shall be machine generated phenolic type with red background and white lettering, affixed to equipment with stainless steel screws (no adhesives allowed). Minimum lettering size to be 1/4" unless otherwise noted or required for legibility.
- B. Provide a placard clearly visible the main service panel to identify all sources of power, with the following wording in 1/4" high lettering per NEC Article 690: "Warning - This Service Is Fed By Multiple Sources Of Power - The Utility Service Main Disconnect, The PV System Main Disconnect, and the Battery System Main Disconnect - All Three Systems Must Be Disconnected To Remove Power From The Main Panel and Facility".
- C. Provide a placard on the PV system input circuit breaker at the main service panel with the following wording in 1/4" high lettering per NEC Article 690: "Warning - Inverter Output Connection - Do Not Relocate This Overcurrent Device".
- D. Provide a placard on the BESS input circuit breaker at the main service panel with the following wording in 1/4" high lettering per NEC Article 690: "Warning - Inverter Output Connection - Do Not Relocate This Overcurrent Device".
- E. Provide a placard on all PV System disconnects with the following wording in 1/4" high lettering per NEC Article 690: "Warning - Electric Shock Hazard - Do Not Touch Terminals - Terminals On Both The Line and Load Sides May Be Energized In The Open Position".
- F. Provide a placard on the Main PV System Disconnect with the following information in 1/4" high lettering per NEC Article 690: "Photovoltaic Power Source Disconnect - Operating Current: XX Amps; Operating voltage: 208 VAC; Maximum System Voltage: 208 VAC; Short-Circuit Current: XXX Amps", where XX is the maximum AC amperes of the installed system and XXX is the maximum short circuit current that can be delivered through that device - usually the

available utility system short circuit current at that location, or, only if noted on the drawings, the maximum short circuit current that the PV system can provide (from all strings in parallel).

- G. Provide a placard at the main panel with the following information in 1/4" High lettering per NEC Article 690: "Caution - Possible Backfeed From Photovoltaic and Battery Power Systems - 208V, XX Amps", where XX is the maximum AC amperes of the installed systems.
- H. Provide a placard on each PV System Inverter with the following information in 1/4" high lettering: "Photovoltaic Power Source Inverter Rating - Operating Current: XX Amps; Operating voltage: XXX VDC; Maximum System Voltage: 600 VDC; Short-Circuit Current: XXXX Amps", where XX is the maximum DC amperes of the installed system, XXX is the operating voltage DC, and XXXX is the short circuit current that the Inverter can provide (from all strings in parallel).
- I. Provide utility-required system directory placard and utility safety switch identification placard as required by local utility company, to identify all system components.

3.2 UTILITY INTERCONNECTION

- A. The PV generation system shall not be interconnected with the Utility's distribution facilities until written authorization from the Utility Company has been obtained. Unauthorized interconnections may result in injury to persons and damage to equipment or property for which the installing contractor and Owner may be liable.

3.3 INSTALLATION STANDARDS

- A. System Installation shall conform to the equipment manufacturers Installation Manual(s) and requirements or guidelines.
- B. All Local, State, and NEC codes shall be observed, including all industry standards related to the installation, operation, and maintenance of photovoltaic power systems.

3.4 TESTING

- A. Photovoltaic modules shall be tested in the factory for design performance and results shall be included in the Operation and Maintenance manuals.
- B. Inverters shall be factory tested for performance and the results shall be included in the Operation and Maintenance manuals.
- C. System testing of the installed photovoltaic array shall be performed on all system strings and recorded in the Operation and Maintenance manuals.
- D. Megger test each array string prior to energizing to establish that no shorts or ground exist at any point on the arrays.
- E. Testing to be performed per CPUC Electric Rule 21 testing procedures and requirements. All testing to be done on "no-cloud" days to avoid system fluctuation by passing clouds. Installer to provide all testing and certification / commissioning.

- F. System start-up procedure shall be as outlined by the Manufacturer's Installation Manual and the Inverter Manual.

END OF SECTION

SECTION 26 32 01 - LITHIUM-ION BATTERY STORAGE

PART 1 - GENERAL

1.1 DESCRIPTION

- A. The Battery Energy Storage System (BESS) design shown in the contract documents and these specifications is a "basis of design", to be used to establish the minimum system performance, sizing, function, and quality.
- B. The BESS (with microgrid capability) is to be provided and installed as Add Alternate No. 5. Refer to the General Conditions and Bid Alternate Descriptions for more information.
- C. The battery energy storage system shall include the following functions and equipment:
 - 1. Shall be interconnected to the facility main service panel to provide energy arbitrage/peak shaving of normal building loads during normal parallel utility operation.
 - 2. Shall include a set of batteries for energy storage, a grid-forming inverter for AC system output, an isolation transformer for interconnection to the 4-wire, 120/208V electric system, all monitoring and control system components, and an isolation contactor for automatic control of the battery system connection to the facility distribution bus.
 - 3. Shall include all required feeder CT's and interconnection wiring and programming to the system controller to monitor the normal utility supply, the diesel generator supply, the PV system supply, and the BESS supply or charging load.
 - 4. Shall allow microgrid operation when normal utility power fails. Microgrid operation shall occur when the main service ATS switches away from the normal input source to either the neutral position or the emergency position.
 - 5. During microgrid operation, the controller shall operate the various system components (generator, ATS, PV, and BESS) to allow proper and safe system operation without the normal utility company connection.
 - 6. The generator and PV system outputs shall remain isolated from each other at all times.
 - 7. The generator may feed the main facility bus, to support loads and charge the batteries, when the PV system contactor is open.
 - 8. The PV system may feed the main facility bus, to support loads and charge the batteries, when the ATS is in the neutral position (no generator power).
 - 9. The BESS inverter output may feed the main facility bus at all times, except as determined by the microgrid control system for specific event scenarios, if required.
 - 10. After a normal power failure event and microgrid operation, the microgrid controller shall automatically reconnect to the normal utility grid when normal power is available again for a pre-determined delay time. This shall occur by transferring the ATS back to the normal supply source when normal utility supply is available and de-energizing the diesel generator. The PV system and BESS shall provide normal parallel utility operation when normal power is restored.
- D. Since energy storage products and availability actively change over short periods of time, there is no guarantee that the specified products will be available at the time of bid and construction. This contractor shall include a complete system bid to equal the basis of design shown in the contract documents, including finalized shop drawings and any substituted products and installation, design parameters, and/or modified design calculations, and equipment or conduit / wiring / protection sizing.
- E. Change orders for differences between the specified products and the products available at the time of bid and construction will not be approved. The intent is that the contractor include a

complete battery energy storage / microgrid system to equal the basis of design, at the time of installation.

1.2 WORK INCLUDED

- A. The Contractor shall provide a complete behind-the-meter Lithium-ion Battery Energy Storage System (BESS), appropriately sized as indicated on the single line diagram. Contractor shall provide all labor, material, equipment, engineering, and maintenance, to design, install, commission, and interconnect the BESS as required herein.
- B. Refer to the General Conditions and Bid Alternate Descriptions for more information.
- C. The BESS will be part of a micro-grid system along with Photovoltaic, Utility, and Generator power systems.
- D. Electricity from the BESS must be provided at 60 Hertz and at the appropriate voltage for electrical interconnection to the Site at 120/208V, 3PH, 4W on the electrical distribution system. The BESS will interconnect at the main switchboard distribution section.
- E. The BESS and associated equipment, shall be provided in self-contained National Electrical Manufacturers Association (NEMA) enclosure(s) rated for the site conditions. BESS enclosures shall be installed on concrete housekeeping pads constructed by Contractor, at the equipment locations as indicated on the site plan. Contractor shall meet all the seismic requirements for equipment mounting as indicated in Section 260500 of the Specifications. Contractor provided thermal conditioning systems shall maintain ambient temperature within warranty requirements.
- F. BESS components and associated ancillary equipment shall have working space clearances required by local code, and electrical circuitry shall be within weatherproof enclosures marked with the environmental rating suitable for the type of environment in compliance with NFPA 70.
- G. The Contractor shall commission the BESS and provide a commissioning report documenting BESS performance during normal grid-tied operations and during grid failure (microgrid operation).
- H. Contractor is responsible for all permits, approvals, environmental compliance, freight, financing, procurement, monitoring, site inspection, billing, and incidentals as necessary to design, construct, and interconnect the complete BESS, described hereinafter.
- I. Contractor shall provide and install fire suppression system and observe guidelines from the Californial Electrical Code (CEC) 2022 and the International Association of Firefighters.

1.3 WARRANTIES

- A. Contractor shall provide 1 year of turnkey BESS maintenance and warranty service. Contractor shall also provide a warranty from the BESS manufacturer for the full life of the installation for consideration by the Owner.
- B. Battery pack performance warranty for at least 10 years after the date of successful completion of commissioning. Battery pack shall retain at least 70% of nominal energy capacity for the earlier of either 10 years after commissioning.

- C. Power conversion system warranty of at least 10 years from successful completion of the commissioning.
- D. Full turnkey system warranty for 1 year from successful completion of the commissioning. Contractor shall respond within 3 days if maintenance is required.

1.4 APPLICABLE GUIDELINES / REGULATIONS / STANDARDS

- A. All work must follow current National Electrical Code requirements:
 - 1. NFPA 855, "Standard for the Installation of Stationary Energy Storage Systems"
- B. Battery cell:
 - 1. UL 1642 "Standard for Lithium Batteries"
- C. Battery module:
 - 1. UL 1973 "Batteries for Use in Light Electric Rail Applications and Stationary Applications"
- D. Battery system:
 - 1. UL 9540 "Energy Storage Systems and Equipment"
 - 2. UL 9540A "Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems"
- E. Grid interconnection standards, as applicable to the project as a whole:
 - 1. Institute of Electrical and Electronics Engineers (IEEE) 1547
 - 2. UL 1741, "Standard for Static Inverters and Charge, Converters, Controllers and Interconnection System Equipment for Use with Distributed Energy Resources"
 - 3. UL 62109-1 "Safety of power converters for use in photovoltaic power systems – Part 1: General requirements"
- F. Other codes and standards that will apply include:
 - 1. UN 38.3 "Certification for Lithium Batteries" (Transportation)
 - 2. American National Standards Institute (ANSI) C12.1 (electricity metering)
 - 3. American Society of Civil Engineers (ASCE)-7 Minimum Design Loads for Buildings and Other Structures
 - 4. IEEE 2030.2, Guide for the Interoperability of Energy Storage Systems Integrated with the Electric Power Infrastructure

1.5 QUALITY ASSURANCE

- A. All components shall be Underwriters' Laboratories labeled.
- B. Contractor Qualification - The contractor shall be approved by the BESS equipment manufacturer(s) to install the BESS materials, where such approvals are required.

1.6 INTERCONNECTION TO ELECTRICAL DISTRIBUTION SYSTEM

- A. It is the sole responsibility of the Contractor to meet the PG&E interconnection requirements. The Contractor shall provide supporting documentation or drawings required by the utility company. Any local codes required by utility interconnection laws and regulations shall be followed by the Contractor.
 - 1. Owner-Owned Electrical Distribution System Requirements:

- a. Modifications or upgrades to the Owner-owned electrical system required to interconnect the BESS are the responsibility of the Contractor (e.g., service panel, generator coordination). The Contractor is responsible for the proper circuit sizing, overcurrent protection, and coordination of the circuit(s) beyond the point of interconnection to the Owner distribution system, including modifications to the site's electrical equipment and circuits. Any needed modifications to the Owner electrical distribution systems must be included in the review and approval process outlined in the submittal section herein.
2. Serving Utility Electrical Distribution Systems Interconnections:
 - a. The Contractor shall provide for interconnection of the behind-the-meter BESS with the Utility-owned electrical distribution system and take actions to ensure that the Owner- and Utility-owned systems are compatible. Modifications or upgrades to the Utility electrical system for interconnection are the responsibility of the Contractor. The Contractor will assume the costs of these modifications. Any modifications to the approved BESS design that would affect the electrical distribution system shall require the written approval of the Owner.
 - b. The Contractor is responsible for complying with all electric utility interconnection requirements, including upgrades, providing all necessary BESS details for the interconnection applications, and funding any required interconnection studies to be performed by or on behalf of the electric utility. The Contractor is responsible for gaining approval from the electric utility for interconnection and any electric utility-required upgrades.

1.7 PERMITS AND LICENSING

- A. The Contractor shall be responsible for:
 1. Preparing all permitting and licensing applications for the project
 2. Paying all fees and complying with all requirements
 3. Providing any supporting documentation, data, and information that may be required for permitting
 4. Coordinating and acting as the primary liaison with permitting and licensing agencies.

1.8 LOCAL FIRE MARSHALL INVOLVEMENT

- A. The Contractor shall involve the fire marshal to provide design approval. The Contractor shall review the design of the system with the fire marshal and also observe design guidelines regarding firefighter access in the International Association of Fire Fighters and the CEC 2022.
- B. The Contractor shall invite local firefighters to a training where firefighters will be shown how the BESS works and how to shut it down, form a fire suppression plan, and shall work with the fire marshal to determine where to post the permanent instructions and required markings. The Contractor shall coordinate with local fire marshals on BESS setback requirements, first responder training, and posted instructions for shutoffs.

1.9 FEDERAL, STATE, AND LOCAL REBATES AND INCENTIVES

- A. The Contractor shall complete and submit in a timely manner all documentation required to qualify each system for available rebates and incentives, if and/or where applicable.
- B. If rebates or incentives are available, the value of these rebates or incentives shall be provided to the Owner in the form of the final bid numbers or rebates / incentives provided after the installation is complete. Rebates and incentives are to be provided to the Owner for their benefit, not the contractors benefit.

1.10 SUBMITTALS

- A. The Contractor shall include BESS submittals, including:
 - 1. A site plan showing the BESS footprint
 - 2. Electrical schematic diagrams (interconnection, system one-line diagrams)
 - 3. Usable energy storage capacity (kWh)
 - 4. Rated power (kW AC)
 - 5. AC:AC efficiency (including auxiliary loads)
 - 6. Cycle life
 - 7. Annual degradation factor
 - 8. Ambient temperature control system
 - 9. Fire protection/suppression system description as required by code
 - 10. Equipment manufacturers and product names
 - 11. Battery Management System provider
 - 12. Maintenance requirements
- B. The Contractor shall identify an appropriate location for the balance of system enclosure, and its related components and environmental control systems that will meet the following criteria:
 - 1. Ease of maintenance and monitoring
 - 2. Efficient operation
 - 3. Low operating losses
 - 4. Secured location and hardware
 - 5. Compatibility with existing facilities
 - 6. Minimum vegetative and landscape impact
- C. All systems (wiring, component, conduits, and connections) shall be suited for conditions for which they are to be installed.

1.11 OPERATIONS AND MAINTENANCE

- A. The Contractor shall conduct annual O&M and continuous monitoring for a period of (5) Years to verify that the BESS is performing as intended per the proposed battery storage strategy and manufacturer recommendations and submit an annual report to Owner. The report shall include:
 - 1. Use case performance
 - 2. Battery outages with root cause summary and start and end time periods.
 - 3. Summary of all O&M operations; repair and replacements
 - 4. Summary of safety incidents, causes and resolutions
 - 5. Contractor shall perform all required maintenance to the BESS and BMS to ensure that the monthly demand reduction is provided.
 - 6. Contractor will provide O&M training and supporting manuals to Owner personnel.
 - 7. Perform updates to the BMS dispatch strategy if required due to rate tariff or major site load profile changes.
 - 8. Ensure that all system firmware is up to date and meets cybersecurity requirements.

1.12 DELIVERY, STORAGE, AND HANDLING

- A. All equipment shall be handled with care so as not to damage the delivered products. All equipment shall be installed in new and neat condition.

- B. Appropriate protective clothing shall be worn when handling the equipment. Such clothing shall include hard hats and steel-toe boots when lifting materials to roof, and insulated gloves when working on an active system.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Acceptable system manufacturers/vendors shall have been in the business of producing and/or installing similar commercial grade energy storage systems for the last 3 years minimum. Manufacturers shall provide their latest line of equipment, meeting all current industry standards.
- B. The minimum performance of the submitted and installed system shall substantially match the physical and electrical equipment characteristics of the basis of design, as shown on the electrical drawings.
- C. Substitute equipment is acceptable when meeting the minimum system performance matching the basis of design, requirements listed herein, and complying with all other technical aspects of the system.

2.2 PERFORMANCE

- A. Grid outage scenario: Integrate the BESS with the PV and generator being provided as part of this project and automatic transfer switch (ATS). Install a microgrid control system and control the existing ATS as necessary to provide electricity to the building upon grid failure.
 - 1. Configure BESS to provide seamless transition between utility power and microgrid. Optimize system to minimize generator use to conserve fuel.
 - 2. The BESS shall be demonstrated to perform on a recurring basis to ensure the Owner's desired site resilience and energy assurance. This interval testing frequency shall occur approximately every twelve months following final acceptance to ensure provision of power to the site in the event of a utility-provided power outage or interruption. This activity requires pre-coordination with the servicing utility in accordance with the approved interconnection agreement, and that coordination shall be led by the Contractor in consultation with the Owner's facilities management staff. This periodic demonstration shall stagger months, seasons, load, time of day, and weather variabilities, but shall be conducted at least every twelve (12) months. Periodic testing will be the responsibility of the BESS supplier for a period of the first (5) years after installation. In the event of an actual utility power interruption exceeding (12) hours, the BESS's performance will satisfy this requested demonstration, thereby resetting the interval to twelve (12) months prior to the next required demonstration. Any deficiencies or anomalies beyond expected design parameters, in an actual event or demonstration event, are to be reported to the Owner within twenty-four (24) hours in writing, with recommended remediation or corrective steps to prevent recurrence.
- B. Demand control energy arbitrage scenario: Integrate the BESS to provide grid electricity usage and demand control/ peak shaving to limit demand (kW) to programmed setting. The BESS shall have a method for forecasting the peak load and automatically dispatching the battery or scheduling the charge/discharge in advance.
 - 1. Time of Use Charge Reduction: Integrate the BESS to discharge during on-peak hours in accordance with the site's rate tariff.

2. A BMS (battery management system) shall be provided to control the charging and discharging of the equipment. The BMS shall be field programmable by connecting with a laptop and viewing/editing on a locally hosted web browser. Programming instructions and setpoints shall be shared with Owner staff.
3. BESS shall maintain a round-trip efficiency greater than 70% AC-in to AC-out (confirmable by operation over a single round-trip cycle from 0% state of charge to 100%, and back to 0% at standard conditions specified by the manufacturer), including thermal losses and auxiliary loads, over 10 years.
4. Contractor shall guarantee annual BESS savings as established in the proposal. Savings will be generated by discharging the battery during on-peak periods, as well as during peak billing demand periods. Contractor shall propose a methodology to calculate optimal savings, subject to review by Owner, and guarantee optimal savings.
5. Operations and Maintenance for the first (5) years of operation.

2.3 THERMAL MANAGEMENT

- A. The Contractor shall provide all components to operate the BESS within acceptable operating temperatures. Provide any thermal management systems and operating strategies required to maintain the BESS and inverter temperatures within manufacturer's recommendations at all times.

2.4 REGISTRATION SEALS

- A. Each submitted final design drawing, calculation document, and specification manual shall be signed and dated by, bear the seal of, and show the State Certificate Number of the Architect or Engineer who prepared the document and is responsible for its preparation.

2.5 COORDINATION OF PROFESSIONAL SERVICES

- A. The Contractor shall be responsible for the professional quality, technical accuracy, and coordination of all investigations, evaluations, drawings, testing, cost estimating, submittals, written reports, construction, operations, and all deliverables, as required by this document or as required to complete the work of this contract.

2.6 COORDINATION OF SUBCONTRACTORS' CREDENTIALS

- A. The Contractor shall ultimately be responsible for the completeness, accuracy, coordination, and submission of all submittals described above. Contractor may delegate the preparation of submittals to subcontractors or suppliers as long as the intent of Sections 2.5, 2.6, and 2.7 of this specification is met.

2.7 GROUNDING

- A. A suitable equipment grounding system shall be designed and installed for the BESS. The grounding system shall provide personnel protection for step and touch potential in accordance with IEEE 80. The system also shall be adequate for the detection and clearing of ground faults within the BESS. The system should be grounded in all anticipated operating modes (e.g., grid-tied and islanded).

2.8 STRUCTURAL

- A. The vendor shall furnish the design for the structural components of the BESS, concrete pads/foundations as required, and conduit required for the complete BESS. All final (Issued for Construction) drawings, specifications, and calculations shall be stamped by a state-licensed Civil/Structural Professional Engineer. The vendor is responsible for Geotechnical surveying if required.

2.9 CONDUIT AND PREVENTING WATER INTRUSION

- A. Conduit routing and fittings must be selected to prevent water intrusion into inverter enclosures, combiner boxes, switchgear, and transformers. Conduits are to connect through the bottom of enclosures and provide fittings to allow water to drain prior to entering the electrical enclosure. Any exterior PVC conduit must be Schedule 80.

2.10 LOCATING EQUIPMENT

- A. Major electrical components, including the inverter, isolation transformer, monitoring/controls, and metering, shall be installed in code-compliant enclosures. BESS location shall be above the 20 year flood plain and surge levels and shall be elevated higher than any pull vaults and conduit not sloped toward any BESS electrical component in order to prevent equipment flooding. BESS shall be located within a thermally conditioned enclosure.

2.11 EXPECTED SERVICE LIFE

- A. Unless noted otherwise, all materials furnished for the project shall have an expected service life of at least 10 years.

2.12 SITE SERVICE CONDITIONS

- A. Materials shall be designed to withstand the year-round temperatures and conditions to which they are exposed (sunlight, heat, humidity, rain, wind, sand/dust, seismic activity, salt air, fog, marine corrosiveness, etc.)

2.13 NEW EQUIPMENT

- A. The Owner shall not accept used, reconditioned, after-market, or grey-market products or equipment. Any offeror supplying used, reconditioned, after-market, or grey-market products may be held responsible for damages to the Owner.

2.14 MARKINGS (LABELING)

- A. Strict conformance to system marking requirements of BESS and their components is crucial for the safety of operators, service personnel, emergency responders, and others. Include all required and desired labeling language in the design drawings for Owner review.
 - 1. Electrical equipment and components used in BESS shall have markings that identify the manufacturer, size, type, ratings, hazard warnings, and other specifications.
 - 2. Labelling shall include posted instructions for tasks that site staff may need to perform, such as system shutdown during an emergency.
 - 3. All disconnects shall be clearly labelled, indicating operating system voltage, current, and system rating.

4. Equipment markings should never be removed and should be able to withstand the environmental conditions in which the equipment is installed (e.g., "UV rated" for outdoor labels, or on an embossed steel placard, designed for outdoor use and fastened with adhesive and rivets).
5. Markings must be visible or easily accessible during and after installation. The Contractor shall be responsible for all field-applied markings as required by local, state, and federal codes.

2.15 DATA ACQUISITION AND MONITORING

- A. Contractor shall provide a turnkey data acquisition and display system that allows the Owner to monitor, diagnose, and track the charging, discharging, and operating data of the BESS. Minimum requirement is the provision of a web-based monitoring and tracking system. Contractor shall provide internet connection to the BESS, distinct from Owner internet. Ensure that the data acquisition and monitoring solution is compliant with Owner cybersecurity. Monitoring and tracking systems shall include a historical database and real time data portal capturing the data in 15-minute intervals. A minimum of 36 months of data shall be stored by the Contractor and be made available for Owner download via the web portal. The data shall, at a minimum, comprise the following information and frequency of collection:
 1. Date, time.
 2. Apparent power (kVA)/phase, real power (kW) and Volts on each phase; recorded in 15-minute intervals.
 3. BESS state of charge
 4. Ambient temperature, hourly average at hourly intervals, either from on-site measurements or a reliable climate data service.
 5. The web-based monitoring system shall report actual system performance and an estimate of expected performance.
 6. The system shall allow Contractor to interact with BESS BMS to update settings and modify setpoints.

2.16 BESS SAFETY

- A. The BESS shall have a Data Acquisition/monitoring/alarm system. System shall be described in the offeror's proposal and shall include:
 1. Full monitoring of electrical power and related operational data, including voltage, current, and system temperature.
 2. Visual and audible alarm if potential safety hazard exists.
 3. Notification when preventive maintenance is needed.
 4. System level alerts shall be provided by the manufacturer over customer interface.
- B. The BESS shall contain protective relays, circuit breakers, or fuses which self-protect the BESS in the case of internal electrical faults. Set and adjust circuit protection devices according to a short circuit and coordination study.
- C. A detailed plan surrounding battery cell thermal runaway detection and mitigation systems in the BESS will be included in the submittal process.
- D. A visible disconnect shall be installed that isolates BESS in accordance with utility interconnection requirements.
- E. All electrical equipment, enclosures, disconnects, and overcurrent devices shall be clearly marked and identified. Markings shall reference the same designations called out in the final design drawings.

- F. A fire detection/suppression system shall be provided as required by code or by manufacturer.
- G. Develop posted instructions for tasks that site staff may need to perform, such as system shutdown during an emergency.

2.17 SEVERE WEATHER

- A. Earthquake Mitigation—Comply with UFC 3-301-01 for requirements related to the foundation, soil stability, and seismic analysis.
- B. Flooding/Other Water Intrusion Considerations
 - 1. Use NEMA 3R rated or better enclosures
 - 2. Coordinate with Civil Engineer to ensure a comprehensive site stormwater management and drainage plan that encompasses location of BESS out of path of stormwater flows.
 - 3. In addition to code requirements, design shall prevent scour of soils in and around BESS to prevent loss of foundation integrity.

PART 3 - EXECUTION

3.1 COMMISSIONING

- A. A commissioning plan shall be provided by Contractor and approved by Owner prior to proceeding with inspection and commissioning:
 - 1. After Owner's utility provides permission (interim for testing) to operate in parallel with the grid, the Contractor shall complete commissioning in accordance with Owner safety and commissioning plans, its own quality-control plan, manufacturer's recommendations, and franchise utility interconnection requirements. The Contractor shall document all performance measurements.
 - 2. The Contractor shall perform all work required for testing, start-up, and commissioning.
 - 3. The BESS shall be commissioned per the Electric Power Resource Institute (EPRI) "ESIC Energy Storage Commissioning Guide", or by a similar standard reviewed and approved by Owner.
 - 4. Contractor shall give Owner ten (10) business days advance notice prior to starting commissioning. An Owner manager or representative shall be present during any or all phases of the start-up, commissioning, and testing activities.
 - 5. The system shall be started-up and tested in accordance with the regulations of the applicable interconnection standards. All possible anticipated modes of operation shall be tested (grid-tied, islanded).
 - 6. The Contractor shall request and coordinate System Acceptance Testing with the utility and Owner for acceptance and certification of the BESS and permission to operate, after commissioning and final inspections are complete.
 - 7. Commissioning shall include the Contractor's Data Acquisition System and enclosure thermal conditioning system.
 - 8. A commissioning report shall be provided to Owner.

3.2 INSPECTIONS

- A. The Project Manager or the delegated Project Manager's Representative may inspect the Systems at any time during construction or after the Systems have been put in operation. The Contractor may be ordered to stop work, or shut the systems down, if unsafe conditions or code violations are noted.

- B. An initial inspection for Substantial Completion will be made when the work is complete to the point that the Contractor is ready to begin starting, testing, and commissioning the system. Following this inspection, the Owner will provide the Contractor with a punch list describing any incomplete work that must be complete prior to the Contractor's request for final inspection.
- C. The Contractor shall give the Project Manager ten (10) business days advance notice, in writing, of the date the work will be ready for initial inspection.
- D. A final inspection of all Systems shall be made only when all construction is complete in accordance with the terms and conditions set forth in the Contract and all punch list items identified in the initial inspection are complete. If, upon examination by the Project Manager and/or Owner inspection personnel, the project is determined not sufficiently completed to have warranted a final inspection, the Contractor may be charged for any additional cost of re-inspection.
- E. The Contractor shall give the Project Manager ten (10) business days advance notice, in writing, of the date the work will be fully completed and ready for final inspection.
- F. The Contractor's request for final inspection will not be approved unless documentation below, at a minimum, has been provided to, and accepted by, the Project Manager, in addition to all other contract requirements:
 - 1. Final as built drawings (record drawings), meeting as-built requirements
 - 2. Preventive maintenance work schedules and procedures
 - 3. Operation and maintenance manuals (electronic and hardcopy)
 - 4. Training manuals (electronic and hardcopy)
 - 5. Equipment documentation and spare parts lists
 - 6. Certificates of Authority Having Jurisdiction (AHJ) inspections
- G. After commissioning is complete, the Contractor shall instruct and train Owner designated personnel on normal system operation and how to shut down the System in the event of an emergency. Owner may reasonably request additional training and Contractor will conduct such training at Contractor expense. Training may take place over multiple sessions to accommodate Owner's staff availability. The training session(s) may be recorded.
- H. As soon as practicable, following final inspection, the Project Manager will inform the Contractor, in writing, of any discrepancies and/or omissions noted at the final inspection. The Project Manager shall also state the time allowable for replacement of material and performance or re-performance of any unsatisfactory work necessary before written notification of System Acceptance Testing.
- I. Upon written notification that all deficiencies identified during the final inspection have been corrected, the Project Manager may schedule a follow up inspection to confirm all correct work is acceptable. Contractor shall then coordinate with utility, Owner, and any other AHJ to schedule System Acceptance Testing.

3.3 SUMMARY OF PROCESS

- A. In summary, inspections and commissioning will follow the following sequence:
 - 1. Initial inspection for Substantial Completion of construction
 - 2. Commissioning may begin concurrently with the initial inspection
 - 3. Final inspection may be scheduled following completion of commissioning and approval of all documents shown in Section 3.2(vi) of this specification

4. Contractor will be responsible for completing any re-work identified in the Final inspection prior to scheduling System Acceptance Testing

3.4 FINAL ACCEPTANCE

- A. The Contractor shall notify the Owner not less than ten (10) business days prior to the anticipated date of System Acceptance Testing. The Owner shall be present at and observe the System Acceptance Testing, at the Owner's sole cost.
 1. The System Acceptance Testing shall be witnessed by utility if required for interconnection approval and the Contractor shall meet the utility's test notification requirements.
 2. In addition to utility and Owner System Acceptance Test standards, the test shall include, at a minimum, the following:
 - a. While connected to grid power, start up the BESS until it achieves the minimum specified performance requirements. The acceptable productive power output will be measured in kW (AC) at the building electrical interconnection point and must be consistent with the specifications for the system.
 - b. Contractor shall demonstrate the operation of one full charge/discharge cycle of the BESS and report on system performance.
 3. Approvals as required by the fire marshal and local electric utility will be a pre-requisite for acceptance and for authorization to energize the system(s).
- B. Upon successful completion of System Acceptance Testing, Contractor shall send a Completion Notice and a copy of the System Acceptance Test report to the Project Manager so that the Owner can complete their final acceptance. Owner shall have ten (10) Business Days after receipt of the Completion Notice to review the System Acceptance Testing results and verify that the System installation is complete, safe, aesthetically acceptable, functional, constructed to all code requirements, does not interfere with Owner or tenant operations, and otherwise meets all other requirements. The Project Manager will notify the Contractor in writing of Final Acceptance.
- C. If any of these requirements are not met, then the Owner shall provide Contractor with a detailed notice of such failure (a "Rejection Notice") within the ten (10) business day period, with details regarding the required remedy (including repeat of either partial or full System Acceptance Testing, if appropriate, at the discretion of the Project Manager or Project Manager's Representative, and the time allowed to complete remedy. The Contractor shall promptly remedy, at Contractor's cost, the items identified in the Rejection Notice and conduct any additionally required System Acceptance Testing (if required by the Rejection Notice) until the System Acceptance Testing indicates that the System meets the contract requirements. In each such case, the Contractor shall send a new Completion Notice to the Owner with a copy of the results of the new System Acceptance Testing as provided above and the foregoing procedures shall be repeated.
- D. Written acceptance shall be final and conclusive except as regards latent defects, fraud, or such gross mistakes as may amount to fraud, or as regards the Owner's right under any warranty or guarantee, subject to the system performance warranty.
- E. In summary, the following requirements must be fulfilled before final acceptance:
 1. System Acceptance Testing has been completed, with the System having operated at specified performance for one full charge/discharge cycle with power production levels consistent with proposed System's estimated production with 100% system availability, measured with applicable instruments and meters.
 2. The System has been approved for interconnected operation by utility (with signed interconnection agreement).

3. Submission of as-built drawings and all documents required prior to final inspection described in Section 3.2 of this specification.
4. Training fulfillment documentation.
5. Commissioning report provided to Owner.

END OF SECTION

SECTION 26 32 14 - EMERGENCY ELECTRIC GENERATOR

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Provide a complete Emergency Generator plus accessories in a single common housing, as noted herein and as indicated on the Drawings.
- B. Provide CBC 2022 compliant seismic installation. See Section 26 05 00 for all certification and submittal requirements.
- C. Manufacturer shall be responsible for providing a complete assembly of all components in one overall housing with base tank, which is certified to comply as an integrated assembly with CBC 2022 requirements.
- D. Manufacturer shall provide technical assistance to Owner in securing all required local Air Quality Management District permits for installation of the Emergency Generator, including review and inclusion of existing engine generators where applicable on the same site in site analysis.
- E. The Contractor shall include a Manufacturer-provided three-year engine-generator maintenance agreement as described herein as part of the bid. The maintenance agreement shall include an annual 2-hour full load test using a portable load bank.

1.2 SUBMITTALS

- A. Refer to Section 26 05 00 for procedure.
- B. Tests and Reports (Test Requirements are detailed in Paragraph 1.4).
 - 1. Provide certified test reports of the following:
 - a. Factory tests.
 - b. Field Tests: Test reports shall include dates performed, method of testing, test results, test interpretation and recommended action.
 - 2. Shop Drawings and Product Data
 - a. The following list includes the required Shop Drawing information that shall be submitted for the generator:
 - 1) Physical dimensions and weights.
 - 2) CBC Special Certificate of Compliance with importance factor of 1.5 for all components and overall assembly.
 - 3) Brake horsepower of engine.
 - 4) Fuel consumption.
 - 5) Cooling requirements.
 - 6) Noise db level. Provide details of acoustical housing and factory testing to prove acoustical housing performance (71dbA at 7 meters).
 - 7) Electrical characteristics of generator, voltage regulator, and battery charger.
 - 8) Load graphs.
 - 9) Control panel.
 - 10) Elevation.
 - 11) Wiring and control diagrams.
 - 12) Engine and generator details, including governor and base fuel tank.
 - 13) Location of available parts and service.

- 14) Confirmation that engine meets the latest required EPA Tier Exhaust Emission Compliance Statement, complies with latest CARB requirements for emergency standby equipment, and complies with the local Air Quality Management District requirements.
 - 15) Details of base tank fill connection, including overfill basin.
 - 16) Details of primary tank venting, primary and secondary tank emergency relief venting, high fuel alarm panel and sensing, and tank labeling.
 - 17) Passive diesel particulate filter.
3. Maintenance and operating instruction manuals, six bound copies, including approved shop Drawings, parts list, list of recommended spare parts, sources of purchase and similar information.

1.3 GUARANTEE

- A. Provide a written guarantee against all defects in materials and application, which prevent proper functioning for one (1) year from date of acceptance of the project.

1.4 TESTS

- A. Certified copies of factory test giving guaranteed performance characteristics to meet the Specifications should be furnished by the Manufacturer. The unit shall be tested at the Manufacturer's plant for performance of all functions including a 2-hour full load test, using 0.8pf reactive to 1.0pf resistive load banks and until all temperatures have been stabilized for at least 30 minutes.
- B. The Manufacturer shall have field tests made of the generator and wiring systems in place by a qualified factory technician. The complete engine generator set with all of its appurtenances shall be tested after installation for all functions, including a 2-hour full load test with full-rated resistive (1.0pf) load bank. The Manufacturer shall supply all equipment necessary for the tests.
- C. Testing of fuel system - see Part 3 of this Section.

1.5 FUNCTION

- A. The emergency generator shall function to start automatically immediately upon power failure of the normal power supply, assume full load within ten (10) seconds and automatically switch into the emergency power system. The unit shall be automatically removed from the line upon resumption of normal power and stopped five (5) minutes later.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. General:
 1. Provide and install a complete emergency power system, including power plant powered by diesel engine driven generator and operated by means of a signal from an automatic transfer switch. The system shall be complete, tested and meet all the functional requirements of a fully automatic emergency power source serving full load power stabilized at rated voltage and frequency within three seconds after normal power source failure.

2. The diesel engine generating set shall be fully automatic and shall be complete with starting and control equipment, base fuel-tank, skid-mounted batteries, charger, acoustical enclosure, and other equipment necessary to provide a complete, fully automatic system.
3. Arrange for the services of a factory erection engineer for checking installation, making specified and all other necessary tests, making initial start, instructing operating personnel in operating unit through all of its functions to ensure that the unit is performing in accordance with the intentions of the Specifications.
4. Generator set shall be provided with a vertical scoop on discharge.
5. Manufacturers: Cummins-Onan, Kohler, Generac, or Caterpillar.
6. Power Plant:
 - a. Rating shall be based on continuous standby power rating of the generator and with capabilities to carry 100% full load without damage to the engine, generator or components, and with capabilities for starting the largest motor scheduled for the standby power system while carrying full connected load at an altitude of 350 feet above sea level. Full load power ratings shall be as indicated on the drawing, continuous standby at 0.8 PF at 120/208V 3PH 4W where indicated on the drawings.
 - 1) Voltage regulation shall be +/-0.5% for any constant load between no load and rated load.
 - 2) Frequency regulation shall be isochronous from steady state no load to steady state rated load.
 - 3) The diesel engine generator set shall be capable of single step load pick up of 100% nameplate KW and power factor, less applicable derating factors, with the engine-generator set at rated operating temperature.
 - 4) Under motor starting conditions the generator set shall be capable of sustaining a minimum of 90% of rated no load voltage with the specified KVA load at near zero power factor applied to the generator set.
 - 5) Maximum transient voltage dip shall not exceed 35 percent below rated on application of the single largest surge load step.
 - b. Provide unit and all accessories in a common weatherproof acoustical enclosure, with a common base for unit capable of skidding into place. Provide sound absorbing insulation on interior of weatherproof enclosure to produce overall sound rating noted above. All accessories, including muffler shall be concealed within common generator enclosure. If a diesel particulate filter is specified, it shall be concealed within the enclosure.
 - c. Provide minimum 18" flexible section in all electrical, fuel and exhaust lines at connection to power plant.
 - d. Provide flexible steel disc coupling to engine-generator.
 - e. Provide lifting brackets.
 - f. Provide a CBC 2019 compliant and certified vibration isolation system. The vibration isolation system shall be designed to withstand the seismic forces from ground motions and installation shall comply with critical equipment (importance factor = 1.5) requirements. CBC conforming vibration isolators shall be provided and installed at the bolt-down locations between the skid and the base fuel tank.
7. Engine:
 - a. The engine shall be 4-cycle design, water-cooled; series turbo charged with after cool, having no inherent unbalanced reciprocating forces. Operating speed shall be 1800 RPM.
 - b. Starting by battery-driven starter. Include cranking sequencer, which shall give three (3) start attempts before locking out over-cranking protection.
 - c. Governor shall be isochronous electronic as required to maintain generated frequency at 60 Hz. at 75% full load within a steady state band-width of (+/-) 0.25%. Frequency shall not vary over 3% from no load to full load. Governors using engine crankcase lube oil will not be acceptable. Governor shall be type EFC, Electronic Isochronous.
 - d. Provide fuel and lubrication systems for diesel engine, complete with replacement element type air cleaner; primary and secondary fuel filter and oil filter; full pressure

lubrication system with positive displacement lube oil pump and spring-loaded bypass valve; lube oil cooler; engine driven fuel transfer pump; extended base-mounted fuel tank, sight gauge, automatic float switch to maintain fuel level, and high-fuel and low-fuel alarm dry contacts for local and remote indication. Provide base fuel tank rupture basin contacts (2 sets) for local and remote alarm. Point of fuel fill at the base tank shall have provisions for spill containment. Provide shop drawings indicating spill containment system.

- e. Provide a cooling system with sufficient capacity for cooling engine when generator is delivering 100% full load for four hours at ambient of 40 degrees C at an altitude of 350 feet. Include water-circulating pump and thermostatic valve to maintain recommended engine temperature; radiator with drain and air vent and fan with protective guard; jacket water corrosion resistant heating elements (wattage as specified by engine manufacturer, rated at 208V 1PH or 3PH). Radiator shall be filled with antifreeze solution of strength as recommended by Manufacturer. Exhaust air shall be discharged vertically, using a scoop design.
- f. All areas within 24 inches of the muffler shall be covered with 6 lb. density mineral wool. All piping shall be schedule 40 black pipe.
 - 1) The exhaust system, piping, and insulation shall be factory supplied and installed. Extend black standard weight iron pipe from the engine with 18" flexible connection between engine and muffler. The muffler shall be installed within the same overall enclosure as the engine-generator.
- g. Provide a unit-mounted battery for engine start, 24-volt DC with a capacity of not less than 160-ampere hours at 20-hour rate. The battery shall also be sized for six starts of 30-second cranking duration each, with ambient 15 degree F, mounted on earthquake-proof tray on pad with all necessary battery cables, hydrometer and enclosure-mounted voltage-regulated battery charger in Nema 3R enclosure with float, taper, and equalize charge settings and with DC voltmeter, DC ammeter and circuit for low voltage alarm. Battery shall be lead-acid type.
- h. Auxiliary switches for over-speed trip and automatic over-speed shut down at a speed 10% greater than the normal specified operating RPM. The engine shall shut down on over-speed, low oil pressure, high oil temperature and high water temperature by means of auxiliary switches, actuating signal lights and alarms.
- i. Temporary batteries may be used for testing, but new, unused batteries shall be furnished after final testing is complete and before acceptance. New batteries shall be used for one generator start to demonstrate adequacy of final battery installation.
- j. Engine exhaust emissions shall meet the latest adopted EPA Tier Exhaust Emission Compliance Statement, the latest CARB requirements for emergency standby applications, and the local Air Quality Management District standards.
- k. Provide a DPF - concealed within enclosure.

B. Generator:

- 1. Rated for continuous standby service, complying with NEMA standards.
- 2. Brushless, balanced 4-pole revolving field type with rotating rectifier exciter mounted on end of shaft, single ball bearing support to starter housing, rotor connected by semi-flexible steel disc coupling to engine flywheel to assure permanent alignment free of injurious tensional vibrations at speeds up to 125% of synchronous. Rated for 105 degrees Centigrade rise.
- 3. Generator insulation shall be in accordance with latest NEMA standards using minimum Class H materials. All insulation system components shall meet NEMA MG1 temperature rise limits for Class H insulation system. Actual temperature rise measured by resistance method at full load shall not exceed 105 degrees centigrade.
- 4. A permanent magnet generator (PMG) shall provide excitation power for immunity to voltage distortion caused by non-linear loads. The PMG shall sustain excitation power for optimum motor starting and to sustain short circuit current at approximately 300% of rated current for not more than 10 seconds.

5. Voltage regulator of static solid state design to give (+/-) 2% regulation from no-load to full load; instantaneous voltage dip less than 20% of rated when full load at rated power factor suddenly applied; and recovery to stable operation of voltage within 1% of rated within four seconds. The voltage regulator shall be of the asynchronous pulse width modulated type and shall be insensitive to severe load-induced waveshape distortion from SCR or thyristor circuits such as those used in UPS and motor speed control equipment. Manual adjustment of (+/-) 5% of normal to be included by a lockable device or screwdriver slot in rheostat shaft. All voltage sensing shall be 3-phase.
 - a. The automatic voltage regulator shall be temperature compensated, solid-state design. The voltage regulator shall control build up of AC generator voltage to provide a linear rise and limit overshoot. The regulator shall include a torque-matching characteristic, which shall use differential rate of frequency change compensation to use the maximum available engine torque and provide optimal transient load response. Regulators, which use a straight line fixed volts per hertz characteristic, are not acceptable.
 - b. Shielding of generator, exciter and regulator to prevent radio frequency interference.
 - c. The generator, exciter, and voltage regulator shall be designed and manufactured by the engine-generator set manufacturer so that the characteristics shall be matched to the torque wave of the engine to provide the fastest possible system recovery from transient load changes and to prevent engine stall during transient overload conditions.
6. Control Equipment:
 - a. Panel mounted with vibration isolators to plant frame. Top of panel shall not exceed 6'-6" above slab with generator mounted to a 30" high base tank on top of a 3" high pad.
 - b. The generator set shall be provided with a microprocessor-based control system, which shall be designed, to provide automatic starting, monitoring and control functions for the generator set. The control system shall also be designed to allow local monitoring and control of the generator set and remote monitoring and control as described in this Specification. The control panel shall be mounted on the generator set.
 - c. The control panel shall be vibration isolated and prototype tested to verify the durability of all components in the system under the vibration conditions encountered.
 - d. The control panel shall be UL508 labeled, CSA282-M1989 certified and meet IEC8528 part 4. All switches, lamps and meters shall be oil-tight and dust-tight and the enclosure door shall be gasket. There shall be no exposed points in the control panel (with the door open) that operate in excess of 50 volts. The controls shall meet or exceed the requirements of Mil-Std 461C part 9 and IEC Std 801.2, 801.3 and 801.5 for susceptibility, conducted and radiated electromagnetic emissions. The entire control shall be tested and meet the requirements of IEEE587 for voltage surge resistance.
 - e. The generator set mounted control panel shall include the following features and functions:
 - 1) Three position control switch labeled RUN/OFF/AUTO: In the RUN position the generator set shall automatically start and accelerate to rated speed and voltage. In the OFF position the generator set shall immediately stop, bypassing all time delays. In the AUTO position the generator set shall be ready to accept a signal from a remote device to start and accelerate to rated speed and voltage.
 - 2) Red "mushroom-head" push-button EMERGENCY STOP switch: Depressing the emergency stop switch shall cause the generator set to immediately shut down and be locked out from automatic restarting.
 - 3) Push-button RESET switch: The RESET switch shall be used to clear a fault and allow restarting the generator set after it has shut down for any fault condition.
 - 4) Push-button PANEL LAMP switch: Depressing the panel lamp switch shall cause the entire panel to be lighted with DC control power. The panel lamps

shall automatically be switched off 10 minutes after the switch is depressed or after the switch is depressed a second time.

- 5) Generator Set Metering: The generator set shall be provided with a metering set with the following features and functions:
 - (a) 2.5-inch, 90-degree scale analog voltmeter, ammeter, frequency meter and kilowatt (KW) meter. These meters shall be provided with a phase select switch and an indicating lamp for upper and lower scale on the meters. Ammeter and KW meter scales shall be color coded in the following fashion: readings from 0-90% of generator set standby rating: green; readings from 90-100% of standby rating: amber; readings in excess of 100%: red.
 - (b) Digital metering set, 0.5% accuracy, RMS type to indicate generator voltage, frequency, output current, output KW, KW-hours and power factor. Generator output voltage shall be available in line-to-line neutral voltages and shall display all three-phase voltages (line to neutral or line to line) simultaneously.
- 6) Generator Set Alarm and Status Indication: The generator set shall be provided with alarm and status indicating lamps to indicate non-automatic generator status and existing alarm and shutdown conditions. The lamps shall be high-intensity LED type. The lamp condition shall be clearly apparent under bright room lighting conditions. The generator set control shall indicate the existence of the following alarm and shutdown conditions on a digital display panel.
 - (a) Low oil pressure (alarm)
 - (b) Low oil pressure (shutdown)
 - (c) Oil pressure sender failure (alarm)
 - (d) Low engine temperature (alarm)
 - (e) High engine temperature (alarm)
 - (f) High engine temperature (shutdown)
 - (g) Engine temperature sender failure (alarm)
 - (h) Low coolant level (alarm or shutdown--selectable)
 - (i) Fail to crank (shutdown)
 - (j) Over-crank (shutdown)
 - (k) Over-speed (shutdown)
 - (l) Low DC voltage (alarm)
 - (m) High DC voltage (alarm)
 - (n) Weak battery (alarm)
 - (o) Low fuel-base tank (alarm)
 - (p) High AC voltage (shutdown)
 - (q) Low AC voltage (shutdown)
 - (r) Under frequency (shutdown)
 - (s) Over current (warning)
 - (t) Over current (shutdown)
 - (u) Short circuit (shutdown)
 - (v) Ground fault (alarm)
 - (w) Under frequency (alarm)
 - (x) In addition, provisions shall be made for indication of two customer-specified alarm or shutdown conditions. The non-automatic indicating lamp shall be red and shall flash to indicate that the generator set is not able to automatically respond to a command to start from a remote location.
- 7) Engine Status Information: The following information shall be available from a digital status panel on the generator set control:
 - (a) Engine oil pressure (psi or kPA)
 - (b) Engine coolant temperature (degrees F or C; both left and right bank temperature shall be indicated on V-block engines)

- (c) Engine oil temperature (degrees F or C)
 - (d) Engine speed (rpm)
 - (e) Number of hours of operation (hours)
 - (f) Number of start attempts
 - (g) Battery voltage (DC volts)
- 8) The Generator Control Panel shall monitor the status of each Automatic Transfer Switch. The monitoring for each Automatic Transfer Switch shall be:
 - (a) ATS Normal Position
 - (b) ATS Emergency Position
- 9) The ATS status for each switch shall also be displayed on the Generator Remote Annunciator.
- f. Control Functions: The control system provided shall also include a cycle cranking system, which allows for user selected crank time, rest time and number of cycles. Initial settings shall be for 3 cranking periods of 15 seconds each with 15-second rest periods between cranking periods.
- g. The control system shall include an idle mode control, which allows the engine to run in idle mode in the RUN position only. In this mode the alternator excitation system shall be disabled.
- h. The control system shall include an engine governor control which functions to provide steady state frequency regulation as noted elsewhere in this Specification. The governor control shall include adjustments for gain; damping and a ramping function to control engine speed and limit exhaust smoke while the unit is starting. The governor control shall be suitable for use in paralleling applications without component changes.
- i. The control system shall include time delay start (adjustable 0-300 seconds) and time delay stop (adjustable 0-600 seconds) functions.
- j. The control system shall include sender failure monitoring logic, which is capable of discriminating between failed senders or wiring components and an actual failure conditions.
- 7. Remote Indicator Panel: The remote annunciator panel shall be surface or flush-mountable with micarta label reading "Engine Operator Conditions," and with the following devices:
 - a. Green pilot light with engraving to indicate "ENGINE RUNNING."
 - b. One audible alarm with silencing switch to indicate engine start failure for any of the following reasons:
 - 1) High water temperature.
 - 2) Low oil pressure.
 - 3) Over-speed.
 - 4) Over-crank.
 - 5) Low battery voltage.
 - 6) Low-level fuel alarm.
 - c. Remote indicator panel shall include an amber light indication for each of the following:
 - 1) Control switch not in auto position.
 - 2) Low water temperature.
 - 3) ATS in normal position (for each ATS).
 - 4) ATS in emergency position (for each ATS).
 - d. Each of the functions listed previously in Paragraphs 2.(a) through (f) shall be indicated by a separate red warning light and each warning light shall be so engraved.
- 8. Fuel System: The fuel system shall conform to NFPA 30 and 37. The fuel system shall be complete and shall consist of a dual filtering system, and engine fuel pump. The engine shall start, operate, and stop on DF-2 fuel.
 - a. Fuel Filtering System: The fuel filtering system shall consist of a strainer, located between the storage tank and the fuel transfer pump, and a duplex fuel filter, located between the engine fuel pump and the engine. The filtering system shall be capable

of removing from the fuel system flakes, dirt, metallic chips, carbon, water, or other foreign matter, which would be harmful to the engine. The filtering system shall be easily accessible for quick-and-easy replacement of the filter element and cleaning of the strainer. Components of the filtering system shall be the standard products of the engine Manufacturer.

- b. Engine Fuel Pump: The engine fuel pump shall be a positive-displacement, engine-driven pump capable of supplying an adequate quantity of fuel to the engine under all operational conditions. Solenoid shut-off valve in fuel line discharge side of pump shall be interlocked with the emergency engine shutdown circuitry.
- c. Fuel Tank: The fuel tank shall be skid mounted under the engine, and shall be capable of providing an immediate fuel supply to the engine fuel pump. The tank shall be no more than 30" high. The tank shall be provided with a flexible tubing suction line to the engine, an excess fuel return line from the engine to the tank, a fuel gauge, a vent, a drainpipe, and high and low-level float switches for activating the low-level alarms. Usable fuel tank capacity shall not exceed 660 gallons - see drawings for exact capacity of fuel tank. The tank shall be a full double wall, U.L. listed design and shall be factory tested in accordance with the requirements of U.L. 142. The outer containment cavity shall be equipped with an emergency leak-detection float switch for the leak detection and alarm system. The tank shall as a minimum include the following:
 - d. Primary tank venting shall be provided and shall be a minimum of 1.25" nominal inside diameter. The primary vent shall be extended thru the roof of the generator enclosure to a height not less than 12 feet above grade. The normal vent shall have a screened mushroom style cap.
 - e. Emergency relief venting shall be provided for both the primary and secondary tanks. Rain protected, open style rupture basins are not acceptable. These vents shall be extended thru the roof of the generator enclosure to a height of not less than 12 feet above grade. The extended emergency vents shall be capped with U.L. listed pressure relief type caps designed to open at 2.5 PSI. Materials used for vent pipes, supply and return piping, valves and fittings shall conform to Article 79, Division VII.
 - f. A 5 gallon capacity overfill basin shall be provided around the fill pipe and shall include a means of draining spills back into the main tank. The fill pipe shall be extended down into the fuel tank and terminate within 6" of the bottom of the tank.
 - g. The high fuel sensor shall be wired to an audible and visual local alarm that shall be activated at 90% of the tank capacity. The alarm panel shall be mounted at the fill location and shall include a reset/silence for the alarm and light that will automatically reset the controller for the next fill. An "off switch" is not acceptable.
 - h. The tank shall include labels to indicate "Diesel Fuel". The lettering shall be 3" high, half-inch stroke, red letters on a white background outlined in red.
 - i. All of the above items shall be submitted for approval. Do not release the fuel tank for manufacture until it has been approved.
- 9. Output Circuit Breakers:
 - a. Provide engine-generator mounted circuit breakers for each automatic transfer switch, and where indicated on the drawings one for a portable load bank, with ratings as noted on the Drawings. Breaker handles shall not exceed 6'-6" above grade when engine-generator is mounted to a 30" high base fuel tank on top of a 3 inch high structural isolation pad.
- 10. Weather Protective Sound Attenuated Enclosure:
 - a. The generator set and accessories, including muffler, shall be completely housed in a weather protective and sound attenuated enclosure. The enclosure shall have a cambered roof to prevent rain accumulation, shall include stainless steel hardware to prohibit rust, and shall include stainless steel retainers to hold doors securely in place. The generator set, enclosure, and sub-base fuel tank shall be U.L. 2200 listed as a package.
 - b. Material used for the enclosure shall be 14-gage steel for panels and 12-gage steel for posts. Three hinged lockable access doors shall be provided on each side, with

hold-open retainers as indicated in Paragraph 1 above. Non-hygroscopic sound insulating materials shall be provided on the interior walls. Rodent barriers shall be provided on inlet and outlet sides. Louvers shall be fixed. The enclosure shall attach to the sub-bases fuel tank. Oil and coolant drains shall be run to the exterior of the enclosure. Interior valves on the oil and coolant lines shall be provided for ease of service.

- c. The enclosure shall completely house the muffler on the generator set.
- d. Provide sheet metal scoop on radiator output, to direct the radiator exhaust air directly up and vertically out of the generator enclosure. Provide screen on output of scoop to prevent dirt, leaves, and bird incursions. Provide drain on floor of scoop to drain off any water. Provide access door in scoop for cleaning of any accumulated debris. Scoop shall be factory-mounted as part of the engine-generator package. Provide detailed shop drawings of scoop with engine-generator submittals.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. The engine-generator set shall be mounted on a rigid steel chassis suitable for installation on seismic isolators.
- B. The Contractor shall assist the Owner in preparing and submitting a permit-to-construct application to the local Air Quality Management District for installation of the engine-generator. The Contractor shall pay for permit and permit approval fees, and the Owner shall pay for annual operational fees. Granting of such permit will require a site-specific screening application followed by an analysis by the local Air Quality Management District.
- C. Load tests shall be run as required in Paragraph 1.4 of this Section to the generator rated load after generator installation is complete. The Manufacturer shall provide auxiliary load banks for full-load testing of the generator. Manufacturer shall schedule the tests with the Owner so that final tests may be witnessed. Verify correct reading and operation of all meters, indicators and controls.
- D. Readings required during both preliminary and final tests requested in Paragraph 1.04 shall be taken and shall include the following:
 - 1. Frequency.
 - 2. Voltage.
 - 3. Current.
 - 4. Wattage.
 - 5. Ambient temperature.
 - 6. Water temperature.
 - 7. Oil pressure and temperature.
 - 8. Protection: Provide protection facilities and procedures to prevent damage and deterioration.
 - 9. Verify utility phase rotation prior to connection of the Generator to the distribution system. Modify the Generator output phase rotation to match the Utility Company.
 - 10. The Manufacturer shall perform the following fuel tank tests at the site after the installation is complete, but before the fuel is delivered. Tests shall be conducted in the presence of the Owner and shall include as a minimum the following:
 - a. Primary tank shall be pneumatically tested at 5 psig for 30 minutes; during which time the connections to the primary tank shall be soap tested.
 - b. Secondary tank shall be pneumatically tested at 3 psig for 30 minutes; during which time the connections to the secondary tank shall be soap tested.

- c. Primary piping shall be pneumatically tested at 150 percent of the maximum operating pressure for 30 minutes; during which time all fittings shall be soap tested.
 - d. Secondary piping, if provided, shall be pneumatically tested at 5 psig for 30 minutes, with soap testing.
 - e. Other secondary containment shall be liquid tight as demonstrated by a 24-hour standing water test.
 - f. Correct operation of the leak detection system shall be demonstrated.
11. The DPF manufacturer's authorized factory representative shall verify correct installation of the DPF and filter at the site prior to engine-generator start-up and shall perform all testing and commissioning of the DPF in conjunction with the engine-generator testing after the installation is complete. Tests shall be conducted in the presence of the Owner.

3.2 INSTRUCTION AND MAINTENANCE

- A. Instruct the Owner's personnel in the proper use, operation and maintenance of the set. Review emergency provisions, including emergency access and procedures to be followed at time of failure in operation and other building emergencies. Train Owner's personnel in the procedures to be followed, checking for the source of an operational failure or malfunction.
- B. Instruct the Owner's personnel in the proper use, operation and maintenance of the DPF, including maintenance and replacement of the filters and periodic use of a portable load bank to maintain DPF in optimum condition.
- C. Maintenance Period: Starting at the date of acceptance of the Work, provide complete systematic inspection and maintenance for the first three years. Furnish trained experts and equipment to check, adjust, lubricate and otherwise maintain the generator set in operation without defects or deterioration. Replace or repair materials and parts, which become defective or deteriorated for any reason.
- D. Furnish a factory-trained Engineer for a minimum of one working day prior to final acceptance of the generator installation, or as needed to satisfy Owner that the system is functioning properly. Testing and training for the new engine-generator installation will take place at non-standard times. Training and testing will take place on weekends, and could be scheduled on holidays and in the middle of the night, at the discretion of the Owner. Provisions shall be made in the bidding for this contract for such scheduling requirements.
- E. Provide 3 year Manufacturer maintenance Contract, for the new engine-generator installation. This shall include two site visits per year and annual load-bank testing as specified in Paragraph 1.1 of this Section.

END OF SECTION

SECTION 26 36 01 - AUTOMATIC TRANSFER SWITCH

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Provide an automatic transfer switch, integral to the main service panel, as indicated on the drawings and as specified herein.

1.2 SUBMITTALS

- A. Refer to Section 26 05 00 and Division 1 for procedure.
- B. Shop Drawings and Product Data, including complete wiring diagrams, including system interconnections.
- C. Test and Test Report for each Automatic Transfer Switch:
 - 1. Visual and Mechanical Inspection:
 - a. Inspect for physical damage.
 - b. Compare equipment nameplate information and connections with single line diagram and report any discrepancies.
 - c. Check switch to ensure positive interlock between normal and alternate sources.
 - d. Check tightness of all cable connections and bus joints.
 - e. Perform manual transfer operations.
 - f. Electrical Tests for Automatic Transfer Switch:
 - 1) Perform insulation resistance tests phase-to-phase and phase-to-ground with switch in both source positions.
 - 2) Set and calibrate in accordance with the Manufacturer's recommendations.
 - (a) Voltage-sensing relays.
 - (b) Transfer time delay relays.
 - (c) Engine shutdown relays.
 - 3) Perform automatic transfer by:
 - (a) Simulating loss of normal power.
 - (b) Return to normal power.
 - 4) Monitor and verify correct operation and timing:
 - (a) Normal voltage-sensing relays.
 - (b) Engine start sequence.
 - (c) Time delay upon transfer.
 - (d) Alternate voltage-sensing relays.
 - (e) Automatic transfer operation.
 - (f) Interlocks and limit switch function.
 - (g) Timing delay and retransfer upon normal power restoration.
 - (h) Engine shutdown feature.
 - (i) Correct functioning of auto-exercising controller.
- D. Maintenance and operating instruction manuals. Submit four bound copies including approved Shop Drawings, parts list, list of recommended spare parts, sources of purchase and similar information.

1.3 REFERENCE STANDARDS

- A. The following Specifications and standards, except as hereinafter modified, are incorporated herein by reference and form a part of this Specification to extend the indicated by the references thereto. Except where specific date is given, issue in effect (including amendments, addenda, revisions, supplements, and errata) on the bid date shall be applicable. In text such Specifications and standards are referred to by basic designation only.
- B. National Fire Protection Association (NFPA)
 - 1. No. 70 National Electrical Code (NEC)
- C. Underwriters' Laboratories, Inc. (UL):
 - 1. No. 1008 Automatic Transfer Switches
 - 2. No. 489 Molded Case Circuit Breakers
- D. National Electrical Manufacturers Association (NEMA):
 - 1. ICS 2.447 Industrial Control and Systems
- E. American National Standards Institute (ANSI):
 - 1. V37-90a Guide for Surge Withstand Capability (SWC) Tests

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. ASCO
- B. Russelectric
- C. Onan
- D. Kohler
- E. Caterpillar

2.2 AUTOMATIC TRANSFER SWITCH, GENERAL

- A. The automatic transfer switch (ATS) shall be 3-pole as indicated on the drawings and rated as indicated on the drawings. Transfer switch shall be listed and labeled under UL-1008 as a device for use on emergency generator systems. ATS shall be integral to the main service panel as indicated on the drawings. ATS shall be the standard product of a company engaged in manufacturing automatic transfer switches for at least 10 years. ATS shall be manufactured so that no rear or side access is required.

2.3 CONSTRUCTION AND PERFORMANCE

- A. Transfer Switch shall consist of completely enclosed multi- pole contact assembly and a separate control logic panel. The contact assemblies shall be operated by a stored energy mechanism, and be energized only momentarily during transfer providing inherently double

throw switching action. Control power for all transfer operations shall be derived from the line side of the source to which the load is being transferred.

- B. Transfer Switch shall be positively interlocked mechanically and electrically to prevent simultaneous closing of both sources under either automatic or manual operation. A neutral position shall not be possible under normal electrical operation except that the switch shall be provided with a Delayed Transition accessory for switching highly inductive loads. Transfer Switch shall have a manual neutral position for load circuit maintenance. A Transfer Switch position indicator shall be visible from the front of the switch to show to which source the transfer switch is connected.
- C. Transfer switch shall be capable of being operated manually under full load conditions. Manual operation shall be accomplished via integrally mounted pushbutton operators located on the face of the contact assemblies. Removable manual operation handles and handles which will move in the event the electrical operator becomes energized while performing a manual transfer operation are not acceptable. The manual operator shall provide the same contact-to-contact transfer time as provided under normal automatic operation to prevent possible flashovers from switching the main contacts slowly. In addition, provisions shall be provided to allow disengagement of the electrical operator during manual operation.
- D. Transfer Switch shall have four extra sets of normally open and normally closed auxiliary contacts, which indicate ATS positions.
- E. A solid state sensing and control logic panel shall be separately mounted from the power-switching portion of the Transfer Switch. The two Sections shall be connected together by control cables with plug-in connectors. The control Section shall be capable of being disconnected from the power Section for maintenance purposes.
- F. The logic circuit shall utilize differential sensing solid-state components mounted on printed circuit boards to accomplish proper operation and to perform functions such as timing and voltage and frequency monitoring. LED's on each PC card shall indicate the proper operation of each function furnished. Construction shall be such that functions cards are individually replaceable without requiring replacement of the complete solid-state package. Cards for plug-in modifications shall be available for field installation with retention of the UL label.
- G. Where indicated on the drawings, ATS shall be provided with power meter, Square D PM850 or equal, with network card for communications interface with Owner's power management system.

2.4 SEQUENCE OF OPERATION

- A. Upon reduction of phase-to-phase voltage of the normal source to 80% of nominal, and after a time delay of 1-90 seconds (adjustable to meet field conditions) to override momentary dips and/or outages, the auxiliary engine start contacts shall close to initiate starting of the Emergency Generator.
- B. After the Generator has reached 90% of nominal voltage and frequency, and after a time delay, ATS shall transfer the load to the Generator. Provide an under-voltage / under-frequency monitor for the Emergency/Standby Source.
- C. When the Normal Source has been restored to 90% of rated voltage, and after a time delay adjustable from 0.5-30 minutes (to insure the integrity of the Normal Power Source), the load shall be retransferred to the Normal Source.

- D. A time delay module shall be provided in the ATS, adjustable 0.5-30 minutes, to delay shutdown of the Emergency/Standby Power Source after retransfers to allow the generator to run unloaded for cool-down.
- E. If the Emergency Generator should fail while carrying the load, transfer to the Normal Power supply shall be made instantaneously upon restoration of the Normal Source to satisfactory conditions.
- F. ATS shall be provided with a Delayed Transition timer, adjustable 0-120 seconds. The ATS shall pause during transfer with both sources disconnected from the load, to allow back-EMF from large inductive loads to decay. Methods, which use relative phase-angle differences to control transfer, are not acceptable.
- G. ATS shall be provided with 4 sets of Auxiliary Contacts to indicate 'Normal' or 'Emergency' Position of the ATS.

2.5 ADDITIONAL ACCESSORIES/SPARE PARTS

- A. Provide an engine exercise timer in the ATS indicated on the drawings which shall automatically start the engine periodically. Timer shall be digital, with lithium battery back up. Provide a selector switch to select exercise with load transfer or without load transfer. The timer shall include a failsafe circuit, such that if the engine fails during exercise the ATS will immediately retransfer to the Normal source.
- B. Provide pilot lights to indicate to which source the load is connected, and to indicate the availability of each power source.
- C. The transfer switch shall be rated for the short circuit currents noted on the Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Refer to Section 26 05 00 for details of Work under this Section, including seismic installation requirements.
- B. Testing: See Section 26 08 00 and Section 26 32 14.

END OF SECTION

SECTION 26 51 01 - LIGHTING

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Luminaires (i.e., lighting fixtures): Refer to the Luminaire Schedule and provide a complete and working facility Lighting System. Catalog numbers in the Luminaire Schedule are design series references and may not represent the exact catalog number as specified or as required for particular installations. Provide complete luminaires to correspond with the number of LEDs, power supply, wattage, mounting hardware, ceiling type, trim, size, and special requirements as specified in the Luminaire Schedule for each luminaire type. Additional features, accessories, and options specified, described, scheduled, or necessary for installation shall be included.
- B. LEDs and power supplies.
- C. Lighting controls, including occupancy sensors. See Section 26 57 00 for Low Voltage Lighting Control System.
- D. Exit and Emergency Egress lighting where indicated and where required.
- E. Supports for outlet boxes and luminaires, including seismic restraint slack wires for recessed luminaires in suspended ceilings per code and backing in walls as required to keep luminaires secure and level.

1.2 INCORPORATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this section.
- B. Section 26 05 00 and 26 27 00 apply to all work in this section.
- C. Division 03: Concrete (Bases for pole-mounted luminaires as noted in Luminaire Schedule).
- D. Division 09: Painting and Finishes (cutting of holes in finished surfaces for recessed luminaires).

1.3 RELATED WORK

- A. Ceiling Access panels where required for access to equipment, outlets, transformers, etc., located above suspended ceilings, sheet rock or plaster ceilings. Coordinate with the Architect and other trades.

1.4 SUBMITTALS

- A. Submit under provisions of Division 1 and Section 26 05 00.
- B. Submit (6) six sets of submittals for review by the project team unless otherwise noted in these specifications. The submittals shall include the following information:
 - 1. Product Index: The following information shall be included in the product index.
 - a. Luminaire Type. The index shall list, in alphabetical order, each luminaire type per the Luminaire Schedule.

- b. Manufacturer's Catalog Number. Outstanding information required to make a complete catalog number shall be clearly identified in the index.
 - c. Where a pole is included with the luminaire, include the catalog number of the pole in addition to that of the luminaire.
 - d. LED Data. Provide the Manufacturer's name for each LED array including wattage, color temperature, lumen output, and color rendering index.
- 2. Manufacturer's literature for every luminaire listed on the Luminaire Schedule.
 - a. Catalog Information:
 - 1) Luminaire Data Sheet: The manufacturer's cut sheet shall include the following:
 - (a) Photometrics: Candlepower distribution curve or table with horizontal readings at 0, 22.5, 45, and 90 degrees and vertical readings from 0 to 180 degrees in 5 degree increments in accordance with the Illuminating Engineering Society published test procedures.
 - (b) Catalog Number Nomenclature
 - (c) Coefficient of Utilization Tables
 - (d) Luminaire Line Drawing
 - (e) Power supply (each type)
- 3. Data sheets for electronic ballasts and power supplies. Indicate luminaire types on applicable ballast/power supply data sheets.
- 4. Data sheets for wallbox controls and other products specified in this section.
- 5. Shop Drawings:
 - a. Provide shop drawings of suspension details for luminaires recessed in, mounted on, or suspended from hung ceilings. Details shall clearly illustrate proposed methods for supporting luminaires independent of the suspended ceiling system.
 - b. Detailed shop drawings of all cove or box mounted luminaires containing the following information:
 - 1) Exact field measured length (clear inside dimension) of cove pocket or box.
 - 2) Exact luminaire length and arrangement of luminaires in cove or box.
 - c. Detailed shop drawings of pendant mounted luminaires constructed with linear metal housings containing the following information.
 - 1) Support mechanism, including swivel canopies.
 - 2) Trim details.
 - 3) Closure piece details.
 - 4) Pattern configurations.
- 6. Samples:
 - a. Provide samples of luminaire trim where "Finish as selected by Architect" is indicated on the Luminaire Schedule. Submit two finish samples, 75 mm x 75 mm (3" x 3") minimum, of all custom color, decorative metal, or anodized aluminum finishes. Samples must be approved in writing by the Architect prior to ordering.

C. For Any Luminaires Substituted For Those Specified:

- 1. Refer to Division 1 - Product Requirements, for all substitution procedures.
- 2. Provide Independent Testing Laboratories, Inc., or equal, photometric test report for each Luminaire type and lamp combination listed on the Luminaire Schedule. Test reports shall be based on Illuminating Engineering Society published test procedures and shall contain polar coordinate candlepower distribution curves in five lateral planes for luminaires with asymmetric distributions and luminaire luminance data for vertical angles above 45 degrees from nadir. Test results shall indicate luminaire efficiency for the lamp and aperture assembly specified. Luminaires with efficiencies more than 2% below the values of specified luminaires are not acceptable and will be rejected.
- 3. Provide photometric calculations for each room or area where a substituted luminaire is proposed. Such calculations shall be made using comprehensive lighting software, such as AGI32, and include point-by-point illuminance values at IES recommended heights, average illuminance, and maximum-to-minimum and average-to-minimum uniformity ratios. Room dimensions, configurations (including sloping ceilings), room surface

reflectances, light loss factors, and heights of suspended luminaires shall match the heights specified in the contract documents.

4. Due to the variety of lumen outputs and light distributions of LED Luminaires, substitutions will require additional review on the part of the Engineer or Architect to ascertain the equivalency of the substituted luminaires. Substitutions will be reviewed to determine their aesthetic, construction, and photometric equivalency to maintain similar design impact and performance in their intended environment. The Engineer and Architect have not included such unknown and unquantifiable review time in their scope of work and are not compensated by the Owner for such services. The Contractor shall reimburse the Engineer and Architect for labor costs to review substitutions.
5. Prior approval does not guarantee final approval by the Engineer. The Contractor shall be responsible for providing luminaires that meet or exceed the quality and performance of the specified products in their entirety. All deviations in quality and performance from the specified products must be listed and individually signed off by the engineer.
6. The Owner reserves the right to reject a proposed substitution based on their agent's professional judgment as to the utility, quality, performance, visual appropriateness, or finish of substitutions.

1.5 OCCUPANCY SENSORS

- A. Equipment Qualification
 1. Wall switch products must be capable of withstanding the effects of inrush current. Submittals shall clearly indicate the method used.
 2. Contractor's work to include all labor, materials, tools, appliances, control hardware, sensor, wire, junction boxes and equipment necessary for and incidental to the delivery, installation and furnishing of a completely operational occupancy sensor lighting control system, as described herein.
 3. Contractor and Contractor's Supplier shall examine all general specification provisions and drawings for related electrical work required as work under Division 26.
 4. Contractor shall coordinate all work described in this section with all other applicable plans and specifications, including but not limited to wiring, conduit, luminaires, HVAC systems and building management systems.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to site and store in unopened cartons in protected location. Inspect products immediately and report all damage accordingly.

1.7 GUARANTEE AND WARRANTIES

- A. All work performed under this section must be guaranteed to be free of defects in products or workmanship for one year after date of acceptance by Owner, unless noted otherwise in General Conditions.
- B. Warranties:
 1. Electronic power supplies must be warranted against failure for at least five years after date of substantial completion.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Provide luminaires as indicated in Luminaire Schedule; if conflict exists between Luminaire Schedule and Specifications, the more stringent requirement shall take precedence.
- B. Provide luminaires new and complete with mounting accessories, junction boxes, trims, and lamps.
- C. Provide products with UL labels appropriate to intended installation conditions, or with labels from other testing laboratories whose results are acceptable to local inspector, showing compliance with UL standards. Labels must be concealed from normal viewing angles.
- D. All products of same type by same manufacturer.

2.2 SOLID STATE LUMINAIRES

- A. Housing, where applicable:
 - 1. Steel bonderized or equal rust protected, or aluminum, rigid construction. Minimum gauge thickness shall be as follows:
 - a. Interior locations: No. 20-gauge steel, No. 16-gauge aluminum.
- B. Finish:
 - 1. Baked enamel finish (except when otherwise specified).
 - a. Concealed interior surfaces (this applies to interior hardware, circuit boards, etc.) matte black.
 - b. Concealed exterior surfaces: matte black.
 - c. Visible surfaces: color and texture as specified below for each luminaire type or as selected.
- C. Light Emitting Diode (LED) requirements:
 - 1. Correlated color temperature (CCT) for phosphor-coated white LEDs must have one of the following designated CCTs, as specified on the Luminaire Schedule, and fall within the following binning standards.
 - a. 2700K defined as 2725 +/- 145K
 - b. 3000K defined as 3045 +/- 175K
 - c. 3500K defined as 3465 +/- 245K
 - 2. Color spatial uniformity shall be limited to variations in chromaticity for different directions (i.e. changes in viewing angle) within 0.004 from the weighted average point on the CIE 1976 (u',v') diagram.
 - 3. Color maintenance shall be limited to a maximum change in chromaticity of 0.007 on the CIE 1976 (u',v') diagram over the lifetime of the product.
 - a. Color rendering index: Color rendering index to be determined using ANSI C78.377-2008 and applicable IESNA standards.
 - b. Laboratory tests must be produced using specific module(s)/array(s) and power supply combination that will be used in production.
 - c. Manufacturers must provide a test report from a laboratory accredited by NVLAP or one of its MRA signatories
 - 4. Lumen depreciation
 - a. Lumen depreciation to be measured using IESNA LM-80-08 and TM-21-11 standard for IES approved method of measuring lumen maintenance of LED light sources.

- b. Phosphor-coated white LED modules/arrays shall deliver at least 70% of initial lumens for a minimum of 50,000 hours when installed in-situ and operated at 100% output and the maximum specified operating temperature.
 - c. Colored LED modules/arrays shall deliver at least 50% of initial lumens for a minimum of 50,000 hours when installed in-situ and operated at 100% output and the maximum specified operating temperature.
 - 5. Acceptable LED manufacturers:
 - a. Cree
 - b. Nichia
 - c. Osram Opto Semiconductors
 - d. Philips Lumileds
 - e. Xicato
- D. Luminaire Efficacy:
 - 1. Luminaire efficiency shall be measured using IESNA LM-79-08 standard for electrical and photometric measurements of solid state lighting products.
 - 2. Manufacturer shall provide published luminaire efficacy, which is defined as luminaire light output divided by luminaire input power measured in a 25 degree Celsius environment. Efficacy shall include power supply, thermal, optical, and luminaire losses.
- E. Thermal Management:
 - 1. Solid state luminaire shall not exceed LED manufacturer's maximum junction temperature requirements when operated in-situ at luminaire manufacturer's maximum ambient operating temperature and 100% light output.
 - 2. Solid state luminaires shall be thermally protected using one or more of the following thermal management techniques:
 - a. Metal core board
 - b. Gap pad
 - c. Internal monitoring firmware
 - 3. Solid state luminaire housing shall be designed to transfer heat from the LED board to the outside environment.
- F. Power Supplies (LED Drivers) requirements:
 - 1. Power factor of 0.90 or greater for primary application
 - 2. Input current shall have Total Harmonic Distortion (THD) of less than 20%.
 - 3. Minimum operating temperature of minus 20 degrees Celsius or below when used in luminaires intended for outdoor applications.
 - 4. Operating frequency equal to or greater than 120 Hz.
 - 5. Operate with sustained input variations of +/- 10% (voltage and frequency) with no damage to the driver.
 - 6. Tolerate sustained open circuit and short circuit output conditions without damage and without need for external fuses or trip devices.
 - 7. Output shall be regulated to +/- 5% across published load range.
 - 8. Class A sound rating.
 - 9. Outputs shall have current limiting protection.
 - 10. Operate LEDs at constant and regulated current levels. LEDs shall not be overdriven beyond the diode manufacturer's specified nominal voltage and current.
 - 11. Inrush currents not exceeding peak currents specified in NEMA 410.
- G. Solid State Lighting Controls:
 - 1. Control interface to dimmable power supplies shall consist of one of the following:
 - a. Line Voltage Dimming. Controls to be rated for magnetic or electronic low voltage transformer operation.

- b. Low voltage (0-10V) control. Controls to be compatible with either current sink or current source operation.
 - 2. Dimmable LED power supplies shall use pulse width modulation (PWM) or constant current reduction (CCR) to regulate power to LEDs.
 - a. PWM power supplies shall have 12-bit or greater resolution to obtain flicker-free operation throughout their dimming range.
 - b. PWM power supplies shall be provided in luminaires that will be dimmed lower than 40% and must maintain consistent color temperature.
 - c. CCR power supplies shall be provided in areas that have strict electromagnetic interference (EMI) requirements, high motion activity, or rotating machinery.
- H. System Installation
 - 1. Hardwired connections to solid state luminaires shall be reverse polarity protected and provide high voltage protection in the event connections are reversed or shorted during the installation process.
 - 2. All solid state luminaires (100% of each lot) shall undergo a minimum eight-hour burn-in test during manufacturing. Solid state lighting installations shall be UL Listed as a low-voltage lighting system including, but not limited to, luminaire, power supply, controller, keypad, and wiring.
- I. Warranty
 - 1. Luminaires, drivers, and controllers for solid state lighting systems shall be covered by a five-year warranty against defects in workmanship or material. Warranty shall include in-warranty service program providing for payment of authorized labor charges incurred in replacement of inoperative in-warranty equipment.

2.3 LUMINAIRE CONSTRUCTION

- A. Sheet metal: materials and thicknesses shall be 20 gauge (0.7 mm or 0.027") min., free of dents, scratches, oil-can, or other defects.
- B. Painted luminaires: exposed weld marks, joints, and seams shall be filled and sanded smooth before finishing.
- C. All edges cleaned and dressed to remove sharp edges or burrs.
- D. Extrusions: 1/10" min. wall thickness, smooth and free of tooling lines, with cast end plates that exactly match extrusion profiles.
- E. Castings: smooth, free of pits, scales, gate marks, or blemishes.
- F. Spinings shall have 1/32" min. thickness, smooth, free of spinning lines or blow-back, with clean edges.
- G. Welds: Follow recommendations of American Welding Society. All welds continuous and free of spatter, residue, or warping.
- H. No light leaks visible in finished room. Ensure that downlight housings mounted in wood slat ceilings are not visible from below. Field paint exterior of housing with high temperature paint if necessary.
- I. Exposed end plates and joiners, with concealed fasteners.

- J. End-to-end mounted luminaires: Verify row configurations and provide joiners, aligning splines, and trims to suit.
- K. Hardware:
 - 1. Steel or aluminum interior luminaires: cadmium-plated hardware.
 - 2. Steel or aluminum exterior luminaires: stainless steel hardware.
 - 3. Stainless steel luminaires: stainless steel hardware.
 - 4. Copper alloy luminaires: brass hardware.
- L. Raceways: Where used for through wiring, luminaires must be approved for use as raceways.

2.4 RECESSED LUMINAIRES

- A. Point-source luminaires: provide pre-wired junction box and thermal protection, and provide slack wires to structure at two diagonal corners.
- B. Troffer luminaires: provide hold-down clip at each luminaire corner, and slack wires to structure as detailed on the drawings. The detail will take precedence.
- C. Verify ceiling construction details and provide luminaire housings and trims to suit.
- D. Non-accessible ceilings: Provide access to junction boxes, ballasts, transformers, and battery packs through luminaire apertures; no access panels in ceiling.
- E. Mounting frames: To prevent rusting, provide galvanized steel or cast aluminum frames for installation in damp locations or in plaster ceilings.
- F. Adjustable luminaires shall be provided with rotation and tilt locking devices.

2.5 PENDANTS

- A. Stem-mounted: 16 mm (5/8") max. o.d. stem with ball swivels at top (and bottom of linear luminaires) to permit 45 degree swing in any direction from vertical. Flat canopy to permit splice inspection after installation. Pendants must permit +/- 13 mm (1/2") threaded vertical adjustment after installation, leaving at least 6 mm (1/4") thread contact at all points.
- B. Provide internal safety cable from luminaire body to stud in outlet box.
- C. Cable-mounted: 1 X 7 strand 3/32" diameter stainless steel aircraft cable, factory crimped, independently tested and verified to exceed 1500 pounds.
 - 1. Verify mounting heights for each luminaire and provide cable lengths and coordinate cord lengths with manufacturer as required prior to ordering luminaires. Provide aircraft cable adjuster nipple with locking jaws.
- D. Supports: Carry luminaire weight to structure and provide horizontal bracing from suspension points to ceiling framing to prevent sideways shifting. Provide diagonal seismic restraint wires per code.

2.6 TRIMS

- A. Trims must fit tightly and be held in by gravity, spring clips, or mechanical fasteners. Trims must not drop out under normal conditions or seismic forces which do not exceed the design criteria of the building.
- B. Aluminum parabolic cones shall be smooth, properly shaped, with Alzak finish in colors as indicated.
 - 1. No hot spots or lamp images visible at angles shallower than lamp shielding angle.
 - 2. Self-flange cones must bend parallel to ceiling and cover ceiling hole without additional trim ring. Unpainted flange, shall have the same finish as cone interior.
 - 3. Cones and louvers for fluorescent luminaires must have permanent anti-iridescence treatment.
- C. Lenses, diffusers, and patterned glass: glass or virgin acrylic as noted, with patterns as noted.
 - 1. Finished thickness 2 mm (1/10") min. unless noted otherwise.
 - 2. Linear runs over 1200 mm (4'-0") long shall be in equal-length pieces.

2.7 FINISHES

- A. Steel Reflectors: Unless otherwise specified, the reflector surface finish shall be of synthetic white enamel or polyester powder coating. Finish shall show no indication of chipping, cracking, flaking or any other sign of loss of adhesion. The initial reflection factor shall be not less than 88 percent averaging 5 randomly selected points on the reflector. After 100 hours of exposure to the radiation of a glass enclosed carbon arc lamp, such as a Fade-O-Meters, the reflectance of the exposed portion shall not be less than 5 percent and finish shall show no appreciable color change. The carbon arc lamp shall be operated at appreciable color change. The carbon arc lamp shall be operated at 13 plus or minus 0.5 amperes at 140 volts. The reflector shall be placed ten inches from the arc and the lamp so ventilated that the temperature of the exposed portion does not exceed 105 degrees F.
- B. Aluminum Reflectors: Reflecting surfaces shall be provided with either a specular or diffuse finish as indicated. Reflection factors shall be not less than 83 percent for specular reflecting surfaces. Each reflecting surface shall be protected by dense coating of oxide weighing not less than 5.0 milligrams per square inch, applied by an anodic process. The reflector shall be given a sealing treatment that will prevent staining of the reflecting surface when subjected to a stain test. All aluminum reflectors & louvers shall be a low iridescent equivalent to that provided by Coil Anodizers.
- C. Non-Reflecting Surfaces: Unless otherwise specified, the finish on all non-reflecting exterior surfaces shall be aluminum oxide or aluminum; white, gray or aluminum paint on steel; nickel or chromium plating on copper alloy. Fastening devices shall be nickel, chromium, cadmium or zinc plated. All painted surfaces shall be free of tears, star marks, blisters, pinholes, chipping and any other defects that may impair appearance or serviceability.

2.8 LAMPS

- A. Relamp luminaires or replace LED boards and power supplies at no cost to owner if lamps or LEDs exhibit color variation, flicker, or burn out within 90 days of substantial completion date.
- B. LEDs:
 - 1. LED quantity, wattage, and color temperature as specified for each LED luminaire.

2. 3500 deg. K color temperature for interior luminaires, 3000 deg. K for exterior luminaires, unless otherwise specified.

2.9 DRIVERS AND TRANSFORMERS

- A. General:
 1. Verify input voltages and match to branch circuit voltages.
 2. Remote drivers or transformers: Provide suitable enclosures and mounting hardware, and install in accessible, ventilated locations.
 - a. Secondary wiring: provide number and size of conductors as required, with 3% max. voltage drop between transformer and last lamp.
 - b. Keep transformers at least 300 mm (12") apart and do not stack vertically.
- B. LED Drivers:
 1. High power factor, thermally-protected.
 2. Compatible with LED lamps being used.
 3. Capable of dimming LED source without perceptible flicker or stroboscopic effects.

2.10 EMERGENCY LIGHTING AND EXIT SIGNS

- A. Emergency lighting:
 1. Provide lighting for paths of egress as required by Code.
- B. Emergency lighting to consist of wall or ceiling mounted (normally off) lamp head units with battery back-up for 90 minutes of egress lighting output.
- C. Exit signs shall be edge or back lit LED, surface-mounted on ceiling or wall, with integral battery packs as described above.
 1. Fabricated aluminum construction, no light leaks around canopy. Plain box, with no decorative trim.
 2. Letters shall be 20mm (3/4") stroke, 150 mm (6") high, with concealed knockouts for left or right arrows, brightness and evenness of illumination per code, green color.
 - a. Battery pack contained in basic luminaire housing. No add-on packs or canopies.
 - b. Green LED lamps located at interior perimeter for indirect illumination of stencil letters.
 - c. Provide finish as specified in the Luminaire Schedule.
 - d. Knock out the arrows as indicated on the plans.
- D. Emergency luminaires supplied by a separate emergency power source.
 1. For luminaires supplied by a separate emergency power source, provide "switched" control of the emergency designated lamps to allow complete "off" control when required by the user. The switched control shall include an automatic transfer feature to automatically turn "on" the emergency designated lamps upon the normal source power failure.
 2. Automatic control function shall be provided using a UL 924 listed relay, LVS Inc. #EPC-A or equal, suitable for mounting in a standard 4" square j-box (min. 2.5" deep). Control relay shall provide automatic diagnostic test feature which shall maintain power to the emergency designated lamps for 15 seconds after the room is switched off via the respective light switch or control relay. Emergency designated lamps shall turn off after the 15 second test period and shall come back on when the control device is turned back on to restore full lighting to the space.
 3. Provide (1) control module per "switched" zone.

2.11 WALL-BOX DIMMERS

- A. Provide dimmer controls as specified on the drawings and in Specifications Section 26 57 00.
- B. Ganging and Labeling:
 - 1. Dimmers and matching switches in same location shall be installed in same gang box.
 - 2. Follow dimmer manufacturer's instructions for gang-box sizes. Do not break off fins on dimmers unless noted otherwise.
 - a. 1+1, 4+1, 7+1 installation: to gang an even number of small devices without breaking off fins, provide multi-gang box as indicated (1, 4, or 7 gangs) and provide additional single-gang box at end, with ears of single box 70 mm (2-3/4") o.c. from last set of ears on multi-gang box.
 - b. When rows of devices are stacked vertically space rows 230 mm (9") o.c. to allow heat dissipation.
 - 3. Provide Lutron multi-gang plates to cover each group of devices.
 - 4. Plates with "-NFB" in catalog number with no fins broken.
 - 5. Plates with "-FB" in catalog number: break off all interior fins on devices, but do not break off outside fins at either end of row.
 - 6. Nova-T: install aligning backplate (provided by Lutron with each multi-gang plate) between wall and devices.
 - 7. Labels: text as indicated 3 mm (1/8") high, all capital letters, engraved on device faceplate, filled with black paint and wiped clean.

2.12 OCCUPANCY SENSORS

- A. General
 - 1. Wall switch sensors shall be capable of detection of occupancy at desktop level up to 300 square feet, and gross motion up to 1000 square feet.
 - 2. Wall switch sensors shall accommodate loads from 0 to 800 watts at 120 volts; 0 to 1200 watts at 277 volts and shall have 180° coverage capability.
 - 3. Wall switch products shall utilize Zero Crossing Circuitry which increases relay life, protects from the effects of inrush current, and increases sensor's longevity.
 - 4. Wall switch sensors shall have no leakage current to load, in manual or in Auto/Off mode for safety purposes and shall have voltage drop protection.
 - 5. Where specified, wall switch sensors shall provide a field selectable option to convert sensor operation from automatic-ON to manual-ON.
 - 6. Where specified, vandal resistant wall switch sensors shall utilize a hard lens with a minimum 1.0mm thickness. Products utilizing a soft lens will not be considered.
 - 7. Passive infrared sensors shall utilize Pulse Count Processing and Digital Signature Analysis to respond only to those signals caused by human motion.
 - 8. Passive infrared sensors shall utilize mixed signal ASIC which provides high immunity to false triggering from RFI (hand-held radios) and EMI (electrical noise on the line), superior performance, and greater reliability.
 - 9. Passive infrared sensors shall have a multiple segmented Lodif Fresnel lens, in a multiple-tier configuration, with grooves-in to eliminate dust and residue build-up.
 - 10. Where specified, passive infrared and dual technology sensors shall offer daylighting foot-candle adjustment control and be able to accommodate dual level lighting.
 - 11. Dual technology sensors shall be corner mounted to avoid detection outside the controlled area when doors are left open.
 - 12. Dual technology sensors shall consist of passive infrared and ultrasonic technologies for occupancy detection. Products that react to noise or ambient sound shall not be considered.

13. Ultrasonic sensors shall utilize Advanced Signal Processing to adjust the detection threshold dynamically to compensate for constantly changing levels of activity and air flow throughout controlled space.
14. Ultrasonic operating frequency shall be crystal controlled at 25 kHz within $\pm 0.005\%$ tolerance, 32 kHz within $\pm 0.002\%$ tolerance, or 40 kHz $\pm 0.002\%$ tolerance to assure reliable performance and eliminate sensor cross-talk. Sensors using multiple frequencies are not acceptable.
15. All sensors shall be capable of operating normally with electronic ballasts, PL lamp systems and rated motor loads.
16. Coverage of sensors shall remain constant after sensitivity control has been set. No automatic reduction shall occur in coverage due to the cycling of air conditioner or heating fans.
17. All sensors shall have readily accessible, user adjustable settings for time delay and sensitivity. Settings shall be located on the sensor (not the control unit) and shall be recessed to limit tampering.
18. In the event of failure, a bypass manual override shall be provided on each sensor. When bypass is utilized, lighting shall remain on constantly or control shall divert to a wall switch until sensor is replaced. This control shall be recessed to prevent tampering.
19. All sensors shall provide an LED as a visual means of indication at all times to verify that motion is being detected during both testing and normal operation.
20. Where specified, sensor shall have an internal additional isolated relay with Normally Open, Normally Closed and Common outputs for use with HVAC control, Data Logging and other control options. Sensors utilizing separate components or specially modified units to achieve this function are not acceptable.
21. All sensors shall have UL rated, 94V-0 plastic enclosures.

B. Circuit Control Hardware - CU

1. Control Units - For ease of mounting, installation and future service, control unit(s) shall be able to externally mount through a 1/2" knock-out on a standard electrical enclosure and be an integrated, self-contained unit consisting internally of an isolated load switching control relay and a transformer to provide low-voltage power. Control unit shall provide power to a minimum of two (2) sensors.
2. Relay Contacts shall have ratings of:
 - a. 13A - 120 VAC Tungsten
 - b. 20A - 120 VAC Ballast
 - c. 20A - 277 VAC Ballast
3. Control wiring between sensors and controls units shall be Class II, 18-24 AWG, stranded U.L. Classified, PVC insulated or TEFLON jacketed cable suitable for use in plenums, where applicable.
4. Minimum acceptable wire gauge from the circuit control hardware relays shall be #14 AWG.

C. Acceptable Manufacturers

1. The Watt Stopper, or Pre-Approved Equal: For pre-approval, provide all the information listed under "submittals" a minimum of ten (10) working days prior to initial bid date.
2. The listing of any manufacturer as "acceptable" does not imply automatic approval. It is the sole responsibility of the Contractor to ensure that any price quotations received and submittals made are for sensors that meet or exceed the specifications and the requirements of the contract documents.
3. Products
 - a. All products shall be Watt Stopper product numbers:
 - 1) Ceiling sensors: WT-605, WT-600, WT-1105, WT-1100, WT-2205, WT-2200, WT-2250, WT-2255, W-500A, W-1000A, W-2000A, W-2000H, WPIR, DT-200, DT-205, CX-100, CX-105, CI-200, CI-205

- 2) Wall sensors: WI-200, WS-120/277, WA-100, WD-170, WD-180, WD-270, WD-280
- 3) Power and Slave Packs: B120E-P, B277E-P, C120E-P, C277E-P, S120/277-P, AT-120, AT-277

PART 3 - EXECUTION

3.1 PREPARATION

- A. Architectural Reflected Ceiling Plans and Elevations shall govern exact location and mounting conditions for all luminaires. Contractor shall coordinate luminaire mounting and compatibility with ceiling construction and other trades.
- B. Coordinate work with other trades. Location of lighting has priority over location of new framing (except major structural members), ducts, diffusers, sprinklers, speakers, smoke detectors, and other obstructions.
- C. If obstructions are encountered which prevent installation of luminaires according to drawings, notify Architect immediately and do not proceed until conflict has been resolved.
- D. Coordinate the location of luminaires in mechanical or unfinished spaces. Locations shown on Drawings may be adjusted by the Contractor to suit conditions. Install luminaires to avoid obstructions and maximize light output, 2100 mm (7'-0") min. mounting height.
- E. In Elevator Machine Rooms, locate the luminaires so that the illumination level at the floor is not less than 200 lx (19 fc). Illuminate areas in front of and behind (if accessible) controllers, machines and other elevator equipment.
- F. In Elevator Pits, locate the luminaires so that the illumination level at the pit floor is not less than 100 lx (10 fc).
- G. Coordinate the location of any exposed conduit used to feed luminaires with the Architect prior to installation.

3.2 INSTALLATION

- A. General:
 1. Contractor shall be responsible for handling and installation of luminaires including all supports, hangers and hardware necessary for a complete installation. Luminaires shall be clean, plumb, level in straight lines, without distortion. Luminaires must be installed so they do not shift during relamping or adjustment. Remedy any light leaks which may develop after installation of recessed or enclosed luminaires.
 2. Install luminaires at locations and heights as indicated, in accordance with luminaire manufacturer's written instructions, applicable requirements of NEC, NECA's "Standard of Installation", NEMA standards, and with recognized industry practices to ensure that luminaires fulfill requirements.
 3. Point-source luminaires shall be located as dimensioned, or in center of tile or on tile joint as drawn; 6 mm (1/4") max. off-center tolerance.
 4. Linear luminaires shall have 3 mm (1/8") max. horizontal or vertical alignment variation in any 5 m (16-ft.) portion of run.
 5. Tighten connectors and terminals, including screws and bolts, in accordance with equipment manufacturer's published torque tightening values for equipment connectors. Where manufacturer's torquing requirements are not indicated, tighten connectors and

terminals to comply with tightening torques specified in UL Std. 486 A and B, and the National Electrical Code.

6. Clean luminaires of dirt and construction debris upon completion of installation. Clean fingerprints and smudges from lenses.
7. Remove and replace luminaires that may have been damaged during construction at no additional cost to the Owner.
8. Protect installed luminaires from damage during remainder of construction period.
9. Provide equipment grounding connections for luminaires as indicated. Tighten connections to comply with tightening torques specified in UL 486 A to assure permanent and effective grounds.
10. Install luminaires, lamps, lenses, etc., after building is enclosed, weather tight and environmental conditions are nominally the same as expected for the complete spaces. All lenses, glass, reflectors, and refractors shall be clean and free of chips, cracks, and scratches.
11. All wall mounted luminaires and all ceiling mounted surface luminaires including exit lights shall be fed through a luminaire Stud/Hickey/Nipple assembly and with provisions to prevent luminaire turning.
12. Installation of exit signs shall be coordinated with other trades to ensure signs are visible as intended.
13. All junction box cover plates for the lighting branch circuit system shall be clearly marked with a permanent ink felt pen identifying the branch circuit and control relay (panel number, circuit number, lighting control cabinet designation and control relay number) contained in the box.
14. Provide permanently affixed adhesive labels with machine printed lettering (min. 1/8" high) at junction boxes serving luminaires that are supplied by (2) electrical sources (i.e. normal and emergency lighting). Label to read "CAUTION - This luminaire is powered by (2) separate sources. The normal power source breaker and the emergency power source breaker must be turned off before servicing this luminaire."

B. Recessed Luminaires:

1. The contractor shall verify the fire rating of the ceiling system and wall in which the luminaires are to be mounted. Where luminaires are installed in fire rated ceilings or walls, provide fire rated enclosures around and over luminaires to maintain ceiling fire rating. No additional cost shall be allowed for failure to include such enclosures and installation in the bid.
2. Holes for Recessed Point-Source Luminaires: Cut holes to follow luminaire housings exactly so no gaps will be visible after trims are installed.
3. Install bottom of housing aligned with finished ceiling.
4. Keep ceiling insulation at least 75 mm (3") away from luminaires. Exception: luminaires with insulated contact (IC) rating shall be permitted to be in contact with insulation.
5. Install trims after painting of spaces. Install trims tightly, with no gaps or light leaks.
6. Seismic restraints: Provide and install slack wires and hold-down clips per code.
7. Wallwashers:
 - a. Orient wallwasher housings according to manufacturer's instructions to maximize brightness on the upper portion of the wall.

C. Ceiling-Mounted and Pendant Luminaires:

1. Provide support for outlet boxes and suspension points so luminaires can be installed securely, including seismic supports per code.
 - a. Luminaire weight less than 23 kg (50 lb) at each suspension point: hang from strap or stud on outlet box, or at non-feed points, provide 1/4"-20 stud projecting 20 mm (3/4") below ceiling.
 - b. Luminaire weight 23 kg (50 lb) or more at each suspension point: hang directly from structure, either independent of outlet box or from stud extending through outlet box to structure, unless the outlet box is listed for not less than the weight to be supported.

Boxes used as the sole support of luminaires weighing more than 50 pounds must be listed and marked by the manufacturer with the maximum weight.

2. Pendants:
 - a. Provide horizontal bracing from suspension points to ceiling framing to prevent sideways shifting.
 - b. Provide diagonal seismic restraint wires above ceiling per code.
 - c. Furnish suspended luminaires with universal joint type hanger canopy (and longitudinal sway adapter at each stem connection point for linear luminaires), to permit 45 degree swivel on 360 degree circle from Nadir at canopy (and 45 degree longitudinal movement at sway adapter).
 - d. Luminaires over 450 mm (18") wide shall be provided with supports at all corners.
 - e. Install pendants plumb and level.
 - f. Verify luminaire weights and provide backing in ceiling as required.

D. Wall-Mounted Luminaires:

1. Mounting heights shown on Drawings are measured from finished floor to centerline of outlet box or recessed housing, unless otherwise noted.
2. Verify luminaire weights and provide backing in wall as required. Luminaires must not droop or tilt away from wall.
3. Wet locations: install sealant between luminaire and outlet box.
4. In circulation areas, wall-mounted luminaires must not project more than 100 mm (4") from wall if mounted above 685 mm (27") and below 2030 mm (80").

3.3 LIGHTING CONTROLS

- A. Install controls so that all operable parts are at 48 inches (1220 mm) maximum height.
- B. Lighting controls to include occupancy sensors in most spaces (for local control) and relay system lighting control for larger common spaces as indicated on the drawings.
- C. Occupancy sensors shall initially be set as follows:
 1. Maximum sensitivity.
 2. Maximum time delay (or 30 minutes).
 3. Manual-on operation.
 4. Automatic off operation.
 5. Aim all adjustable sensors to properly cover room areas.
- D. Timeclock System shall initially be set to control the low voltage relays as per the Relay Panel Schedule LCP.
 1. Assign all interior relays to an automatic off sweep, with flick warn (except those noted as "NL"). Off time shall be set to an Owner-determined time in the evening, after dark or normal business operations.
 2. Off signals to may originate from BAS system, which shall be inter-connected to the Lighting Control System where indicated on the drawings.
 3. Assign all interior relays noted as "NL" to be on 24 hours per day. No automatic relay operation.
 4. Assign "after hours" and "Weekend / Holiday" hours to match normal business calendar and times.
 5. All interior relays shall be allowed to be overridden by use of the local dataline switches for a maximum of 2 hours (per Title 24) when used after hours or on Weekends / Holidays. If used during these times, automatic shut-off shall re-activate at the end of the 2-hour period.

6. All interior relays shall be allowed to be overridden by use of the local dataline switches when used during normal business hours. Standard timeclock operation shall resume with the next scheduled timeclock function for each relay.
7. Assign all exterior relays for automatic on operation with the astro-dial feature, set to 30 minutes before sunset. Latitude = 37.5 degrees North / Longitude = 122 degrees West.
8. Assign exterior relays noted as "astro-on, astro-off" for automatic off operation with the astro-dial feature, set to 30 minutes after sunrise. Latitude and Longitude as noted above.
9. Assign exterior relays noted as "astro-on, timeclock-off" for automatic off operation with the normal timeclock feature, set to an owner determined time in the late evening.
10. Assign exterior relays noted as "NL" or "On All Night" for astro-dial operation, for automatic on 30 minutes before sunset and automatic off 30 minutes after sunrise.

3.4 DELIVERY, STORAGE, & HANDLING:

- A. Deliver luminaires in factory-fabricated containers or wrappings, which properly protect luminaires from damage. Inspect luminaires immediately upon delivery to ensure correct shipment without damage.
- B. Store luminaires in original packaging. Store inside well-ventilated area protected from weather, moisture, soiling, extreme temperatures, humidity, laid flat and blocked off ground.
- C. Handle luminaires carefully to prevent damage, breaking, and scoring of finishes. Do not install damaged units or components; replace with new. Protection wrapping on louvered (parabolic) luminaires shall not be removed until luminaires are ready for operation.

3.5 SEQUENCING AND SCHEDULING:

- A. General:
 1. Coordinate with other work including wires/cables, electrical boxes and fittings, and raceways, to properly interface installation of luminaires with other work.
 2. Sequence lighting installation with other work to minimize possibility of damage and soiling during remainder of construction.

3.6 PROJECT CLOSEOUT

- A. Clean luminaires and remove plaster and paint spatters.
- B. Clean fingerprints and dust from downlight reflectors. Refer to manufacturer's instructions.
- C. Verify that luminaires and controls are working at time of final acceptance by Owner.
 1. Repair or replace lighting control devices that are inoperable.
 2. Repair or replace LED modules or entire LED luminaires that are inoperable.
 3. Repairs and/or replacements shall be at no additional cost to the Owner.
- D. Test emergency lighting system for 90 minutes in presence of Owner's representative, check each luminaire for proper operation at end of 90-minute test, then recharge for 24 hours and briefly test each luminaire again for proper operation.
- E. Install and aim adjustable lighting as directed by Architect.
 1. Provide personnel, lifts, ladders, and walkie-talkies as required.
 2. Aiming will occur at night, outside of normal working hours, at times as approved by the Architect.

- F. Prepare two copies of a Lighting Systems Maintenance Manual consisting of the following in a hardcover binder. Deliver to Architect. After review, Architect will deliver one copy to Owner.
1. One complete set of approved submittals, including product data and shop drawings.
 2. Luminaire cleaning instructions, including chemicals to be used or avoided.
 3. Instructions for code-required testing and maintenance of emergency lighting system.

END OF SECTION

SECTION 26 56 01 - SITE LIGHTING

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Luminaires
- B. LEDs
- C. Power Supplies/Drivers
- D. Poles
- E. Pole bases
- F. Controls and wiring

1.2 SYSTEM DESCRIPTION

- A. Furnish all labor, materials, apparatus, tools, equipment transportation, temporary construction and special or occasional services as indicated on the Drawings or described in these Specifications and as required to make a complete working site lighting system.
- B. Illumination levels shall be in accordance with recommendations by the Illuminating Engineering Society (IES).

1.3 INCORPORATED DOCUMENTS

- A. Section 26 05 00 and Section 26 51 01 apply to all work in this Section.

1.4 SUBMITTALS

- A. Catalog Information:
 - 1. Luminaire (each type) with photometric pattern.
 - 2. Contactors.
 - 3. Ballast or Driver (each type)
 - 4. Poles.
 - 5. Brackets.
- B. Shop Drawings.
- C. Manufacturer's Recommendations: Provide two copies before material is used.
 - 1. PVC conduit joints and junctions.
 - 2. Solvent welding directions.
 - 3. Pole bases.
- D. Laboratory Test: Determine soil density relationships for compaction of backfill material in accordance with ASTM D1557, Method D.

PART 2 - PRODUCTS

2.1 MATERIAL AND EQUIPMENT

- A. Provide new materials and equipment unless otherwise specifically indicated or specified. Materials shall be listed by Underwriter's laboratories, Inc. (U.L.) and bear evidence of such approval where applicable.
- B. Luminaires: Site luminaires shall be weatherproof. Reflectors and refractors shall provide the light configuration indicated and conforming to IES recommendations.
- C. Luminaires and poles shall be finished in epoxy enamel designed to withstand the effects of salt spray. Lens shall be securely attached to the lens frame for security during maintenance and relamping.
- D. Lighting Contactors: NEMA ICS 2. Electrically operated, magnetically held unit in NEMA enclosure, rated poles and ratings as indicated on Drawings. Units shall have silver alloy double breaker contacts and coil clearing contacts and shall require no arcing contacts. On-off selector switch.
- E. Poles, Brackets, Pole Bases and Attachments: Shall be rated for service with wind velocities of 100 mph considering the force exerted by the wind on the maximum exposure of the fixture luminaire selected.
- F. Poles shall be anchor base type round, height and style as indicated, finished to match luminaire, complete with handhole and gasketed cover, anchor bolts with leveling and locking screws, grounding connection, and matching base cover.
- G. Concrete pole bases shall be cast-in-place reinforced concrete as indicated with anchor bolts and conduit entries as per manufacturer. Concrete shall be rated 3,000 PSI at 28 day test.
- H. Concrete:
 - 1. Concrete for electrical requirements shall be at least 3,000 psi concrete with 1-inch maximum aggregate conforming to the requirements of Division 3 for Cast-In-Place concrete.

2.2 SOLID STATE LUMINAIRES

- A. Housing, where applicable:
 - 1. Steel bonderized or equal rust protected, or aluminum, rigid construction. Minimum gauge thickness shall be as follows:
 - a. Interior locations: No. 20-gauge steel, No. 16-gauge aluminum.
- B. Finish:
 - 1. Baked enamel finish (except when otherwise specified).
 - a. Concealed interior surfaces (this applies to interior hardware, circuit boards, etc.) matte black.
 - b. Concealed exterior surfaces: matte black.
 - c. Visible surfaces: color and texture as specified below for each luminaire type or as selected.
 - d. Exterior luminaire finish: refer to Luminaire Schedule.

- C. Light Emitting Diode (LED) requirements:
1. Correlated color temperature (CCT) for phosphor-coated white LEDs must have one of the following designated CCT's and fall within the following binning standards.
 - a. 3500K defined as 3465 +/- 245K
 2. Color spatial uniformity shall be limited to variations in chromaticity for different directions (i.e. changes in viewing angle) within 0.004 from the weighted average point on the CIE 1976 (u',v') diagram.
 3. Color maintenance shall be limited to a maximum change in chromaticity of 0.007 on the CIE 1976 (u',v') diagram over the lifetime of the product.
 - a. Color rendering index: Color rendering index to be determined using ANSI C78.377-2008 and applicable IESNA standards.
 - b. Laboratory tests must be produced using specific module(s)/array(s) and power supply combination that will be used in production.
 - c. Manufacturers must provide a test report from a laboratory accredited by NVLAP or one of its MRA signatories
 4. Lumen depreciation
 - a. Lumen depreciation to be measured using IESNA LM-80-08 standard for IES approved method of measuring lumen maintenance of LED light sources.
 - b. Phosphor coated white LED module(s)/array(s) shall deliver at least 70% of initial lumens for a minimum of 50,000 hours when installed in-situ and operated at 100% output and the maximum specified operating temperature.
 - c. Colored LED module(s)/array(s) shall deliver at least 50% of initial lumens for a minimum of 50,000 hours when installed in-situ and operated at 100% output and the maximum specified operating temperature.
 5. Acceptable LED manufacturers:
 - a. Cree
 - b. Nichia
 - c. Osram Opto Semiconductors
 - d. Philips Lumileds
- D. Luminaire Efficacy:
1. Luminaire efficiency shall be measured using IESNA LM-79-08 standard for electrical and photometric measurements of solid state lighting products.
 2. Manufacturer shall provide published luminaire efficacy, which is defined as luminaire light output divided by luminaire input power measured in a 25 degree Celsius environment. Efficacy shall include power supply, thermal, optical, and luminaire losses.
- E. Thermal Management:
1. Solid state luminaire shall not exceed LED manufacturer's maximum junction temperature requirements when operated in-situ at luminaire manufacturer's maximum ambient operating temperature and 100% light output.
 2. Solid state luminaires shall be thermally protected using one of more of the following thermal management techniques:
 - a. Metal core board
 - b. Gap pad
 - c. Internal monitoring firmware
 3. Solid state luminaire housing shall be designed to transfer heat from the LED board to the outside environment.
- F. Power Supply/Driver requirements:
1. Power factor of 0.90 or greater for primary application
 2. Input current shall have Total Harmonic Distortion (THD) of less than 20%.

3. Minimum operating temperature of minus 20 degrees Celsius or below when used in luminaires intended for outdoor applications.
 4. Output operating frequency to be equal to or greater than 120 Hz.
 5. Operate with sustained input variations of +/- 10% (voltage and frequency) with no damage to the driver.
 6. Tolerate sustained open circuit and short circuit output conditions without damage and without need for external fuses or trip devices.
 7. Output shall be regulated to +/- 5% across published load range.
 8. Class A sound rating.
 9. Outputs shall have current limiting protection.
 10. Operate LEDs at constant and regulated current levels. LEDs shall not be overdriven beyond the diode manufacturer's specified nominal voltage and current.
 11. Inrush currents not exceeding peak currents specified in NEMA 410.
- G. Solid State Lighting Controls:
1. Control interface to dimmable power supplies shall consist of one of the following:
 - a. Line Voltage Dimming. Controls to be rated for magnetic or electronic low voltage transformer operation.
 - b. Low voltage (0-10V) control. Controls to be compatible with either current sink or current source operation.
 - c. DMX control
 2. Dimmable LED power supplies shall use pulse width modulation (PWM) to regulate power to LEDs
 - a. Dimmable power supplies shall have 12-bit or greater resolution to obtain flicker-free operation throughout the dimming range.
- H. System Installation
1. Hardwired connections to solid state luminaires shall be reverse polarity protected and provide high voltage protection in the event connections are reversed or shorted during the installation process.
 2. All solid state luminaires (100% of each lot) shall undergo a minimum eight-hour burn-in test during manufacturing. Solid state lighting installations shall be UL Listed as a low-voltage lighting system including, but not limited to, luminaire, power supply, controller, keypad, and wiring.
- I. Warranty
1. Luminaires, drivers, and controllers for solid state lighting systems shall be covered by a minimum five-year warranty against defects in workmanship or material. Warranty shall include in-warranty service program providing for payment of authorized labor charges incurred in replacement of inoperative in-warranty equipment.

2.3 LUMINAIRE CONSTRUCTION

- A. Sheet metal: materials and thicknesses shall be 20 gauge (0.7 mm or 0.027") min., free of dents, scratches, oil-can, or other defects.
- B. Painted luminaires: exposed weld marks, joints, and seams shall be filled and sanded smooth before finishing.
- C. All edges cleaned and dressed to remove sharp edges or burrs.
- D. Extrusions: 1/10" min. wall thickness, smooth and free of tooling lines, with cast end plates that exactly match extrusion profiles.

- E. Castings: smooth, free of pits, scales, gate marks, or blemishes.
- F. Spinings shall have 1/32" min. thickness, smooth, free of spinning lines or blow-back, with clean edges.
- G. Welds: Follow recommendations of American Welding Society. All welds continuous and free of spatter, residue, or warping.
- H. No light leaks visible. Field paint exterior of housing with high temperature paint if necessary.
- I. Exposed end plates and joiners, with concealed fasteners.
- J. Hardware:
 - 1. Steel or aluminum exterior luminaires: stainless steel hardware.
 - 2. Stainless steel luminaires: stainless steel hardware.
 - 3. Copper alloy luminaires: brass hardware.
- K. Raceways: Where used for through wiring, luminaires must be approved for use as raceways.

PART 3 - EXECUTION

3.1 INSTALLATION:

- A. Refer to Section 26 27 00, Part 2.2, for wiring and splicing requirements.
- B. Underground cable installation shall conform to National Electrical Code except as otherwise specified or indicated.
- C. Contractor Damage: The Contractor shall promptly cause repairs to be made to any indicated utility lines or systems damaged by his operation.
- D. Under roads and paved areas, ducts shall be EPC-80-PVC polyvinyl chloride conduit.
- E. Cables shall be in one piece without splices between connections except where the distance exceeds the lengths in which the cable is furnished.
- F. Bends in cables shall have an inner radius of not less than 12 times the cable diameter.
- G. Horizontal slack of approximately 3 feet shall be left in the ground on each end of cable runs, on each side of connection and at all points where connections are to be made above ground level.
- H. Earthwork: Earthwork for electrical requirements shall conform to the requirements of Division 31.
- I. Coordinate work with other trades. Pre-ship anchor bolts and templates for use in preparing bases for installation. After leveling luminaires, pack grout between mounting plate and concrete footing. Provide weep holes to prevent accumulation of moisture inside pole base.

3.2 TESTS

- A. Test under provisions of Division 1, Section 26 08 00, and Section 26 51 01.
- B. The Owner shall be notified at least three working days in advance of the Contractor's proposed date of the tests to permit scheduling, and to permit witnessing of the tests. The Contractor shall furnish the Owner with three copies of the results of the tests.
- C. Circuits: The Contractor shall test each circuit, all controllers, and components of the system for proper operation. The Contractor shall furnish the Owner with three copies of the test results.
- D. Compaction Tests: Backfill shall be tested for compaction in accordance with ASTM D1556.
- E. Operating Test: Contractor shall operate the system in the presence of the Owner proving the proper operation.

END OF SECTION

SECTION 26 57 00 - LOW VOLTAGE LIGHTING CONTROL SYSTEM

PART 1 - GENERAL

1.1 SYSTEM DESCRIPTION

- A. Furnish all labor, materials, apparatus, tools, equipment transportation, temporary construction and commissioning services as indicated on the Drawings or described in these Specifications and as required to make a complete working facility lighting control system.
- B. Integrated Low Voltage Lighting Control System:
 - 1. The low voltage lighting control system shall consist of switches, digital occupancy sensors, daylighting sensors, room controllers, relay panels, wireless network bridges, wireless border routers, and LMCS system configuration software.
 - 2. The system shall accept program changes from the LMCS system configuration software for date and time, location, holidays, event scheduling, button binding and group programming.
- C. Requirements are indicated in Section 26 27 00 for raceways and electrical boxes and fittings required for installation of control equipment and wiring.
- D. Provide CBC 2013 compliant seismic installation. See Section 26 05 00 for all certification and submittal requirements.

1.2 INCORPORATED DOCUMENTS

- A. Sections 26 05 00, 26 27 00, 26 51 01 and 26 56 01 apply to all Work in this Section.
- B. Section 25 55 00 - Integrated Automation Control of HVAC- Integrated Automation. Building integrator shall provide integration of the lighting control system with Building Automation Systems.
- C. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.3 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing of centralized and distributed lighting control systems with a minimum of 10 years documented experience.
- B. Installer Qualifications: Company certified by the manufacturer and specializing in installation of networked lighting control products with minimum three years documented experience.
- C. System Components: Demonstrate that individual components have undergone quality control and testing prior to shipping.
- D. System Support: Factory applications engineers shall be available for telephone support.
- E. NEC Compliance: Comply with NEC as applicable to electrical wiring Work.
- F. NEMA Compliance: Comply with applicable portions of NEMA standards pertaining to types of electrical equipment and enclosures.

- G. UL Approvals: Remote panels are to be UL listed under UL 916 Energy Management Equipment and/or UL 508 Industrial Control Equipment.
- H. CSA Approvals: Remote panels are to be CSA listed.
- I. FCC Emissions: All assemblies are to be in compliance with FCC emissions Standards specified in Part 15 Subpart J for commercial and residential application.
- J. All System components shall be California Title 24 compliant, where applicable.

1.4 SUBMITTALS

- A. Submit under provisions of Section 01 33 00 and 26 05 00.
- B. Product Data: Manufacturer's data sheets on each product to be used, including:
 - 1. Catalog sheets and specifications.
 - 2. Ratings, configurations, standard wiring diagrams, dimensions, colors, service condition requirements, and installed features.
 - 3. Storage and handling requirements and recommendations.
 - 4. Installation instructions.
- C. Shop Drawings: Wiring diagrams a for the various components of the System specified including:
 - 1. Composite wiring and/or schematic diagram of each control circuit as proposed to be installed.
 - 2. Show location of all devices, including at minimum sensors, load controllers, and switches/dimmers for each area on reflected ceiling plans.
 - 3. Provide room/area details including products and sequence of operation for each room or area. Illustrate typical acceptable room/area connection topologies.
 - 4. Network riser diagram including floor and building level details. Include network cable specification. Illustrate points of connection to integrated systems. Coordinate integration with mechanical and/or other trades.
- D. Manufacturer's Certificates: Certify products meet or exceed specified requirements.
- E. Closeout Submittals:
 - 1. Project Record Documents: Record actual installed locations and settings for lighting control devices.
 - 2. Operation and Maintenance Manual:
 - a. Include approved Shop Drawings and Product Data.
 - b. Include Sequence of Operation, identifying operation for each room or space.
 - c. Include manufacturer's maintenance information.
 - d. Operation and Maintenance Data: Include detailed information on device programming and setup.
 - e. Include startup and test reports.

1.5 MANUFACTURERS

- A. Integrated Low Voltage Lighting Control System:

1. The basis of the specified system is the Watt Stopper Digital Lighting Management (DLM) or an equal. Any other system to be considered must submit descriptive information 10 days prior to bid.
- B. Prior approval does not guarantee final approval by the electrical engineer. The contractor shall be completely responsible for providing a system meeting this specification in its entirety. All deviations from this specification must be listed and individually signed off by the engineer.
- C. The Owner reserves the right to reject a proposed substitution based on his agent's professional judgment as to the utility, visual appropriateness, or finish of substitutions.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Delivery: Deliver materials in Manufacturer's original, unopened, undamaged packages with intact identification labels.
- B. Storage and Protection: Store products in a clean, dry space in original manufacturer's packaging in accordance with manufacturer's written instructions until ready for installation

1.7 GUARANTEE AND WARRANTIES

- A. All Work performed under this Section must be guaranteed to be free of defects in products or workmanship for one year after date of acceptance by Owner, unless noted otherwise in General Conditions.
- B. Products Warranty: Manufacturer shall provide a 5 year limited warranty on products within this installation, except where otherwise noted, and consisting of a one for one device replacement.

PART 2 - PRODUCTS

2.1 DISTRIBUTED DIGITAL LIGHTING CONTROL SYSTEM

- A. Equipment Required: Lighting Control and Automation system as defined under this section covers the following equipment.
 1. Digital Lighting Management (DLM) local network: Free topology, plug-in wiring system (Cat 5e) for power and data to room devices.
 2. Wireless Network Bridge and Border Router: Provides Wireless Network Bridges that automatically create BACnet objects for all DLM devices on their local network (room) and communicate that information over a standalone wireless mesh 6LoWPAN network to a Border Router. The Border Router manages the formation and communication of the mesh network, and provides an ethernet network connection to upstream intelligent devices, such as a Segment Manager.
 3. Programming and Configuration Software: Optional PC-native application capable of accessing DLM control parameters within a room, for the local network, via a USB adapter, or globally, for many segment networks simultaneously, via BACnet/IP communication.
- B. Local Network LMRJ-Series: DLM local network is a free topology lighting control physical connection and communication protocol designed to control a small area of a building.
 1. Digital room devices connect to the local network using pre-terminated Cat 5e cables with RJ-45 connectors, which provide both data and power to room devices. Systems that utilize RJ-45 patch cords but do not provide serial communication data from individual end devices are not acceptable.

2. If manufacturer's pre-terminated Cat5e cables are not used for the installation each cable must be individually tested and observed by authorized service representative following installation.

2.2 DIGITAL SWITCHES / DIMMERS

A. Description

1. Two-way infrared (IR) transceiver for use with personal and configuration remote controls. Low voltage momentary pushbutton switches in 1, 2, 3, 4, 5 and 8 button configuration. Wall switches shall include the following features
2. Removable buttons for field replacement with engraved buttons and/or alternate color buttons. Button replacement may be completed without removing the switch from the wall.
3. Configuration LED indicator light on each switch that blinks to indicate data transmission.
4. Load/Scene Status LED indicator light on each switch button with the following characteristics:
 - a. Bi-level LED
 - b. Dim locator level indicates power to switch
 - c. Bright status level indicates that load or scene is active
5. Dimming switches shall include seven bi-level LEDs to indicate load levels using 14 steps.
6. All digital parameter data programmed into an individual wall switch shall be retained in non-volatile FLASH memory within the wall switch itself. Memory shall have an expected life of no less than 10 years.
7. Two RJ-45 ports for connection to DLM local network.
8. Multiple digital wall switches may be installed in a room by simply connecting them to the free topology DLM local network. No additional configuration shall be required to achieve multi-way switching.
9. Switches shall be constructed of non-breakable Lexan on all exposed parts and shall include a matching screwless Lexan wall plate.
10. Individual buttons shall be custom engraved with a maximum of 15 characters (including spaces) on two lines, where shown on plans.
11. Multiple digital switches wired to control the same relay or relay group shall indicate the same status automatically.

- B. WattStopper product numbers: LMSW-101, LMSW-102, LMSW-103, LMSW-104, LMSW-105, LMSW-108, LMDM-101.

C. Description

1. Intelligent digital switching shall be operating on the DLM Category 5e local network. Switches shall be available in single, dual, quad, or octal (1-button, 2-button, 4-button, or 8-button) designs. All devices shall mount in a standard single-gang box
2. Each button in a switch can be individually programmed. Programming is done by the LMCT-100 handheld configuration tool. Each button can control any one of the following options:
 - a. Any individual relay in any single panel.
 - b. Any group of relays in any single panel.
 - c. Any group of relays in the system.

2.3 DIGITAL OCCUPANCY SENSORS

A. Description

1. Digital Occupancy Sensors - Self-configuring, digitally addressable and calibrated occupancy sensors with LCD display and two-way active infrared (IR) communications.

2.4 DIGITAL PHOTOSENSORS

A. Description

1. Digital Photosensors - Single-zone closed loop, multi-zone open loop and dual-loop daylighting sensors with two-way active infrared (IR) communications can provide switching, bi-level, tri-level or dimming control for daylight harvesting.

2.5 DIGITAL DIMMING / SWITCHING ROOM CONTROLLERS

A. Description

1. Digital controllers for lighting and plug loads automatically bind the room loads to the connected devices in the space without commissioning or the use of any tools. Room and plug load controllers shall be provided to match the room lighting and plug load control requirements. The controllers will be simple to install, and will not have dip switches or potentiometers, or require special configuration for standard Plug n' Go applications. The control units will include the following features.
2. Automatic room configuration to the most energy-efficient sequence of operation based upon the devices in the room.
3. Simple replacement - Using the default automatic configuration capabilities, a room controller may be replaced with an off-the-shelf.
4. Multiple room controllers connected together in a local network must automatically prioritize each room controller, without requiring any configuration or setup, so that loads are sequentially assigned using room controller device ID's from highest to lowest.
5. Device Status LEDs to indicate:
 - a. Data transmission
 - b. Device has power
 - c. Status for each load
 - d. Configuration status
 - e. Quick installation features including:
 - 1) Standard junction box mounting
 - 2) Quick low voltage connections using standard RJ-45 patch cable
 - f. Based on individual configuration, each load shall be capable of the following behavior on power up following the loss of normal power:
 - 1) Turn on to 100%
 - 2) Remain off
 - 3) Turn on to last level
 - g. Each load shall be configurable to operate in the following sequences based on occupancy:
 - 1) Auto-on/Auto-off (Follow on and off)
 - 2) Manual-on/Auto-off (Follow off only)
 - h. The polarity of each load output shall be reversible, via digital configuration, so that on is off and off is on.
 - i. BACnet object information shall be available for the following objects:
 - 1) Load status
 - 2) Electrical current
 - 3) Total watts per controller
 - 4) Schedule state - normal or after-hours
 - 5) Demand response control and cap level
 - 6) Room occupancy status
 - 7) Total room lighting and plug loads watts
 - 8) Total room watts/sq ft
 - 9) Force on/off all loads
 - j. UL 2043 plenum rated

- k. Manual override and LED indication for each load
 - l. Dual voltage (120/277 VAC, 60 Hz), or 347 VAC, 60 Hz (selected models only). 120/277 volt models rated for 20A total load, derating to 16A required for some dimmed loads (forward phase dimming); 347 volt models rated for 15A total load; plug load controllers carry application-specific UL 20 rating for receptacle control.
 - m. Zero cross circuitry for each load
 - n. All digital parameter data programmed into an individual room controller or plug load controller shall be retained in non-volatile FLASH memory within the controller itself. Memory shall have an expected life of no less than 10 years.
- B. On/Off Room Controllers shall include:
- 1. One or two relay configuration
 - 2. Efficient 150 mA switching power supply
 - 3. Three RJ-45 DLM local network ports with integral strain relief and dust cover
 - 4. WattStopper product numbers: LMRC-101, LMRC-102
- C. On/Off/Dimming enhanced Room Controllers shall include:
- 1. Real time current monitoring
 - 2. Multiple relay configurations
 - a. One, two or three relays (LMRC-21x series)
 - b. One or two relays (LMRC-22x series)
 - c. Efficient 250 mA switching power supply
 - d. Four RJ-45 DLM local network ports with integral strain relief and dust cover
 - e. One dimming output per relay
 - 1) 0-10V Dimming - Where indicated, one 0-10 volt analog output per relay for control of compatible ballasts and LED drivers. The 0-10 volt output shall automatically open upon loss of power to the Room Controller to assure full light output from the controlled lighting. (LMRC-21x series)
 - 2) Line Voltage, Forward Phase Dimming - Where indicated, one forward phase control line voltage dimming output per relay for control of compatible two-wire or three-wire ballasts, LED drivers, MLV, forward phase compatible ELV, neon/cold cathode and incandescent loads. (LMRC-22x series)
 - 3) Each dimming output channel shall have an independently configurable minimum and maximum calibration trim level to set the dimming range to match the true dynamic range of the connected ballast or driver.
 - 4) The LED level indicators on bound dimming switches shall utilize this new maximum and minimum trim.
 - 5) Each dimming output channel shall have an independently configurable minimum and maximum trim level to set the dynamic range of the output within the new 0-100% dimming range defined by the minimum and maximum calibration trim.
 - 6) Calibration and trim levels must be set per output channel.
 - 7) Devices that set calibration or trim levels per controller are not acceptable.
 - 8) All configuration shall be digital. Devices that set calibration or trim levels per output channel via trim pots or dip-switches are not acceptable.
 - f. Each load shall have an independently configurable preset on level for Normal Hours and After Hours events to allow different dimmed levels to be established at the start of both Normal Hours and After Hours events.
 - g. Fade rates for dimming loads shall be specific to bound switch buttons, and the load shall maintain a default value for any bound buttons that do not specify a unique value.
 - h. The following dimming attributes may be changed or selected using a wireless configuration tool:
 - 1) Establish preset level for each load from 0-100%
 - 2) Set high and low trim for each load

- 3) Set lamp burn in time for each load up to 100 hours
 - i. Override button for each load provides the following functions:
 - 1) Press and release for on/off control
 - 2) Press and hold for dimming control
 - j. WattStopper product numbers: LMRC-211, LPMC-212, LPMC-213, LMRC-221, LMRC-222
- D. Plug Load Room Controllers shall include:
- 1. One relay configuration with additional connection for unswitched load
 - 2. Configurable additive time delay to extend plug load time delay beyond occupancy sensor time delay (e.g. a 10 minute additive delay in a space with a 20 minute occupancy sensor delay ensures that plug loads turn off 30 minutes after the space is vacated).
 - 3. Factory default operation is Auto-on/Auto-off, based on occupancy
 - 4. Real time current monitoring of both switched and un-switched load (LMPL-201 only)
 - 5. Efficient switching power supply
 - a. 150mA (LMPL-101)
 - b. 250mA (LMPL-201)
 - c. RJ-45 DLM local network ports
 - 1) Three RJ-45 ports (LMPL-101)
 - 2) Four RJ-45 ports (LMPL-201)
 - d. WattStopper product numbers: LMPL-101, LMPL-201.

2.6 LMCP LIGHTING CONTROL PANELS AND LMZC ZONE CONTROLLER

- A. Hardware: Provide LMCP lighting control panels in the locations and capacities as indicated on the Drawing and schedules. Each panel shall be of modular construction and consist of the following components:
- 1. Enclosure/Tub shall be NEMA 1, sized to accept an interior with 1 - 8 relays, 1 - 24 relays and 6 four-pole contactors, or 1 - 48 relays and 6 four-pole contactors.
 - 2. Cover shall be configured for surface or flush wall mounting of the panel as indicated on the plans. LMCP panel cover shall have a hinged and lockable door with restricted access to line voltage section of the panel.
 - 3. Interior assembly shall be supplied as a factory assembled component specifically designed and listed for field installation. Interior construction shall provide total isolation of high voltage (Class 1) wiring from low voltage (Class 2) wiring within the assembled panel. Interior assembly shall include intelligence boards, power supply, DIN rails for mounting optional Class 2 control devices, and individually replaceable latching type relays. Panel interiors shall include the following features:
 - a. Removable, plug-in terminal blocks with connections for all low voltage terminations.
 - b. Individual terminal block, override pushbutton, and LED status light for each relay.
 - c. Direct wired switch inputs associated with each relay shall support 2-wire momentary switches only.
 - d. Digital inputs (four RJ-45 jacks) shall support 1-, 2-, 3-, 4-, and 8-button digital switches; digital IO modules capable of receiving 0-5V or 0-10V analog photocell inputs; digital IO modules capable of receiving momentary or maintained contact closure inputs or analog sensor inputs; digital daylighting sensors; and digital occupancy sensors. Inputs are divided into two separate digital networks, each capable of supplying 250mA to connected devices.
 - e. True relay state shall be indicated by the on-board LED and shall be available to external control devices and systems via BACnet.
 - f. Automatically sequenced operation of relays to reduce impact on the electrical distribution system when large loads are controlled simultaneously.
 - g. Group and pattern control of relays shall be provided through a simple keypad interface from a handheld IR programmer. Any set of relays can be associated with a

- group for direct on/off control or pattern (scene) control via a simple programming sequence using the relay override pushbuttons and LED displays for groups 1-8 or a handheld IR programmer for groups 1-99.
- h. Relay group status shall be provided through LED indicators for groups 1-8 and via BACnet for groups 1-99. A solid LED indicates that the last group action called for an ON state and relays in the group are on or in a mixed state.
4. Single-pole latching relays with modular plug-in design. Relays shall provide the following ratings and features:
- a. Electrical:
 - 1) 30 amp ballast at 277V
 - 2) 20 amp ballast at 347V
 - 3) 20amp tungsten at 120V
 - 4) 30 amp resistive at 347V
 - 5) 1.5 HP motor at 120V
 - 6) 14,000 amp short circuit current rating (SCCR) at 347V
 - 7) Relays shall be specifically UL 20 listed for control of plug-loads
 - b. Mechanical:
 - 1) Replaceable, 1/2 inch KO mounting with removable Class 2 wire harness.
 - 2) Actuator on relay housing provides manual override and visual status indication, accessible from Class 2 section of panel.
 - 3) Dual line and load terminals each support two #14 - #12 solid or stranded conductors.
 - 4) Tested to 300,000 mechanical on/off cycles.
5. Isolated low voltage contacts provide for true relay status feedback and pilot light indication.
6. Power supply shall be a multi-voltage transformer assembly with rated power to supply all electronics, occupancy sensors, switches, pilot lights, and photocells as necessary to meet the project requirements. Power supply to have internal over-current protection with automatic reset and metal oxide varistor protection.
7. Where indicated, lighting control panels designated for control of emergency lighting shall be provided with factory installed provision for automatic bypass of relays controlling emergency circuits upon loss of normal power. Panels shall be properly listed and labeled for use on emergency lighting circuits and shall meet the requirements of UL924 and NFPA 70 - Article 700.
8. Integral system clock shall provide scheduling capabilities for panel-only projects without DLM segment networks or BAS control.
- a. Each panel shall include digital clock capability able to issue system wide automation commands to up to 11 other panels for a total of 12 networked lighting control panels. Clock shall provide capability for up to 254 independent schedule events per panel for each of the ninety-nine system wide channel groups.
 - b. Clock capability of each panel shall support the time-based energy saving requirements of applicable local energy codes.
 - c. Clock module shall provide astronomic capabilities, time delays, blink warning, daylight savings, and holiday functions and will include a battery back up for clock function and program retention in non-volatile FLASH memory. Clocks that require multiple events to meet local code lighting shut off requirements shall not be allowed.
 - d. Clock capability of each panel shall operate on a basis of ON/OFF or Normal Hours/After Hours messages to automation groups that implement pre-configured control scenarios. Scenarios shall include:
 - 1) Scheduled ON / OFF
 - 2) Manual ON / Scheduled OFF
 - 3) Astro ON / OFF (or Photo ON / OFF)
 - 4) Astro and Schedule ON / OFF (or Photo and Schedule ON / OFF)
 - e. User interface shall be a portable IR handheld remote control capable of programming any panel in the system (LMCT-100)

- f. Clock capability of each panel shall employ non-volatile memory and shall retain user programming and time for a minimum of 10 years.
 - g. Schedules programmed into the clock of any one panel shall be capable of executing panel local schedule or Dark/Light (photocell or Astro) events for that panel in the event that global network communication is lost. Lighting control panels that are not capable of executing events independently of the global network shall not be acceptable.
- 9. Lighting control panel can operate as a stand-alone system, or can support schedule, group, and photocell control functions, as configured in a Segment Manager controller, via a segment network connection.
- 10. Lighting control panel shall support digital communications to facilitate the extension of control to include interoperation with building automation systems and other intelligent field devices. Digital communications shall be RS485 MS/TP-based using the BACnet protocol.
 - a. Panel shall have provision for an individual BACnet device ID and shall support the full 222 range (0 - 4,193,304). The device ID description property shall be writable via the network to allow unique identification of the lighting control panel on the network.
 - b. Panel shall support MS/TP MAC addresses in the range of 0 - 127 and baud rates of 9600k, 38400k, 76800k, and 115.2k bits per second.
 - c. Lighting control relays shall be controllable as binary output objects in the instance range of 1 - 64. The state of each relay shall be readable and writable by the BAS via the object present value property.
 - d. Lighting control relays shall report their true on/off state as binary input objects in the instance range of 1 - 64.
 - e. The 99 group Normal Hours/After Hours control objects associated with the panel shall be represented by binary value objects in the instance range of 201 - 299. The occupancy state of each channel group shall be readable and writable by the BAS via the object present value property. Commanding 1 to a channel group will put all relays associated with the channel into the normal hours mode. Commanding 0 or NULL shall put the relays into the after hours mode.
 - f. Setup and commissioning of panel shall not require manufacturer-specific software or a computer. All configuration of the lighting control panel shall be performed using standard BACnet objects or via the handheld IR programming remote. Provide BACnet objects for panel setup and control as follows:
 - 1) Binary output objects in the instance range of 1 - 64 (one per relay) for on/off control of relays.
 - 2) Binary value objects in the instance range of 1 - 99 (one per channel) for normal hours/after hours schedule control.
 - 3) Binary input objects in the instance range of 1 - 64 (one per relay) for reading true on/off state of the relays.
 - 4) Analog value objects in the instance range of 101 - 199 (one per channel group) shall assign a blink warn time value to each channel. A value of 5 shall activate the blink warn feature for the channel and set a 5-minute grace-time period. A value of 250 shall activate the sweep feature for the channel and enable the use of sweep type automatic wall switches.
 - g. Description property for all objects shall be writable via the network and shall be saved in non-volatile memory within the panel.
 - h. BO and BV 1 - 99 objects shall support BACnet priority array with a relinquish default of off and after hours respectively. Prioritized writes to the channel BV objects shall propagate prioritized control to each member relay in a way analogous to the BACnet Channel object described in addendum aa. (<http://www.bacnet.org/Addenda/Add-135-2010aa.pdf>)
 - i. Panel-aggregate control of relay Force Off at priority 2 shall be available via a single BV5 object. Force On at priority 1 shall be available via a single BV4 object.
 - j. Lockout of all digital switch buttons connected to a given panel shall be command-able via a single BV2 object. The lock status of any connected switch station shall be represented as BV101-196.

11. In addition to the LMCP Relay Panels, an LMZC Zone Controller panel shall be available for zero-relay applications. The panel is designed for applications where LMFC-011 Fixture Controllers or other distributed load controllers are used to switch and/or dim the controlled loads. Key similarities to and differences from the LMCP panel design shall include:
 - a. Use the same intelligence board as the LMCP relay panel.
 - b. Shall not include relay driver boards or relays.
 - c. Have a removable interior section to facilitate installation, and a Tub/Cover. Cover is for surface mounting applications only.
 - d. Tub shall have two interior KOs to allow installation of LMPB-100 Power Boosters. Each installed Power Booster can provide an additional 150 mA for either of the two available DLM local networks provided by the LMZC.
 - e. All programming and networking (whether DLM Local Network and/or Segment Network) capabilities in the LMZC Zone Controller shall be similar to capabilities for LMCP relay panels, except for functions designed for panel-mounted HDR relays.
12. To aid in project start up, if LMFC Fixture Controllers are connected to an LMZC Zone Controller, Plug n' Go automatic configuration will establish a unique sequence of operation so that all LMFC-controlled fixtures will turn on to 50 percent output when any digital occupancy sensor detects motion.
13. WattStopper Product Number: Relay Panels: LMCP8, LMCP24 or LMCP48, Zone Controller: LMZC-301.

2.7 CONFIGURATION TOOLS

- A. A wireless configuration tool facilitates optional customization of DLM local networks using two-way infrared communications, while PC software connects to each local network via a USB interface.
- B. Features and functionality of the wireless configuration tool shall include but not be limited to:
 1. Two-way infrared (IR) communication with DLM IR-enabled devices within a range of approximately 30 feet.
 2. High visibility organic LED (OLED) display, pushbutton user interface and menu-driven operation.
 3. Must be able to read and modify parameters for room controllers, occupancy sensors, wall switches, daylighting sensors, network bridges and relay panels, and identify room devices by type and serial number.
 4. Save up to eight occupancy sensor setting profiles, and apply profiles to selected sensors.
 5. Temporarily adjust light level of any load(s) on the local network, and incorporate those levels in scene setting. Set room mode for testing of Normal Hours (NH) and After Hours (AH) parameter settings.
 6. Adjust or fine-tune daylighting settings established during auto-configuration, and input light level data to complete configuration of open loop daylighting controls.
 7. Set room mode for testing of Normal Hours (NH) and After Hours (AH) parameter settings.
 8. Verify status of building level network devices.
- C. WattStopper Product Numbers: LMCT-100, LMCI-100/LMCS-100

2.8 DLM WIRELESS NETWORK (ROOM TO ROOM NETWORK).

- A. Communication between the Wireless Network In-Room Bridges LMBC-650 and the Wireless Network Border Router and Master Border Router LMBR-650 shall meet the following specifications:
 1. The communication between the Wireless Bridges and the Border Router shall be via a standalone wireless mesh network that does not require interface with any other wireless

network in the space. The mesh network shall allow communication between all rooms as long as they are within 100' of another connected room.

2. The Wireless Bridges shall communicate over a 6LoWPAN 2.4 GHz IEEE 802.15.4 network and use AES 128 bit Key Encryption for network security.
3. In addition to IEEE IPV6, the Border Router shall have available Constrained Application Protocol (CoAP), Routing Protocol for Low Power Networks (RPL), and Stateless Multicast RPL Forwarding (SMRF).
4. The wireless protocol shall allow BACnet communication to be transported transparently between the Network Bridge and any front end BAS device.
5. Features of the Wireless Border Router shall be:
 - a. The Wireless Border Router shall manage the formation and configuration of the 6LoWPAN wireless mesh network, and provide connectivity via wired 10/100 Ethernet to a local area network that may include a LMSM Segment Manager or Building BAS System.
 - b. Border Router shall provide key information about the health of the mesh network in the form of signal quality, device status, network status, and other real-time network information such as energy monitoring.
 - c. The LMBR-650 shall have dual internal omni-directional antennas with diversity to ensure reliable communication with Wireless Network Bridges, and provide a user interface for set up and configuration.
 - d. Include an internal MicroSD card and a Real-time clock with supercap back-up. Border Router shall get power for operation via a 120V outlet (in non-plenum applications) and a dedicated DLM LMPB-100 Power Booster connected to a Cat 5e to DC barrel connector (for plenum applications).

2.9 NETWORK BRIDGE

- A. The network bridge module connects a DLM local network to a BACnet-compliant segment network for communication between rooms, relay panels and a segment manager or BAS. Each local network shall include a network bridge component to provide a connection to the local network room devices. The network bridge shall use industry standard BACnet MS/TP network communication and an optically isolated EIA/TIA RS-485 transceiver.
 1. The network bridge shall be provided as a separate module connected on the local network through an available RJ-45 port.
 2. Provide Plug n' Go operation to automatically discover room devices connected to the local network and make all device parameters visible to the segment manager via the segment network. No commissioning shall be required for set up of the network bridge on the local network.
 3. The network bridge shall automatically create standard BACnet objects for selected room device parameters to allow any BACnet-compliant BAS to include lighting control and power monitoring features as provided by the DLM room devices on each local network. BACnet objects will be created for the addition or replacement of any given in-room DLM device for the installed life of the system. Products requiring that an application-specific point database be loaded to create or map BACnet objects are not acceptable. Systems not capable of providing BACnet data for control devices via a dedicated BACnet Device ID and physical MS/TP termination per room are not acceptable. Standard BACnet objects shall be provided as follows:
 - a. Read/write the normal or after hours schedule state for the room
 - b. Read the detection state of each occupancy sensor
 - c. Read the aggregate occupancy state of the room
 - d. Read/write the On/Off state of loads
 - e. Read/write the dimmed light level of loads
 - f. Read the button states of switches
 - g. Read total current in amps, and total power in watts through the room controller

- h. Read/write occupancy sensor time delay, PIR sensitivity and ultrasonic sensitivity settings
- i. Activate a preset scene for the room
- j. Read/write daylight sensor fade time and day and night setpoints
- k. Read the current light level, in footcandles, from interior and exterior photosensors and photocells
- l. Set daylight sensor operating mode
- m. Read/write wall switch lock status
- n. Read watts per square foot for the entire controlled room
- o. Write maximum light level per load for demand response mode
- p. Read/write activation of demand response mode for the room
- q. Activate/restore demand response mode for the room

B. WattStopper product numbers: LMBC-300

2.10 SEGMENT MANAGER

- A. For networked applications, the Digital Lighting Management system shall include at least one segment manager to manage network communication. It shall be capable of serving up a graphical user interface via a standard web browser utilizing either unencrypted TCP/IP traffic via a configurable port (default is 80) or 256 bit AES encrypted SSL TCP/IP traffic via a configurable port (default is 443).
- B. Each segment manager shall have integral support for at least three segment networks. Segment networks may alternately be connected to the segment manger via external routers and switches, using standard Ethernet structured wiring. Each router shall accommodate one segment network. Provide the quantity of routers and switches as shown on the plans.
- C. Operational features of the Segment Manager shall include the following:
 - 1. Connection to PC or LAN via standard Ethernet TCP/IP via standard Ethernet TCP/IP with the option to use SSL encrypted connections for all traffic.
 - 2. Easy to learn and use graphical user interface, compatible with Internet Explorer 8, or equal browser. Shall not require installation of any lighting control software to an end-user PC.
 - 3. Log in security capable of restricting some users to view-only or other limited operations.
 - 4. Automatic discovery of DLM devices and relay panels on the segment network(s). Commissioning beyond activation of the discovery function shall not be required to provide communication, monitoring or control of all local networks and lighting control panels.
 - 5. After discovery, all rooms and panels shall be presented in a standard navigation tree format. Selecting a device from the tree will allow the device settings and operational parameters to be viewed and changed by the user.
 - 6. Ability to view and modify room device operational parameters. It shall be possible to set device parameters independently for normal hours and after hours operation including sensor time delays and sensitivities, and load response to sensor including Manual-On or Auto-On.
 - 7. Ability to set up schedules for rooms and panels, view and override current status of panel channels and relays, and assign relays to groups. Schedules shall automatically set controlled zones or areas to either a normal hours or after hours mode of operation. Support for a minimum of 100 unique schedules, each with up to four time events per day. Support for annual schedules, holiday schedules and unique date-bound schedules.
 - 8. Ability to group rooms and loads for common control by schedules, switches or network commands.
 - 9. Ability to monitor connected load current and display power consumption for areas equipped with room controllers incorporating the integral current monitoring feature.

10. Provide capabilities for integration with a BAS via BACnet protocol. At a minimum, the following points shall be available to the BAS via BACnet IP connection to the segment manager: room occupancy state; room schedule mode; room switch lock control; individual occupancy sensor state; room lighting power; room plug-load power; load ON/OFF state; load dimming level; panel channel schedule state; panel relay state; and Segment Manager Group schedule state control.
 11. The Segment Manager shall allow access and control of the overall system database via Native Niagara AX FOX connectivity. Systems that must utilize a Tridium Niagara controller in addition to the programming, scheduling and configuration server are not acceptable.
- D. Segment Manager shall support multiple DLM rooms as follows:
1. Support up to 120 network bridges and 900 digital in-room devices (LMSM-3E).
 2. Support up to 300 network bridges and 2,200 digital in room devices, connected via network routers and switches (LMSM-6E).
- E. Segment Manager shall provide single point of integration to the Building Management System (BMS) via BACnet/IP. The lighting control system shall provide an export table to transfer room/device data to the BMS system. Data provided in the export table will be confirmed in the pre-installation meeting referred to in Section 3.02-B. Time based scheduling shall remain a function of the lighting control system unless all scheduling is coordinated with the building management system. The export table is limited to the BACnet objects listed in section 2.08-A-3. The lighting control manufacture is recommending the following BACnet objects be included as a minimum standard in the export table:
1. Occupancy state of the room
 2. On/Off state of loads in each room
 3. Total power in watts through the load controller
 4. Read watts per square foot for the entire controlled room
 5. Current light level in foot-candles from interior and exterior photosensors
- F. WattStopper Product Numbers: LMSM-3E, LMSM-6E, NB-ROUTER, NB-SWITCH, NB-SWITCH-8, NB-SWITCH-16.

2.11 WIRELESS NETWORK BRIDGES AND BORDER ROUTER

- A. Wireless Network Bridges connect to a DLM local network (room) and use IEEE 802.15.4 6LoWPAN for communication between rooms and to a Border Router that oversees the formation and configuration of the wireless network. Each local network shall include a wireless network bridge that connects to the other DLM devices on the local network, and a group of Wireless Bridges shall connect to a Border Router.
- B. Features of the Wireless Network Bridges shall be as follows:
1. Network bridge shall be provided as a separate module connected on the local network through an available RJ-45 port.
 2. Wireless Bridges provide Plug n' Go operation to automatically discover room devices connected to the local network and make all device parameters visible to the Border Router via the wireless network. No commissioning shall be required for set up of the network bridge on the local network.
 3. Wireless Bridges shall incorporate dual internal omni-directional antennas with diversity to provide wide and robust communication, and so the antennas will be protected against accidental contact with other objects in the space.
 4. Two LEDs shall be included on the bridge to provide feedback about the local network (red) and wireless network (blue) health.

5. Wireless Network bridge shall automatically create standard BACnet objects for selected DLM devices to allow any BACnet-compliant BAS to include lighting control and power monitoring features as provided by the DLM devices on each local network. BACnet objects will be created for the addition or replacement of any given DLM device for the installed life of the system. Products requiring that an application-specific point database be loaded to create or map BACnet objects are not acceptable. Systems not capable of providing BACnet data for control devices via a dedicated BACnet Device ID per room are not acceptable. Standard BACnet objects shall be provided as follows:
 - a. Read/write the normal or after hours schedule state for the room
 - b. Read the detection state of each occupancy sensor
 - c. Read the aggregate occupancy state of the room
 - d. Read/write the On/Off state of loads
 - e. Read/write the dimmed light level of loads
 - f. Read the button states of switches
 - g. Read total current in amps, and total power in watts through the load controller
 - h. Read/write occupancy sensor time delay, PIR sensitivity and ultrasonic sensitivity settings
 - i. Activate a preset scene for the room
 - j. Read/write daylight sensor fade time and day and night setpoints
 - k. Read the current light level, in foot-candles, from interior and exterior photosensors and photocells
 - l. Set daylight sensor operating mode
 - m. Read/write wall switch lock status
 - n. Read watts per square foot for the entire controlled room
 - o. Write maximum light level per load for demand response mode
 - p. Read/write activation of demand response mode for the room
 - q. Activate/restore demand response mode for the room
- C. Features of the Wireless Border Router shall be:
1. The Wireless Border Router shall manage the formation and configuration of the 6LoWPAN wireless mesh network, and provide connectivity via wired 10/100 Ethernet to a local area network that may include a LMSM Segment Manager or Building BAS System.
 2. Border Router shall provide key information about the health of the mesh network in the form of signal quality, device status, network status, and other real-time network information such as energy monitoring.
 3. The LMBR-600 shall have dual internal omni-directional antennas with diversity to ensure reliable communication with Wireless Network Bridges, and provide a user interface for set up and configuration.
 4. Include an internal MicroSD card and a Real-time clock with supercap back-up. Border Router shall get power for operation via a 120V outlet (in non-plenum applications) and a dedicated DLM LMPB-100 Power Booster connected to a Cat 5e to DC barrel connector (for plenum applications).
- D. Communication between the Wireless Network Bridges and the Border Router
1. The communication between the Wireless Bridges and the Border Router shall be via a standalone wireless mesh network that does not require interface with any other wireless network in the space. The mesh network shall allow communication between all rooms as long as they are within 100' of another connected room.
 2. The Wireless Bridges shall communicate over a 6LoWPAN 2.4 GHz IEEE 802.15.4 network and use AES 128 bit Key Encryption for network security.
 3. In addition to IEEE IPV6, the Border Router shall have available Constrained Application Protocol (CoAP), Routing Protocol for Low Power Networks (RPL), and Stateless Multicast RPL Forwarding (SMRF).

4. The wireless protocol shall allow BACnet communication to be transported transparently between the Network Bridge and any front end BAS devices such as the LMSM Segment Manager.
- E. The Master Border Router shall provide single point of integration to the Building Management System (BMS) via BACnet/IP. The lighting control system shall provide an export table to transfer room/device data to the BMS system. Data provided in the export table will be confirmed in the pre-installation meeting referred to in Section 3.02-B. Time based scheduling shall remain a function of the lighting control system unless all scheduling is coordinated with the building management system. The export table is limited to the BACnet objects listed in section 2.08-A-3. The lighting control manufacture is recommending the following BACnet objects be included as a minimum standard in the export table:
 1. Occupancy state of the room
 2. On/Off state of loads in each room
 3. Total power in watts through the load controller
 4. Read watts per square foot for the entire controlled room
 5. Current light level in foot-candles from interior and exterior photosensors
- F. WattStopper product numbers: Wireless Network Bridge LMBC-650, Wireless Border Router LMBR-650.

2.12 PROGRAMMING, CONFIGURATION AND DOCUMENTATION SOFTWARE

- A. PC-native application for optional programming of detailed technician-level parameter information for all DLM products, including all parameters not accessible via BACnet and the handled IR configuration tool. Software must be capable of accessing room-level parameter information locally within the room when connected via the optional LMCI-100 USB programming adapter, or globally for many segment networks simultaneously utilizing standard BACnet/IP communication.
- B. Additional parameters exposed through this method include but are not limited to:
 1. Occupancy sensor detection LED disable for performance and other aesthetic spaces where blinking LEDs present a distraction.
 2. Six occupancy sensor action behaviors for each controlled load, separately configurable for normal hours and after hours modes. Modes include: No Action, Follow Off Only, Follow On Only, Follow On and Off, Follow On Only with Override Time Delay, Follow Off Only with Blink Warn Grace Time, Follow On and Off with Blink Warn Grace Time.
 3. Separate fade time adjustments per load for both normal and after hours from 0 - 4 hours.
 4. Configurable occupancy sensor re-trigger grace period from 0 - 4 minutes separate for both normal hours and after hours.
 5. Separate normal hours and after hours per-load button mode with modes including: Do nothing, on only, off only, on and off.
 6. Load control polarity reversal so that on events turn loads off and vice versa.
 7. Per-load DR (demand response) shed level in units of percent.
 8. Load output pulse mode in increments of 1second.
 9. Fade trip point for each load for normal hours and after hours that establishes the dimmer command level at which a switched load closes its relay to allow for staggered On of switched loads in response to a dimmer.
- C. Generation of reports at the whole file, partial file, or room level. Reports include but are not limited to:
 1. Device list report: All devices in a project listed by type.
 2. Load binding report: All load controller bindings showing interaction with sensors, switches, and daylighting.

3. BACnet points report: Per room Device ID report of the valid BACnet points for a given site's BOM.
 4. Room summary report: Device manifest for each room, aggregated by common BOM, showing basic sequence of operations.
 5. Device parameter report: Per-room lists of all configured parameters accessible via hand held IR programmer for use with O&M documentation.
 6. Scene report: All project scene pattern values not left at defaults (i.e. 1 = all loads 100%, 2 = all loads 75%, 3 = all loads 50%, 4 = all loads 25%, 5-16 = same as scene 1).
 7. Occupancy sensor report: Basic settings including time delay and sensitivity(ies) for all occupancy sensors.
- D. Network-wide programming of parameter data in a spreadsheet-like programming environment including but not limited to the following operations:
1. Set, copy/paste an entire project site of sensor time delays.
 2. Set, copy/paste an entire project site of sensor sensitivity settings.
 3. Search based on room name and text labels.
 4. Filter by product type (i.e. LMRC-212) to allow parameter set by product.
 5. Filter by parameter value to search for product with specific configurations.
- E. Network-wide firmware upgrading remotely via the BACnet/IP network.
1. Mass firmware update of entire rooms.
 2. Mass firmware update of specifically selected rooms or areas.
 3. Mass firmware upgrade of specific products.
- F. WattStopper Product Number: LMCS-100, LMCI-100

PART 3 - EXECUTION

3.1 PREPARATION

- A. Do not begin installation until measurements have been verified and work areas have been properly prepared.
- B. If preparation is the responsibility of another installer, notify Architect of unsatisfactory preparation before proceeding.

3.2 PRE-INSTALLATION MEETING

- A. Manufacturer shall provide a factory authorized representative to provide a functional overview of the lighting control system prior to products being installed.
 1. Discuss functionality and integration of all products per design requirements.
 2. Confirm location of occupancy sensors and photocells as required.
 3. Confirm low voltage control wires meet specification.
 4. Explain adjustment options and verify specification requirements for each device.
- B. Convene meeting two weeks prior to commencing work of this section to coordinate integration of lighting control system to building management system. Meeting to be attended by electrical contractor, mechanical contractor, general contractor, system installer if not electrical contractor, factory authorized representative and IT manager from facility.
 1. Review integration requirements in this section of specification
 2. Confirm responsibilities of network terminations for electrical and mechanical system
 3. Review information to be exchanged between system and delivery method of data
 4. Identify scope of the integration including but not limited to the following use cases:

- a. Scheduling
- b. Demand Response
- c. Normal hours vs. after hours adjustments of device parameters
- d. Identify points in export table to monitor and control
 - 1) Occupancy State
 - 2) Power usage in room
 - 3) Daylight status/footcandle level

3.3 INSTALLATION

- A. Install system in accordance with the approved system shop drawings and manufacturer's instructions.
- B. Install no wireless devices within 10'-0" radius of the border router, including Wattstopper wireless network bridges and third-party devices such as wi-fi access points, security systems, RFID tagging equipment, etc. Coordinate this requirement with the general contractor and all trades.
- C. Install all room/area devices using manufacturer's factory-tested Cat 5e cable with pre-terminated RJ-45 connectors.
 - 1. If pre-terminated cable is not used for room/area wiring, each field-terminated cable shall be tested following installation and testing results submitted to the Manufacturer's Representative for approval prior to proceeding with the Work.
 - 2. If fixtures have internal DLM Control Modules, ensure that they are also connected with Cat 5e cable.
 - 3. Install all room to room network devices using manufacturer-supplied LM-MSTP network wire or wireless devices. Network wire substitution is not permitted and may result in loss of product warranty.
 - 4. Low voltage wiring topology must comply with manufacturer's specifications.
 - 5. Route network wiring as indicated on the Drawings as closely as possible. Document final wiring location, routing and topology on as built drawings.
- D. All line voltage connections shall be tagged to indicate circuit and switched legs.
- E. Test all devices to ensure proper communication.
- F. Calibrate all sensor time delays and sensitivity to guarantee proper detection of occupants and energy savings. Adjust time delay so that controlled area remains lighted while occupied.
- G. Provide written or computer-generated documentation on the configuration of the system including room by room description including:
 - 1. Sensor parameters, time delays, sensitivities, and daylighting setpoints.
 - 2. Sequence of operation, (e.g. manual ON, Auto OFF. etc.)
 - 3. Load Parameters (e.g. blink warning, etc.)
- H. Post start-up tuning - Adjust sensor time delays and sensitivities to meet the Owner's requirements 30 days from beneficial occupancy. Provide a detailed report to the Architect / Owner of post start-up activity.
- I. Tighten all panel Class I conductors from both circuit breaker and to loads to torque ratings as marked on enclosure UL label.

- J. All Class II cabling shall enter enclosures from within low-voltage wiring areas and shall remain within those areas. No Class I conductors shall enter a low-voltage area.
- K. Run separate neutrals for any phase dimmed branch load circuit. Different types of dimming loads shall have separate neutral.
- L. Verify all non-panel-based lighting loads to be free from short circuits prior to connection to room controllers.

3.4 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing. Notify Architect and Manufacturer in writing a minimum of 3 weeks prior to system start-up and testing.
- B. Tests and Inspections: Manufacturer's service representative shall perform the following inspections and prepare reports.
 - 1. Verify Class I and II wiring connections are terminated properly by validating system performance.
 - 2. Set IP addresses and other network settings of system front end hardware per facilities IT instructions.
 - 3. Verify / complete task programming for all switches, dimmers, time clocks, and sensors.
 - 4. Verify that the control of each space complies with the Sequence of Operation.
 - 5. Correct any system issues and retest..
- C. Provide a report in table format with drawings, or using a software file that can be opened in the manufacturer's system software including each room or space that has lighting control installed. Indicate the following:
 - 1. Date of test or inspection.
 - 2. Loads per space, or Fixture Address identification.
 - 3. Quantity and Type of each device installed
 - 4. Reports providing each device's settings.

3.5 SYSTEM STARTUP

- A. The Manufacturer shall provide a factory authorized technician to commission and confirm proper installation and operation of all system components.
- B. Contractor shall provide system documentation after the equipment has been installed:
 - 1. Lighting control operational summary sheet.
 - 2. Programming record sheet.
 - 3. System Installation and Operation Manual shall be provided to the owner.

3.6 TRAINING

- A. Manufacturer shall provide factory authorized application engineer to train Owner personnel in the operation and programming of the lighting control system for the first (2) two days of occupancy; then (1) one week later, and again (1) month later.

3.7 PRODUCT SUPPORT AND SERVICE

- A. Factory telephone support shall be available at no cost to the Owner following acceptance. Factory assistance shall consist of assistance in solving application issues pertaining to the control equipment.

3.8 TESTS

- A. Test under provisions of Section 26 05 00 and 26 08 00.
- B. The Owner shall be notified at least three working days in advance of the Contractor's proposed date of the tests to permit scheduling, and to permit witnessing of the tests. The Contractor shall furnish the Owner with three copies of the results of the tests.
- C. Circuits: The Contractor shall test each circuit, all controllers, and components of the system for proper operation. The Contractor shall furnish the Owner with three copies of the test results.
- D. Operating Test: Contractor shall operate the system in the presence of the Owner proving the proper operation of the system and all components.

END OF SECTION